



2024
Publications
Archive

ENERQA's 2024 Publications Archive

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Contents

Foreword	2
Introduction	3
1. The Evolution of Climate Change Understanding and Response.....	6
2. Climate Forcers: The Hidden Drivers of Global Warming	8
3. Weathering the Storm: Climate Resilience in Supply Chains	10
4. Climate Change and War: A Complex Interplay -How Conflicts in Ukraine, Palestine, and Sudan in 2024 are Shaped by and Impact the Global Climate Crisis	13
5. Greenhouse Gases and Climate Change: An Overview	16
6. Breathing vs. Burning: The Carbon Footprint Contrast	20
7. GHG EMISSIONS – The Burden on Our Planet.....	22
8. Exploring the Rainbow of Hydrogen Technology: A Path to Sustainable Energy and Climate Resilience.....	24
9. Green Credit Lines in the Gulf Cooperation Council: Evaluating Opportunities, Overcoming Challenges, and Assessing the Impact on Energy Transition in Qatar	33
10. Supercritical Water Technology (SCWT) - A Sustainable Solution for Waste Management in Qatar ...	54
11. Driving Climate Action Through Renewable Energy Finance: Insights from an Expert	58
12. I-RECs - A catalyst for renewable energy investment in Qatar	62
13. The Role of Artificial Intelligence in Environmental Sustainability	64
14. The Hidden Costs of Your Burger and Pizza: What’s Really at Stake?	70
15. Practical Tips for Reducing Food Waste at Home: A Step Towards Sustainability.....	72
16. Sustainable Tourism.....	75
17. The Hidden Link Between Cigarette Smoking and Climate Change	78
18. Empowering Communities Through Social Sustainability: A Call to Action	80
19. The Origins of Urban Greening.....	84
20. Environmental Impacts of Mercury Use in Artisanal Gold Mining in Africa (Sudan- case): The Amplifying Effect of Torrential Rains and Floods.....	89
21. Understanding the DPSIR, DPCER, DPSWR, and DPSEIR Frameworks in analyzing the interactions between human activities and the environment	95
22. DPSIR Framework for Sustainable Aquaculture in the Red Sea Region (KSA): A Climate Adaptation and Mitigation Perspective	98
23. The Imperative for ESG Readiness Tools: Unlocking Benefits for Companies, Institutions, and Society.....	103
24. Scope 4 Emissions: The Concept of Avoided Emissions	107
Authors Biographies	111

Foreword

As we reflect on the year 2024, it is clear that the urgency of addressing climate change and sustainability challenges has never been greater. At ENERQA, we are proud to contribute to this global effort by offering thought leadership, actionable insights, and innovative solutions that empower decision-makers to create meaningful change.

This publication archive is a testament to our dedication to knowledge sharing and collaboration. It captures a year of groundbreaking work in renewable energy, energy efficiency, climate resilience, and environmental sustainability. Each report, article, and case study within these pages reflects the expertise and commitment of our team and partners, as well as the growing momentum in transitioning toward a more sustainable future.

The year 2024 also marked significant milestones for ENERQA. From forging partnerships with key stakeholders to spearheading initiatives that align with global goals like the Paris Agreement and the United Nations Sustainable Development Goals, we have continually strived to turn challenges into opportunities. We are especially proud of the role our publications play in inspiring innovation and driving action across sectors.

This archive is more than just a collection of publications—it is a call to action. It embodies our belief that knowledge is the foundation of change and that collaboration is the key to overcoming the challenges ahead. As you explore this archive, we hope it sparks ideas, fosters partnerships, and motivates progress in your own spheres of influence.

Thank you for being a part of this journey toward a more sustainable and equitable future.

Sincerely,

Dr. Quosay A. Ahmed

Technical Director

ENERQA

Introduction

Welcome to the *ENERQA Publications Archive for 2024*, a curated collection of 24 publications created by a team of about 10 dedicated authors between December 2023 and December 2024. This archive reflects the breadth of ENERQA's contributions through articles, conference papers, social media publications, white papers, and other forms of content. These works address critical challenges in energy, climate change, mining, environment, and sustainability, providing valuable insights to guide action and innovation in these fields.

At ENERQA, we believe that knowledge sharing is the cornerstone of meaningful change. As part of our corporate social responsibility, these publications embody our commitment to raising awareness, fostering informed dialogue, and driving solutions to some of the most pressing global issues. Many of the featured works were inspired by real-life challenges encountered by our team, making them deeply rooted in practical experiences and actionable solutions.

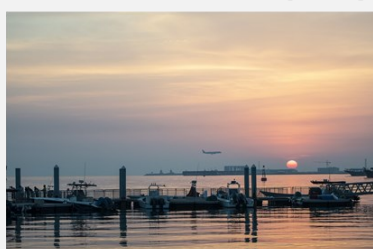
To ensure clarity and accessibility, this archive is organized into four key categories:

1. **Climate Science and Impacts:** Exploring the drivers of climate change, its wide-ranging effects, and strategies for resilience and adaptation.
2. **Energy, Technology, and Finance:** Highlighting innovations in energy systems, emerging technologies, and financial mechanisms to support the global energy transition.
3. **Environment and Society:** Examining the interconnections between environmental sustainability and social well-being, with a focus on actionable solutions for a greener future.
4. **Frameworks and Methodologies:** Providing analytical tools and frameworks to guide sustainable practices and decision-making processes.

Climate Science and Impacts



Environment and Society



Energy, Technology & Finance



Frameworks and Methodologies



What You'll Find in This Archive

Insights Driven by Challenges: Many of the featured publications address real-world problems encountered by ENERQA's team, providing practical, innovative solutions grounded in experience.

Diverse Content: From technical white papers to accessible social media articles, the archive offers materials tailored to a range of audiences, from industry experts to the general public.

Core Focus Areas: Topics span critical domains, including renewable energy, climate adaptation, sustainable mining, energy efficiency, and environmental stewardship.

Global and Regional Relevance: While rooted in ENERQA's projects, these publications address issues of global importance, offering insights that extend beyond borders.

Why Explore This Archive?

Stay Informed: Gain a comprehensive understanding of the latest trends, challenges, and opportunities in sustainability and energy sectors.

Learn from Real-Life Applications: Discover actionable solutions inspired by real challenges faced by ENERQA's team, offering unique perspectives and practical guidance.

Access Expert Knowledge: Benefit from the expertise of ENERQA's team and partners, whose contributions span diverse platforms and areas of specialization.

Be Inspired: Find motivation to take action through examples of innovative solutions, success stories, and forward-thinking ideas.

Whether you are a policymaker, academic, corporate leader, or someone passionate about building a sustainable future, this archive offers a wealth of knowledge to inform, inspire, and empower you. We invite you to delve into this collection and join us in making energy systems cleaner, communities more resilient, and our planet a better place for future generations. Let this archive serve as a catalyst for collaboration and progress as we collectively address the challenges and opportunities of our time.

Climate Science and Impacts

1. The Evolution of Climate Change Understanding and Response
2. Climate Forcers: The Hidden Drivers of Global Warming
3. Weathering the Storm: Climate Resilience in Supply Chains
4. Climate Change and War: A Complex Interplay
5. Greenhouse Gases and Climate Change: An Overview
6. Breathing vs. Burning: The Carbon Footprint Contrast
7. GHG Emissions – The Burden on Our Planet



The Evolution of Climate Change Understanding and Response

By: Mohamed M. Ahmed and Quosay A. Awad

Abstract

The field of climate science has undergone significant transformation over the past century, driven by pivotal discoveries, technological advancements, and international cooperation. This article traces the historical milestones in climate change research, policy development, and global temperature anomalies, emphasizing the interplay between science and policy. A key figure is provided to visualize global temperature anomalies relative to the 1961–1990 baseline and major events in climate change awareness.

Introduction

Climate change has progressed from an obscure scientific concept to a central focus of global policy. Beginning with early evidence of rising global temperatures, the scientific community has built a robust understanding of anthropogenic climate impacts. Simultaneously, policy frameworks have evolved, culminating in agreements like the Paris Accord, aiming to mitigate climate impacts and adapt to inevitable changes.

Historical Overview and Major Discoveries

The trajectory of climate change science is marked by key milestones:

1. **1938:** Guy Callender demonstrated rising global temperatures for the first time, setting the foundation for modern climate studies.
2. **1958:** Charles David Keeling's measurements linked fossil fuel combustion to rising atmospheric CO₂ levels*.
3. **1967:** The first climate change modeling was conducted, revealing the potential magnitude of anthropogenic impacts.
4. **1985:** The discovery of the ozone hole highlighted the interaction between human activities and atmospheric chemistry.
5. **1988:** The establishment of the Intergovernmental Panel on Climate Change (IPCC) ushered in a new era of research and international collaboration.

Global Temperature Anomalies

Figure 1 illustrates global temperature anomalies from 1935 to 2015, highlighting major events in climate science. Anomalies are presented relative to the 1961–1990 baseline. Early 20th-century temperatures show minimal deviation, while late 20th and early 21st centuries exhibit sharp increases, particularly after 1990.

* Charles David Keeling, a geochemist at the Scripps Institution of Oceanography, pioneered the precise measurement of atmospheric CO₂ in the late 1950s. His work began with a simple question: could CO₂ levels in the atmosphere be accurately measured and tracked? This pursuit led him to develop groundbreaking methods for high-precision measurements using an infrared gas analyzer.

In 1958, Keeling established a measurement station at the Mauna Loa Observatory in Hawaii. This remote location, away from industrial pollution, was chosen to capture a baseline measurement of atmospheric CO₂. Initial readings were groundbreaking: they revealed an annual cycle of CO₂ levels fluctuating with plant growth in the Northern Hemisphere. Over time, the data revealed an unequivocal trend—CO₂ concentrations were steadily increasing.

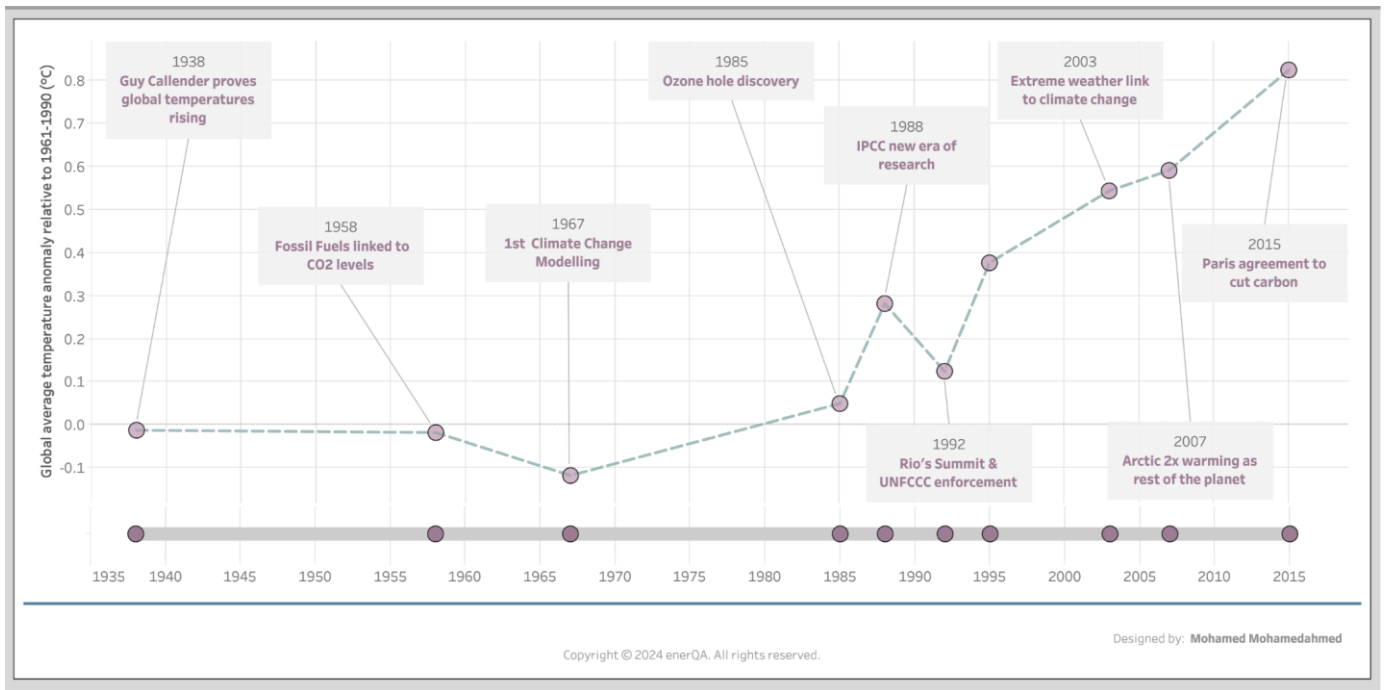


Figure 1: Global Temperature Anomalies Relative to 1961–1990 and Climate Milestones

The Role of Policy

Policy responses have mirrored scientific advancements:

- **1992:** The United Nations Framework Convention on Climate Change (UNFCCC) provided a legal basis for global cooperation.
- **1995:** The Kyoto Protocol and the first Conference of Parties (COP) in Berlin solidified commitments to reduce emissions.
- **2015:** The Paris Agreement marked a turning point, uniting nations in efforts to cap global temperature rise to below 2°C.

In addition to other global actions, such as:

- 1- Montreal Protocol
- 2- Extreme weather preparedness
- 3- Carbon pricing and emission trading schemes
- 4- Net Zero Commitments

Challenges and Future Directions

Despite progress, challenges persist. Current trajectories suggest the need for more aggressive mitigation strategies to avoid catastrophic outcomes. Emphasis on adaptation, equity, and innovation will be vital for future climate resilience.

Economic and Political Resistance: Fossil fuel-dependent economies and industries often resist climate policies.

Equity Concerns: Balancing the responsibilities and capacities of developed and developing nations remains contentious.

Funding Gaps: Climate finance commitments often fall short of promised levels, hindering mitigation and adaptation efforts in vulnerable countries.

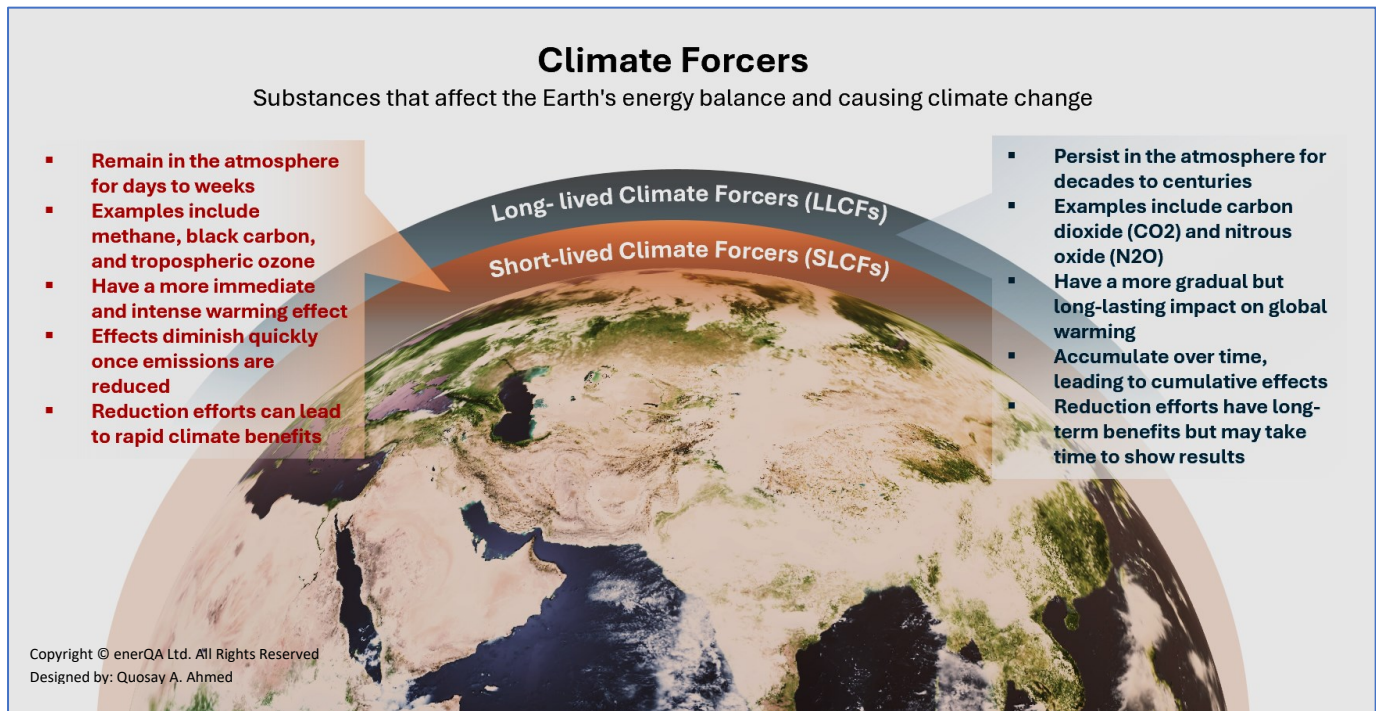
References:

1. Intergovernmental Panel on Climate Change (IPCC). *Climate Change: The IPCC Scientific Assessment*. Cambridge University Press, 1990.
2. UNFCCC. *The United Nations Framework Convention on Climate Change*. UN, 1992.
3. Paris Agreement. *Adopted Framework for Global Climate Mitigation*. UNFCCC, 2015.
4. Keeling, C. D. (1960). *The Concentration and Isotopic Abundances of Carbon Dioxide in the Atmosphere*. *Tellus*, 12(2), 200–203.
5. Intergovernmental Panel on Climate Change (IPCC). (2001). *Climate Change 2001: The Scientific Basis*. Cambridge University Press.

Climate Forcers: The Hidden Drivers of Global Warming

By: Quosay A. Ahmed

We often hear about carbon dioxide as the primary reason behind global warming, but there's more to the story. A variety of substances, known as climate forcers, are playing a significant role in heating or cooling our planet. Let's delve into the world of climate forcers and understand how they're contributing to the climate crisis.



Climate forcers come in two main types: short-lived and long-lived.

1- Long-Lived Climate Forcers (LLCFs):

Long-lived climate forcers are greenhouse gases that remain in the atmosphere for an extended period, ranging from several decades to centuries. The most well-known LLCFs include:

Carbon Dioxide (CO₂): The primary greenhouse gas emitted through human activities, particularly the burning of fossil fuels and deforestation. CO₂ can remain in the atmosphere for hundreds to thousands of years.

Methane (CH₄): Although methane is also considered a short-lived climate forcer due to its shorter atmospheric lifetime (about 12 years), its long-term impact on climate is significant because it is much more effective at trapping heat compared to CO₂.

Nitrous Oxide (N₂O): Emitted from agricultural activities and industrial processes, nitrous oxide has a lifetime of about 114 years.

Fluorinated Gases: These include hydrofluorocarbons (HFCs), perfluorocarbons (PFCs), and sulfur hexafluoride (SF₆), which are synthetic gases used in various industrial applications. They can remain in the atmosphere for thousands of years.

LLCFs contribute to long-term climate change by trapping heat in the atmosphere, leading to a gradual increase in global temperatures. Their persistent nature means that even if emissions were to stop today, their effects would continue to influence the climate for many years to come.

2- Short-Lived Climate Forcers (SLCFs)

Short-lived climate forcers are substances that have a much shorter atmospheric lifetime, typically ranging from a few days to a few decades. Despite their short lifespan, SLCFs can have a significant impact on the climate and air quality. Key SLCFs include:

Black Carbon: A component of particulate matter (PM) produced from incomplete combustion of fossil fuels, biofuel, and biomass. Black carbon absorbs sunlight and warms the atmosphere but only remains in the atmosphere for days to weeks.

Tropospheric Ozone (O₃): Unlike stratospheric ozone, which protects us from harmful UV radiation, tropospheric ozone is a pollutant formed by the reaction of sunlight with pollutants like nitrogen oxides (NO_x) and volatile organic compounds (VOCs). It has a lifetime of a few weeks.

Hydrofluorocarbons (HFCs): While some HFCs are long-lived, others have shorter lifetimes and are used as refrigerants and in other industrial applications.

SLCFs can have both warming and cooling effects on the climate. For example, black carbon has a warming effect, while some aerosols can reflect sunlight and cool the atmosphere. The impact of SLCFs is more immediate compared to LLCFs, making them a critical target for short-term climate mitigation efforts.

In addition to the main short- and long-lived forcers we discussed earlier, here are some others:

Water Vapor (H₂O)

Impact: Water vapor is the most abundant greenhouse gas, but it behaves differently than CO₂ or methane. It amplifies the effects of other greenhouse gases because, as the atmosphere warms due to CO₂ or methane, more water evaporates into the air. This extra water vapor traps more heat, creating a feedback loop that increases global warming.

Duration: It has a short life cycle, cycling out of the atmosphere through precipitation within days, but it plays a critical role in the overall climate system.

Aerosols (Reflective Particles)

Impact: Aerosols, like sulfate particles, reflect sunlight back into space, which cools the Earth. They are often produced by industrial pollution, burning of fossil fuels, and volcanic eruptions. While they have a cooling effect, this does not counterbalance the warming from greenhouse gases.

Duration: Aerosols have a short lifespan in the atmosphere (days to weeks), but their cooling effects can mask some of the warming caused by greenhouse gases.

Clouds

Impact: Clouds can act as both climate forcers and climate dampeners. Low clouds tend to reflect sunlight, having a cooling effect, while high clouds trap heat, contributing to warming. Their overall effect on climate is complex and depends on various factors such as altitude and type.

References:

1. IPCC AR5 (2014) - *Climate Change 2013: The Physical Science Basis*. Available at: IPCC
2. Friedlingstein, P., et al. (2019). "Global carbon budget 2019." *Earth System Science Data*, 11(4), 1783-1838.
3. Ramanathan, V., & Carmichael, G. (2008). "Global and regional climate changes due to black carbon." *Nature Geoscience*, 1(4), 221-227.
4. Monks, P. S., et al. (2015). "Tropospheric ozone and its precursors from the urban to the global scale from air quality to short-lived climate forcer." *Atmospheric Chemistry and Physics*, 15(15), 8889-8973.
5. WMO (2018). "Scientific Assessment of Ozone Depletion: 2018." Available at: WMO
6. IPCC AR6 (2021) - *Climate Change 2021: The Physical Science Basis*. Available at: IPCC
7. IPCC AR5 (2014) - *Climate Change 2013: The Physical Science Basis*. Available at: IPCC

Weathering the Storm: Climate Resilience in Supply Chains

By: Abdelaziz M. A. Ahmed

Supply chains, the intricate networks that connect businesses and consumers worldwide, have become increasingly vital to modern economies. They ensure the smooth flow of goods and services, underpinning our daily lives. In recent decades, supply chains have evolved into sophisticated systems, a marvel of human ingenuity and cooperation. However, this complex interconnectedness also makes them vulnerable to a growing threat: climate change.

Supply Chains and Climate Change

Supply chains are a major contributor to climate change due to their reliance on fossil fuels for transportation, energy consumption, and manufacturing processes. According to the Carbon Disclosure Project (CDP), of the 18,600 companies that disclosed their carbon emission data through the CDP in 2022, **scope 3 emissions** (often referred to as supply chain emissions) are, on average, 11.4 times greater than **scope 1 and 2 emissions** (often referred to as operational emissions) [1]. Additionally, the production and consumption of goods and services within supply chains contribute to emissions through energy use, deforestation, and waste generation. Not only are supply chains significant contributors to climate change through greenhouse gas emissions, but they are also increasingly impacted by its adverse effects. A study from the University of Oxford estimates that climate change is already jeopardizing global trade and economic activity, putting a staggering US\$81 billion of trade and US\$122 billion of economic activity at risk annually. Climate-related events such as extreme weather, rising sea levels, and natural disasters pose significant risks to supply chain operations. For example, the summer of 2022 brought low water levels to the Rhine River, causing severe delays to shipping arrivals and departures. Some vessels were forced to sail with cargoes at just 25% of capacity as Europe endured its worst drought in 500 years [2].

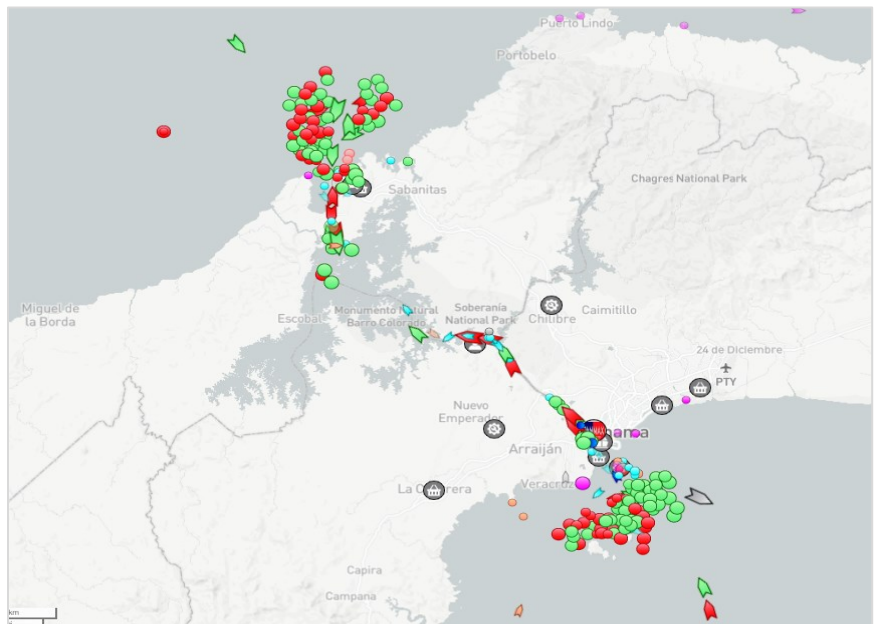


Figure 2: Another example of how recent climate events disrupt supply chains: Ship Buildup at the Panama Canal due to drought and unpredictable weather patterns in 2023 [15].

Mitigation and building resilience

To mitigate climate change and enhance supply chain resilience, organizations must take proactive steps. One crucial strategy is to reduce greenhouse gas emissions throughout the supply chain. This can be achieved by transitioning to renewable energy sources, improving energy efficiency, and adopting sustainable practices. Additionally, organizations can support suppliers who are also committed to reducing their environmental impact.

Building resilience against climate-related supply chain risks is equally important. Supply chains must become more diversified, collaborative, and adaptable. Long-term planning and risk-based management are essential to proactively address climate-related challenges. This can be achieved using the following strategies:

- Avoiding over-reliance on single regions or suppliers.
- Sharing data across the supply chain to enable better monitoring, prediction, and collective action in response to extreme weather patterns.
- Investing in early warning systems to allow for timely responses and mitigation efforts.
- Identifying vulnerable points in supply chains and specific risks posed by climate change such as vulnerable waterways and ports, endangered transportation infrastructure, unprotected manufacturing and production facilities, and agricultural production threatened by droughts and extreme weather.

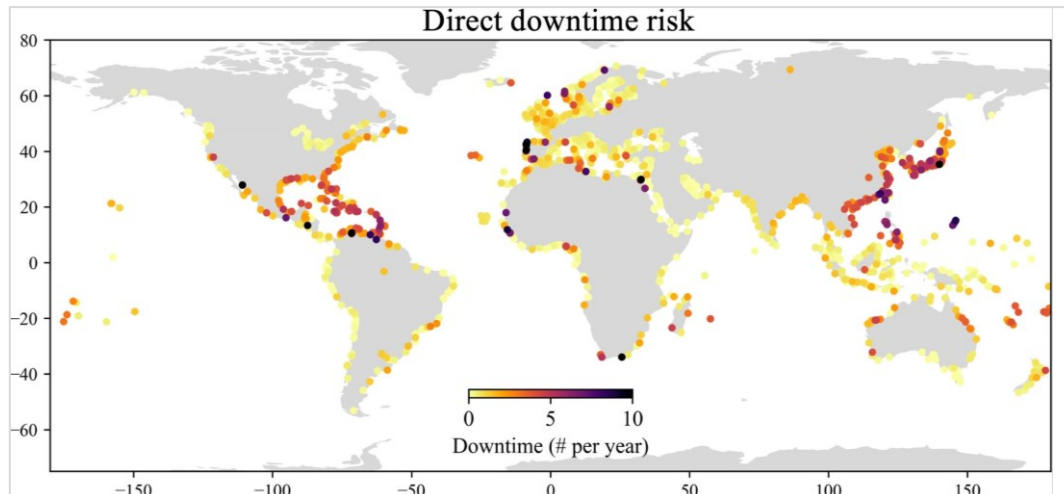


Figure 3: Identifying vulnerabilities in supply chains: The direct downtime risk (annual expected days per year) at seaports [2].

In summary, climate change poses a significant threat to global supply chains. By understanding the risks and taking proactive measures, organizations can mitigate their own environmental impact and build resilience to climate-related disruptions. Supply chain companies and industries heavily reliant on supply chains should be at the forefront of efforts to reduce their carbon footprint. Like a smoker who realizes the negative impact of their habit on themselves and others, supply chain organizations should prioritize carbon reduction. This not only safeguards their own operations but also helps create a more sustainable and resilient global economy.

- [1] "Scoping Out: Tracking Nature Across The Supply Chain," CDP, 2023.
- [2] E. Thomson, "Droughts are creating new supply chain problems. This is what you need to know," World Economic Forum, 25 October 2023. [Online]. Available: <https://www.weforum.org/agenda/2023/10/drought-trade-rivers-supply-chain/>. [Accessed 25 September 2024].
- [3] Google, "Google," [Online]. Available: (DeepMind AI reduces energy used for cooling Google data centers by 40%).
- [4] Tesla, "Tesla," [Online]. Available: Best Tesla Powerwall in 2024: Here.
- [5] BCG, [Online]. Available: <https://www.bcg.com/publications/2023/how-ai-can-speedup-climate-action?utm>.
- [6] IBM, [Online]. Available: <https://newsroom.ibm.com/2023-03-14-IBM-launches-RFP-to-help-accelerate-global-water-management-solutions-for-vulnerable-populations?utm>.
- [7] Seoul. [Online]. Available: <https://english.seoul.go.kr/seoul-to-increase-recycling-rate-to-79-by-2026/?utm>.
- [8] Microsoft. [Online]. Available: <https://news.microsoft.com/apac/2021/10/11/microsoft-ai-for-earth-named-as-one-of-100-outstanding-biodiversity-positive-practices-and-actions-around-the-world>.
- [9] A. M. Research. [Online]. Available: Artificial Intelligence (AI) Market Size, Share, Trends – 2033).
- [10] I. o. S. Studies, 14 October 2024. [Online]. Available: <https://instituteofsustainabilitystudies.com/insights/lexicon/what-are-scope-4-emissions/>.
- [11] W. B. C. f. S. D. (WBCSD), Guidance on Avoided Emissions: Helping business drive innovations and scale solutions toward Net Zero, Geneva, Amsterdam, New Delhi, London, New York, Singapore: World Business Council for Sustainable Development (WBCSD), 2023.
- [12] W. R. INSTITUTE, "WORLD RESOURCED INSTITUTE," 05 November 2013. [Online]. Available: <https://www.wri.org/insights/do-we-need-standard-calculate-avoided-emissions>.
- [13] M. D. G. G. C. L. M. A. Lauren Shwisberg, "How to Build Clean Energy Portfolios A Practical Guide to Next-Generation Procurement Practices," Rocky Mountain Institute, Basalt, CO, 81621 USA, 2020.
- [14] P. A. [Online]. Available: <https://plana.earth/glossary/scope-4-emissions>.

- [15] P. R. Attack, "Panama Canal Authority blames queues on drought and draft cut," *Ship-Technology.com*, 11 August 2023. [Online]. Available: <https://www.ship-technology.com/news/panama-canal-authority-blames-queues-drought-draft-cut/>. [Accessed 25 September 2024].
- [16] J. Verschuur, E. E. Koks and J. W. Hall, "Systemic risks from climate-related disruptions at ports," *Nature Climate Change*, vol. 13, no. 8, p. 804–806, 2023.
- [17] L. Woetzel, D. Pinner, H. Samandari, H. Engel, M. Krishnan, C. Kampel and J. Graabak, "Could climate become the weak link in your supply chain?," *McKinsey Global Institute*, 6 August 2020. [Online]. Available: <https://www.mckinsey.com/capabilities/sustainability/our-insights/could-climate-become-the-weak-link-in-your-supply-chain>. [Accessed 25 September 2024].
- [18] "Climate-Related Supply Chain Risk: Are You Prepared? - FTI Strategic Communications," *FTI Consulting*, 15 December 2022. [Online]. Available: <https://fticomunications.com/climate-related-supply-chain-risk-are-you-prepared/>. [Accessed 25 September 2024].
- [19] G. G. P. (. Protocol). [Online]. Available: <https://ghgprotocol.org/standards-guidance>.

Climate Change and War: A Complex Interplay -How Conflicts in Ukraine, Palestine, and Sudan in 2024 are Shaped by and Impact the Global Climate Crisis

By: Dr. Islam M. Awad

In 2024, the world faces ongoing conflicts in Ukraine, Palestine, and Sudan, which are not only tragic in human terms but also profoundly interconnected with the global climate crisis. As global temperatures rise and environmental degradation worsens, climate change acts as both a trigger and amplifier of conflict. At the same time, wars damage ecosystems, disrupt environmental governance, and increase greenhouse gas (GHG) emissions, intensifying the environmental toll.



Destruction in Ukraine's thermal power

Source: State Emergency Service of Ukraine/Handout via REUTERS

Climate Change as a Conflict Trigger

Climate change has long been recognized as a "threat multiplier," intensifying underlying tensions, particularly in regions already facing political instability and resource scarcity. For example, Sudan has experienced prolonged droughts that severely impact agriculture, increasing competition over resources like water and fertile land. These conditions can foster disputes between communities, particularly in rural areas, fuelling the ongoing conflict (World Resources Institute, 2021).

Similarly, in Ukraine, the war is disrupting access to critical energy sources, including clean energy infrastructure. Ukraine has been on a path toward expanding renewable energy; however, the war has halted many projects, making the energy transition harder and deepening reliance on fossil fuels (IEA, 2022).

News Spotlight: War's Devastating Environmental Impact

Recent news reports show how wars in 2024 are exacerbating environmental degradation. For instance, the war in Ukraine has led to the destruction of key energy infrastructure, including the Zaporizhzhia Nuclear Power



An informal refugee settlement in Tina, Sudan, on Oct. 22, 2023.

Source: Jérôme Tubiana photos.

Plant, raising concerns of catastrophic contamination (BBC News, 2024). In Palestine, the bombardment of cities has destroyed natural habitats and damaged water supplies, while in Sudan, deforestation and desertification are rapidly increasing due to uncontrolled migration and urban sprawl as millions flee the violence (Al Jazeera, 2024).

These crises illustrate that while war is an immediate human disaster, the long-term environmental consequences can linger for decades, adding stress to already fragile ecosystems.

Wars Escalate Greenhouse Gas Emissions

One often overlooked consequence of war is its impact on GHG emissions. A recent study published by the Journal of Environmental Science and Policy revealed that military operations, including the movement of troops and equipment, fuel massive carbon emissions. In Ukraine, the burning of fossil fuels to power military operations and replace damaged infrastructure has pushed carbon emissions back up, negating progress made in recent years (Environmental Science and Policy, 2023).

Additionally, damage to Ukraine's forests and agricultural land has reduced carbon sinks, while the destruction of renewable energy plants has forced a temporary shift to more carbon-intensive energy sources, further compounding the issue (UNEP, 2023).



Water-scarce areas in Palestine, Gaza.

Source: Abed Rahim Khatib – Anadolu Agency on February 19, 2024

Climate Refugees: The Intersection of Conflict and Migration

Wars create large-scale displacement, and when combined with climate impacts, the situation worsens. In Sudan, nearly 4 million people have been displaced by the ongoing conflict, exacerbating environmental stress in neighboring countries where refugee camps are located. The lack of sustainable infrastructure in these camps leads to deforestation and water pollution, which not only damages the environment but also undermines the quality of life for displaced people (UNHCR, 2024). According to the *World Bank*, climate-induced migration is expected to increase dramatically if climate action and conflict resolution are not prioritized in global policy discussions.

Sudan's Extreme Weather in 2024: A Double Blow of Floods and Droughts

In 2024, Sudan faced a devastating combination of floods and droughts, further exacerbating the ongoing humanitarian crisis caused by conflict. Climate change has intensified the severity and frequency of extreme weather in the region. According to recent reports, Sudan experienced one of the worst floods in decades, displacing thousands of people, destroying infrastructure, and contributing to the spread of waterborne diseases (UN, 2024). At the same time, large parts of the country suffered from prolonged droughts, crippling agricultural production and deepening food insecurity.

The interaction between extreme weather events and ongoing conflict has created a "climate-conflict nexus." With resources already scarce due to war, floods and droughts have amplified tensions among communities, particularly in rural areas. Water sources have become unreliable, and competition over fertile land has increased, leading to further displacement and violence (World Resources Institute, 2024).

Additionally, the environmental damage caused by these extreme weather events is immense. Floodwaters have destroyed homes, crops, and infrastructure, while the drought has led to land degradation, making recovery even more difficult. This situation has placed further strain on refugee camps and humanitarian efforts already struggling to cope with the impacts of conflict (UNHCR, 2024).



Source: AFP PHOTO / ASHRAF SHAZLY.

Linking Science to Solutions: Building Climate Resilience in Post-Conflict Recovery

Scientific research emphasizes the need for integrating climate resilience into peace-building and recovery efforts in conflict zones. Rebuilding efforts should prioritize sustainable energy solutions such as solar and wind, as well as water conservation and reforestation projects. For example, Ukraine's post-war recovery could serve as a model for green reconstruction by focusing on renewable energy and infrastructure (World Bank, 2023).

In Palestine, efforts to manage water sustainably through modern technologies like desalination could help prevent future conflicts, while reforesting degraded land in Sudan could create jobs, reduce emissions, and restore ecosystems (FAO, 2023).

The conflicts in Ukraine, Palestine, and Sudan demonstrate that climate change is not just an environmental issue but a security challenge. As wars exacerbate environmental degradation and climate change triggers resource conflicts, we must address both together. Integrating climate resilience into post-conflict recovery will not only prevent future wars but also contribute to the global fight against climate change.

References

1. Al Jazeera. (2024). Sudan's Environmental Crisis Amid Conflict. <https://www.aljazeera.com/news/2024/03/sudan-conflict-environment>
2. BBC News. (2024). Ukraine War and the Threat to Nuclear Facilities. <https://www.bbc.com/news/world-ukraine-2024>
3. Environmental Science and Policy. (2023). War and Emissions: The Carbon Cost of Conflict. *Journal of Environmental Science and Policy*, 12(3), 98-110.
4. Food and Agriculture Organization (FAO). (2023). Reforesting Conflict Zones: A Path to Climate Resilience. <https://www.fao.org/sudan-reforestation>
5. International Energy Agency (IEA). (2022). Ukraine's Energy Crisis: Reversing the Renewable Transition. <https://www.iea.org/ukraine-crisis-report>
6. United Nations Environment Programme (UNEP). (2023). Environmental Impact of War: Ukraine and Beyond. <https://www.unep.org/resources/ukraine-war-impact>
7. United Nations High Commissioner for Refugees (UNHCR). (2024). Sudan: Conflict, Displacement, and Environmental Impact. <https://www.unhcr.org/sudan-2024>
8. World Bank. (2023). Building Back Better: Climate Resilience in Post-War Ukraine. <https://www.worldbank.org/reports/ukraine-recover>

Greenhouse Gases and Climate Change: An Overview

By: Quosay A. Ahmed

Introduction

Greenhouse gases (GHGs) are atmospheric constituents that absorb and emit infrared radiation, contributing to the greenhouse effect and global warming. The most significant GHGs include carbon dioxide (CO₂), methane (CH₄), and nitrous oxide (N₂O). Human activities such as the burning of fossil fuels, industrial processes, and deforestation have dramatically increased the concentration of these gases, leading to a rise in global average temperatures [1] [2]

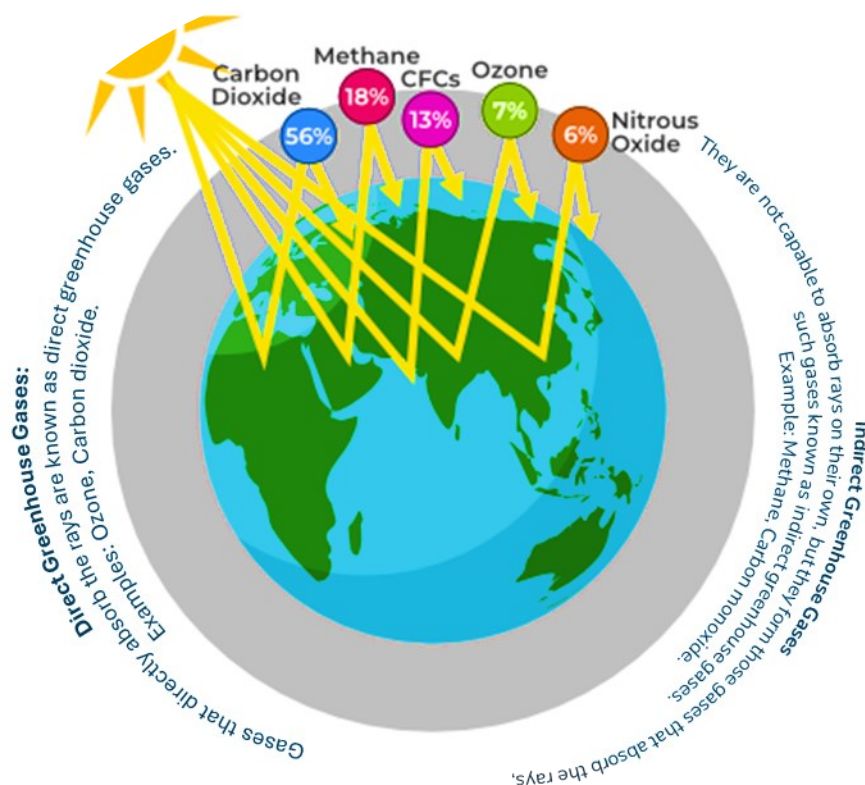


Fig-1: The concept of global warming caused by direct and indirect greenhouse gases

Scientific Consensus on Climate Change

The Intergovernmental Panel on Climate Change (IPCC), the leading international authority on climate science, has consistently emphasized the overwhelming evidence linking human activities to climate change. In its Fifth Assessment Report (AR5), the IPCC concluded that it is "extremely likely" that human influence has been the dominant cause of observed warming since the mid-20th century [3] [4]. This conclusion reflects the consensus of thousands of scientists worldwide and underscores the urgency of mitigating anthropogenic emissions.

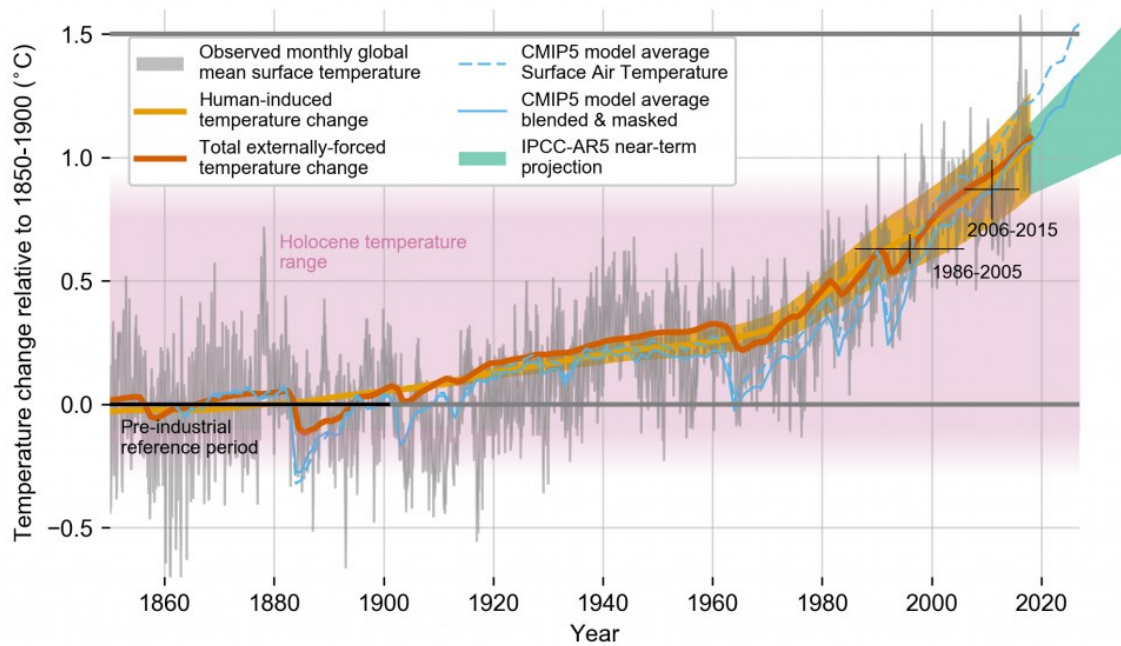


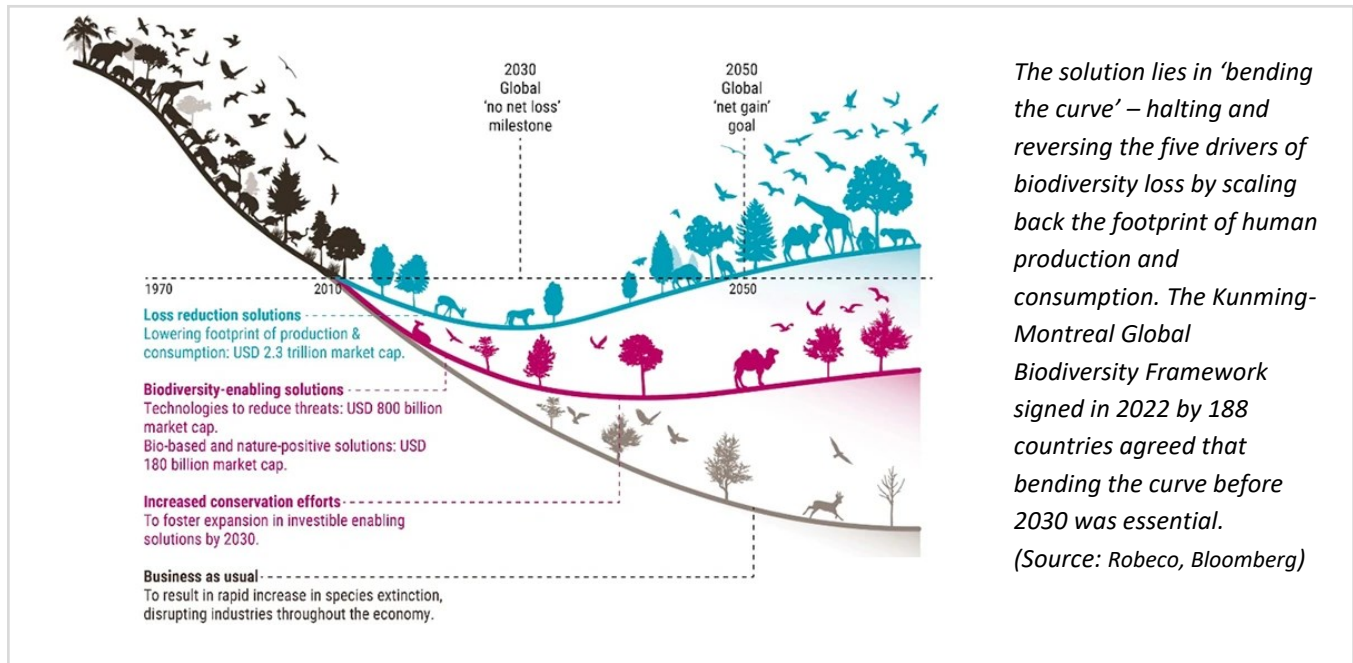
Fig-2: Earth average temperature change over the past 150 years (Source: IPCC – Chater 1- AR5)

Impacts of Climate Change

The consequences of climate change are already evident and are expected to intensify in the coming decades. These include:

- **Rising Sea Levels:** Coastal communities and ecosystems face significant threats from inundation and erosion due to melting ice caps and thermal expansion of seawater [5] [6] .
- **Extreme Weather Events:** Increased frequency and intensity of heatwaves, droughts, floods, and wildfires are disrupting lives and livelihoods [7] [8] .
- **Changes in Precipitation Patterns:** While some regions face water scarcity, others are experiencing heightened risks of flooding [9] .
- **Biodiversity Loss:** Climate change is leading to habitat destruction and ecosystem disruptions, resulting in the loss of species [10] [11] .





The Paris Agreement

In response to the global climate crisis, the Paris Agreement was adopted in 2015 by 195 Parties at the 21st Conference of the Parties (COP21) under the United Nations Framework Convention on Climate Change (UNFCCC). Its primary goal is to limit global warming to well below 2°C, preferably to 1.5°C, above pre-industrial levels [12] .

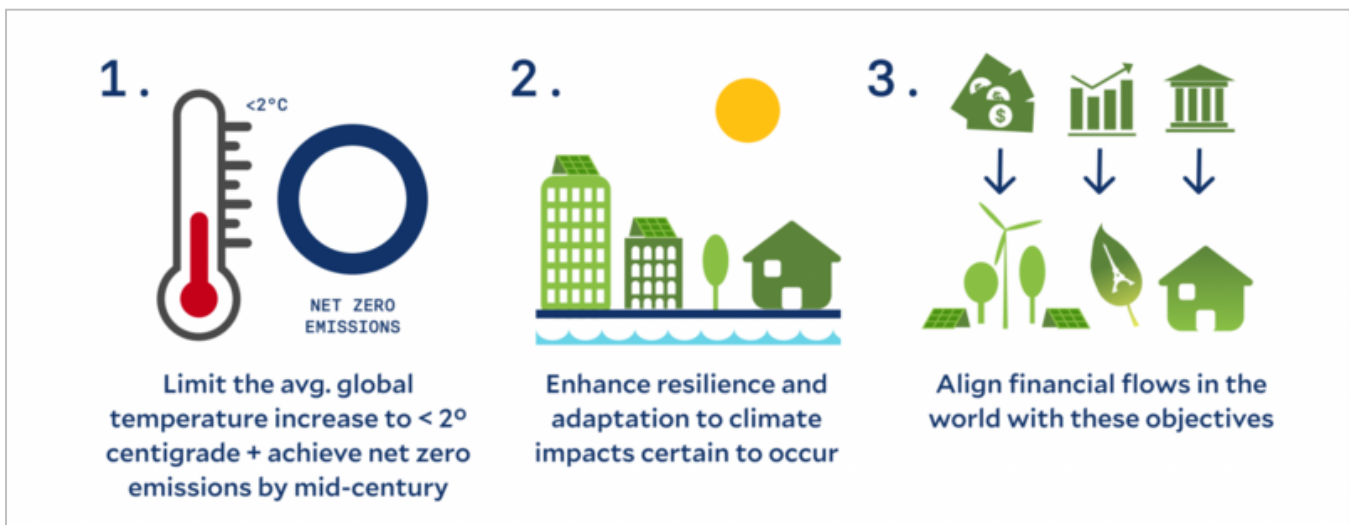


Fig-3: Explanation of Paris Agreement (After Yale Sustainability- www.sustainability.yale.edu/explainers/yale-experts-explain-paris-climate-agreement)

Key Provisions of the Paris Agreement

The Paris Agreement sets forth several critical measures to address climate change:

1. **Nationally Determined Contributions (NDCs):** Countries must outline and update their plans to reduce GHG emissions every five years, aiming for the highest possible ambition [13] .
2. **Long-Term Goal:** Achieving global net-zero GHG emissions by 2050 is a central target [14] .
3. **Financial Support:** Developed nations pledged to provide \$100 billion annually to help developing countries implement their climate strategies [12] [15] .
4. **Transparency Framework:** Parties are required to report their progress and undergo international review to ensure accountability [13] .

Implications of the Paris Agreement

The Paris Agreement represents a historic milestone in global climate governance. By establishing a framework for collective action, it has spurred countries to adopt ambitious national climate plans. However, the current trajectory of global emissions highlights the gap between commitments and actions. According to the UNFCCC, the latest NDCs are projected to result in global GHG emissions of 52.4 Gt CO₂ eq by 2030, a reduction of only 2.6% compared to current levels [16] [17] .

This sobering statistic underscores the need for more ambitious targets and accelerated action. Transforming global energy systems, enhancing financial support, and adopting innovative technologies are critical to bridging this gap and achieving the goals of the Paris Agreement [18] [19] .

References

1. NASA, 2023. What are Greenhouse Gases?. Available at: <https://climate.nasa.gov>.
2. United Nations, 2021. Climate Action: Greenhouse Gases. Available at: <https://www.un.org/en/climatechange>.
3. Intergovernmental Panel on Climate Change (IPCC), 2014. Fifth Assessment Report (AR5): Synthesis Report. Available at: <https://www.ipcc.ch/report/ar5/syr/>.
4. Cook, J. et al., 2016. Consensus on Consensus: A Synthesis of Consensus Estimates on Human-Caused Global Warming. *Environmental Research Letters*, 11(4), p.048002.
5. World Meteorological Organization (WMO), 2023. Sea Level Rise and its Implications. Available at: <https://public.wmo.int>.
6. IPCC, 2021. Sixth Assessment Report (AR6): Impacts, Adaptation, and Vulnerability. Available at: <https://www.ipcc.ch/report/ar6/wg2/>.
7. Emanuel, K., 2020. Tropical Cyclones and Climate Change. *Annual Review of Earth and Planetary Sciences*, 48, pp. 355-389.
8. Diffenbaugh, N. S., & Burke, M., 2019. Global Warming has Increased Global Economic Inequality. *Proceedings of the National Academy of Sciences*, 116(20), pp. 9808-9813.
9. UN Water, 2022. Climate Change and Water. Available at: <https://www.unwater.org>.
10. Thomas, C. D. et al., 2004. Extinction Risk from Climate Change. *Nature*, 427(6970), pp. 145-148.
11. WWF, 2022. Living Planet Report 2022. Available at: <https://www.worldwildlife.org/reports>.
12. UNFCCC, 2015. Paris Agreement. Available at: <https://unfccc.int/process-and-meetings/the-paris-agreement>.
13. UNEP, 2022. Emissions Gap Report 2022. Available at: <https://www.unep.org>.
14. Rogelj, J. et al., 2015. Paris Agreement Climate Proposals Need a Boost to Keep Warming Well Below 2°C. *Nature*, 534, pp. 631-639.
15. OECD, 2023. Climate Finance Provided and Mobilised by Developed Countries in 2020. Available at: <https://www.oecd.org>.
16. UNFCCC, 2023. Synthesis Report on the Aggregate Effect of NDCs. Available at: <https://unfccc.int>.
17. Climate Action Tracker, 2023. Paris Agreement Compatible Pathways. Available at: <https://climateactiontracker.org>.
18. International Energy Agency (IEA), 2022. Net Zero by 2050: A Roadmap for the Global Energy Sector. Available at: <https://www.iea.org/reports/net-zero-by-2050>.
19. World Bank, 2023. Scaling Climate Finance for Low-Carbon Development. Available at: <https://www.worldbank.org>.

Breathing vs. Burning: The Carbon Footprint Contrast

By: Quosay A. Ahmed

Introduction

The human body, a marvel of biological engineering, produces carbon dioxide (CO₂) as a natural byproduct of respiration. While essential for life, this CO₂ emission often gets overshadowed by the staggering amounts of CO₂ released through human activities, particularly the burning of fossil fuels. This article delves into the shades of these two sources of CO₂, comparing their magnitudes and implications for our planet.

All living organisms contribute to atmospheric CO₂ levels through respiration, a fundamental biological process. Respiration involves the breakdown of organic molecules (like glucose) to release energy, producing CO₂ as a byproduct. Here's how different organisms contribute:

1. **Humans and Animals:** Through cellular respiration, humans and animals inhale oxygen and exhale CO₂. The amount of CO₂ produced depends on metabolic activity, which is influenced by factors like physical activity, diet, and body size. On average, a human exhales about 1 kilogram of CO₂ daily (Houghton, 2007).
2. **Plants:** Plants contribute to atmospheric CO₂ during respiration, particularly at night when photosynthesis ceases. However, plants act as net carbon sinks because they absorb more CO₂ for photosynthesis than they emit (Raven et al., 2021).
3. **Microorganisms:** Decomposers, such as fungi and bacteria, break down organic matter, releasing CO₂. This process is vital for nutrient recycling in ecosystems but also adds to natural CO₂ emissions (Schlesinger & Bernhardt, 2013).
4. **Oceans:** Marine life, including plankton, contributes to the carbon cycle. While some marine organisms emit CO₂, oceans also absorb significant amounts of CO₂ from the atmosphere, acting as a natural buffer (Sabine et al., 2004).

Natural CO₂ emissions from respiration and decomposition are immense, estimated at about 550 gigatons per year. However, these emissions are balanced by natural absorption processes, such as photosynthesis and oceanic uptake (IPCC, 2021). In contrast, human activities add approximately 36 gigatons of CO₂ annually—seemingly small relative to natural emissions but entirely unbalanced, leading to a net increase in atmospheric CO₂ concentrations (Friedlingstein et al., 2022).

CO₂ from Breathing: A Drop in the Bucket

The average person exhales roughly 336 kilograms of CO₂ annually (note that the average annual CO₂ exhaled per person can range from approximately 190 kg to 450 kg, with a wider range due to various influencing factors), a considerable amount when viewed in isolation. However, this CO₂ is part of the natural carbon cycle. The carbon we breathe out originates from the plants we consume, which, in turn, absorb CO₂ from the atmosphere during photosynthesis. Thus, our respiratory CO₂ is considered "carbon neutral" and doesn't contribute to the net increase of CO₂ in the atmosphere.

Factors Influencing CO₂ Exhaled:

Age: Children generally have lower CO₂ emissions due to smaller lung capacity and lower metabolic rates compared to adults.

Exercise: Physical activity significantly increases CO₂ production and exhalation due to higher energy demands and faster breathing rates.

Weather/Seasons: Hot weather can increase respiration rate, while cold weather might decrease it slightly.

Altitude: Higher altitudes have lower air pressure, leading to increased respiration rates to compensate for less oxygen per breath.

Individual Variations: Metabolism, body size, and overall health can influence CO₂ production and exhalation rates.

CO₂ from Human Activities: The Elephant in the Room

In contrast, the CO₂ emitted through human activities, primarily the burning of fossil fuels for energy, transportation, and industrial processes, is the main driver of climate change. These activities release CO₂ that was previously locked away for millions of years, disrupting the natural carbon cycle and leading to a rapid increase in atmospheric CO₂ concentrations.

The disparity between CO₂ emissions from breathing and human activities is evident when examining per capita emissions across different regions are shown in the above figure, which highlights that CO₂ emissions from human activities overshadow those from breathing, especially in developed regions like North America and Europe.

Therefore, while individual actions like reducing our carbon footprint through lifestyle changes are important, addressing the climate crisis necessitates systemic shifts away from fossil fuels towards renewable energy sources and sustainable practices. The comparison between CO₂ from breathing and human activities underscores the urgent need for global cooperation in mitigating anthropogenic emissions to safeguard our planet's future.

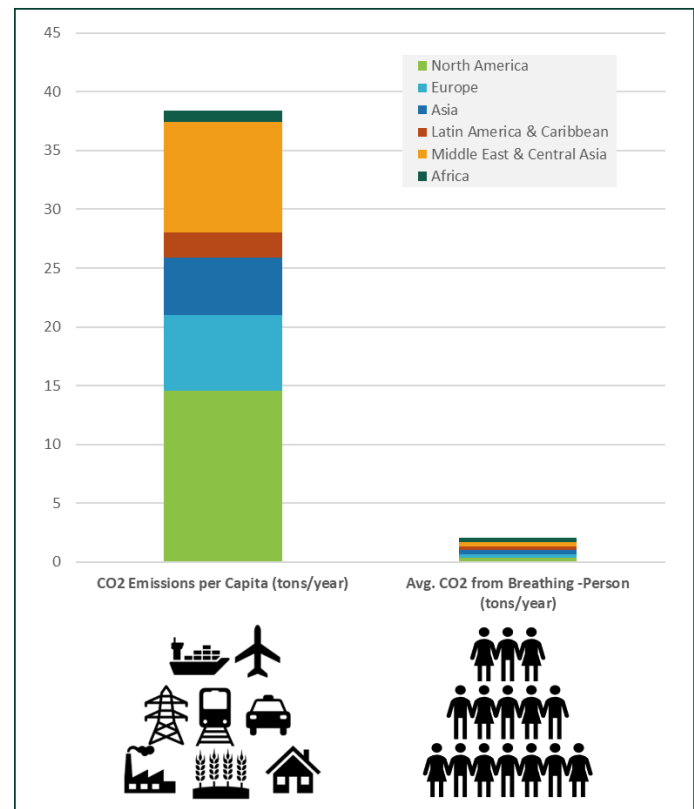


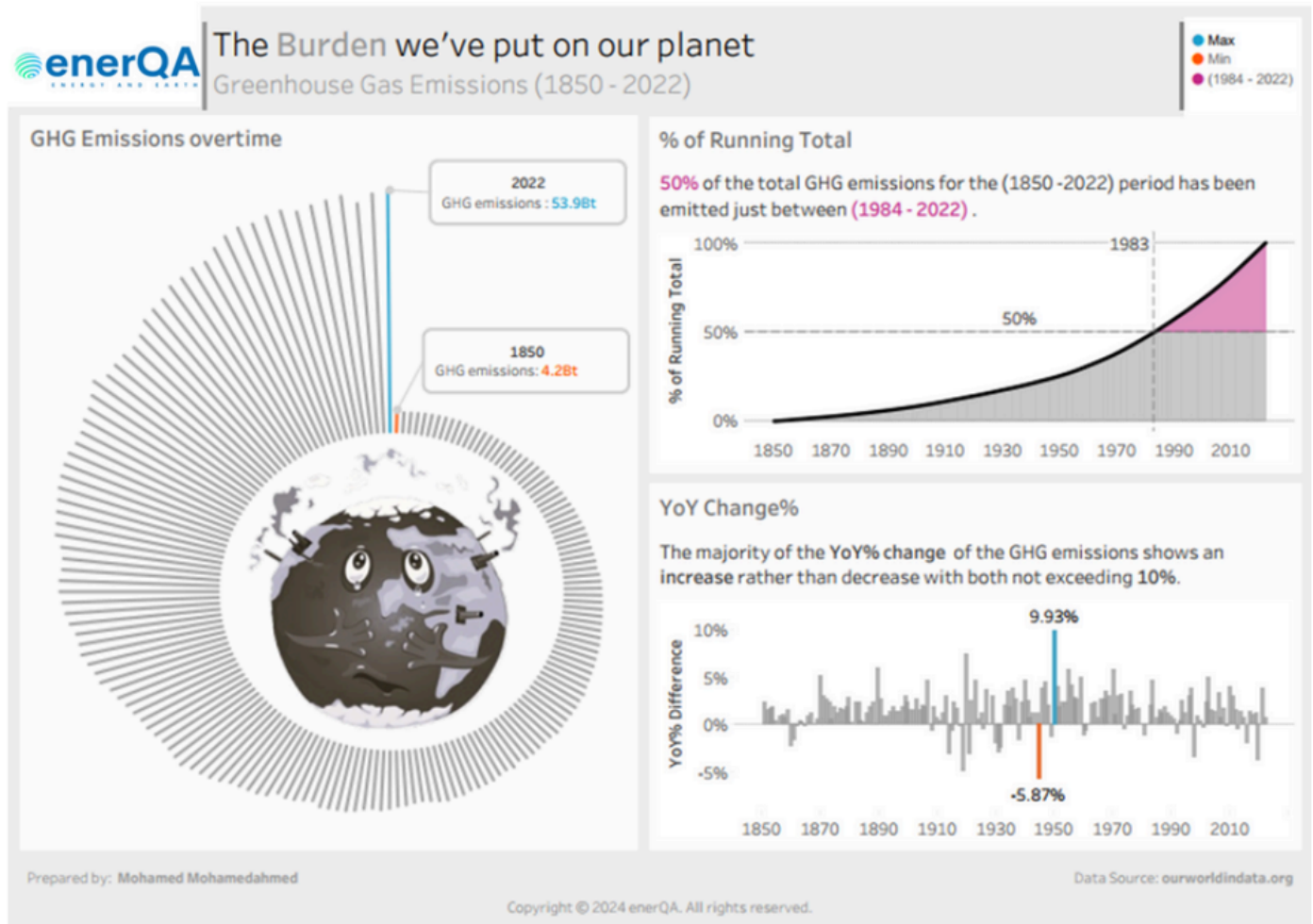
Fig-1: Global CO₂ Emissions: A Disparity in Responsibility
(Author's illustration)

References:

1. Boden, T. A., Marland, G., & Andres, R. J. (2017). *Global, Regional, and National Fossil-Fuel CO₂ Emissions*. Carbon Dioxide Information Analysis Center. DOI: 10.3334/CDIAC/00001_V2017
2. FAO. (2020). *Global Forest Resources Assessment 2020*. Food and Agriculture Organization of the United Nations. Retrieved from <https://www.fao.org>
3. Friedlingstein, P., et al. (2022). *Global Carbon Budget 2022*. Earth System Science Data. DOI: 10.5194/essd-14-1917-2022
4. Houghton, J. (2007). *Global Warming: The Complete Briefing*. Cambridge University Press.
5. IPCC. (2021). *Climate Change 2021: The Physical Science Basis*. Intergovernmental Panel on Climate Change. DOI: 10.1017/9781009157896
6. Keeling, C. D., et al. (2005). *Atmospheric CO₂ Records from Sites in the SIO Air Sampling Network*. Carbon Dioxide Information Analysis Center. DOI: 10.3334/CDIAC/atg.035
7. NOAA. (2023). *Trends in Atmospheric Carbon Dioxide*. National Oceanic and Atmospheric Administration. Retrieved from <https://www.noaa.gov>
8. Raven, P. H., Evert, R. F., & Eichhorn, S. E. (2021). *Biology of Plants* (8th ed.). W. H. Freeman.
9. Sabine, C. L., et al. (2004). *The Oceanic Sink for Anthropogenic CO₂*. *Science*, 305(5682), 367–371. DOI: 10.1126/science.1097403
10. Schlesinger, W. H., & Bernhardt, E. S. (2013). *Biogeochemistry: An Analysis of Global Change*. Academic Press.
11. Smith, P., et al. (2014). *Agriculture, Forestry and Other Land Use (AFOLU)*. In *Climate Change 2014: Mitigation of Climate Change*. Cambridge University Press.
12. *Our World in Data: CO₂ and Greenhouse Gas Emissions*
13. NRDC: *Do We Exhale Carbon?*
14. *GLOBE Scientists' Blog: Release of carbon dioxide by individual humans*
15. *CBM Journal: The impact of human and livestock respiration on CO₂ emissions from 14 global cities*

GHG EMISSIONS – The Burden on Our Planet

BY: Mohamed M. Ahmed



Greenhouse gases (GHG) emissions are a serious issue for our world. It has been demonstrated that in recent decades, the rise has accelerated. It is evident that, overall, the trend has been upward over the past 170 years, from 4.2 to approximately 54 billion tons of CO₂ equivalent (CO₂eq) (an increase of 1183%), between 1850 and 2022. However, the last 40 years of the same period have accounted for more than 50% of all emissions. Put otherwise, a little 23% of that timeframe accounted for over half of the emissions during the same timeframe.

In the meantime, there has been an average annual change of +2% across those years. Additionally, positive shifts have greatly exceeded the negative ones; with the maximum Year-On-Year (YoY) percentage being just below 10% from 1949 to 1950; 4 years following the end of the second world war.

On the contrary, the greatest decrease in YoY% was observed between 1944 and 1945, towards the end of World War II, and it was by 5.87%. This was followed by the 4.99% in 1919, one year after the end of World War I, whilst the third largest decline happened more recently in 2020 during the Covid-19 outbreak.

To sum up, the overall trend reflects the large impact of human activities on the amounts of GHG emissions, as it is directly affected by the major events associated with biggest switches in YoY% change during the period.

Energy, Technology and Finance

1. **Exploring the Rainbow of Hydrogen Technology: A Path to Sustainable Energy and Climate Resilience**
2. **Green Credit Lines in the Gulf Cooperation Council: Evaluating Opportunities, Overcoming Challenges, and Assessing the Impact on Energy Transition in Qatar**
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Exploring the Rainbow of Hydrogen Technology: A Path to Sustainable Energy and Climate Resilience

By: Quosay A. Ahmed and Reem Almlik

Abstract

Hydrogen technologies are gaining significant attention to decarbonize energy-intensive sectors and contribute to global climate goals. This literature review examines the diverse “rainbow” of hydrogen production methods—gray, blue, green, turquoise, pink, yellow, and white—each characterized by unique production processes, emissions profiles, and suitability across various economic and geographic contexts. The paper traces the historical development of these technologies and provides a comparative analysis, focusing on technical feasibility, cost implications, and environmental impact, with an emphasis on the applicability of hydrogen solutions in developing regions, particularly in the MENA and Africa.

Each hydrogen type offers specific benefits and limitations. While gray hydrogen remains widely used, its high emissions make it unsuitable for long-term climate strategies. Blue and green hydrogen, by incorporating carbon capture and renewable power sources respectively, offer low-carbon alternatives that align better with sustainability goals. Emerging technologies, such as turquoise and white hydrogen, add further dimensions but require additional development and market readiness. By linking these technologies to climate change mitigation, the paper highlights hydrogen’s potential role in reducing global emissions and provides recommendations for strategic adoption based on regional strengths and resources. This review underscores the need for tailored policies, investment in infrastructure, and international collaboration to maximize hydrogen’s role in a sustainable energy transition.

1. Introduction

Hydrogen (H_2) is the most abundant element in the universe, constituting approximately 75% of all normal matter by mass. It is found extensively in stars, gas clouds, and even in the vast emptiness of interstellar space, where it plays a critical role in cosmic evolution and stellar formation (Carroll & Ostlie, 2017). On Earth, hydrogen primarily occurs in combination with oxygen in water (H_2O) and within various organic compounds in living organisms and fossil fuels (Glasstone, 2013).

Hydrogen is often discussed as a clean energy source due to its high energy content. However, molecular hydrogen (H_2) is not naturally abundant on Earth because of its high reactivity, which causes it to bond readily with other elements. To use it as an energy carrier, hydrogen must be produced by separating it from compounds such as water or hydrocarbons through processes like electrolysis or steam methane reforming (SMR) (Sharma & Ghoshal, 2015).

As an energy source, hydrogen’s high energy content per unit weight—120 megajoules per kilogram—is nearly three times that of gasoline (Züttel, 2003). When hydrogen is burned or used in fuel cells, it produces only water as a byproduct, making it a promising candidate to replace fossil fuels, which release greenhouse gases (GHGs) and pollutants (Turner, 2004). Due to these properties, hydrogen is increasingly viewed as a pivotal element in the transition toward a clean energy economy (U.S. Department of Energy, 2020).

The current hydrogen economy is tightly connected to the fossil fuel industry, as approximately 96% of hydrogen is produced using fossil fuels like natural gas through SMR, which releases significant CO_2 emissions (International Energy Agency, 2019). To reduce the carbon footprint of hydrogen production, alternatives such as “green hydrogen”—produced via water electrolysis powered by renewable energy sources—are being developed as essential steps toward decarbonizing the hydrogen supply chain (Staffell et al., 2019).

Hydrogen's versatility also makes it ideal for hard-to-decarbonize sectors, including heavy-duty transportation (e.g., trucking, shipping, and aviation), where it can serve directly as a fuel or in derivative forms like ammonia (Tollefson, 2021). In agriculture, hydrogen is essential for synthesizing ammonia (NH_3) through the Haber-Bosch process, which consumes large amounts of hydrogen, thus linking agricultural costs and sustainability to the hydrogen economy (Smil, 2001). Additionally, hydrogen is crucial in industrial processes such as oil refining and methanol production, underscoring its deep integration into the energy and industrial sectors (Speight, 2020).

Hydrogen's relevance extends beyond Earth as well. It is a primary fuel in space exploration, where its high energy density is utilized in liquid hydrogen rocket propellants (NASA, 2021). The potential for hydrogen as a clean fuel on Earth and its essential role in space exploration underscore its versatility and critical importance for future energy strategies.

2. Historical Evolution of Hydrogen Production

The production of hydrogen dates back to the early 19th century, when scientists first recognized hydrogen as a distinct element. Initially, hydrogen production was focused on extracting it from water through electrolysis, a process discovered by British scientists William Nicholson and Anthony Carlisle in 1800. This method was revolutionary at the time but required significant energy input, limiting its practical applications (Smil, 2017).

The industrial production of hydrogen began with the development of steam methane reforming (SMR) in the 1920s, which uses high-temperature steam to extract hydrogen from methane gas. SMR became a dominant method due to its efficiency and the widespread availability of natural gas, leading to what is now termed "gray hydrogen"—hydrogen produced from fossil fuels without carbon capture (Rostrup-Nielsen, 2017). Gray hydrogen is associated with high carbon dioxide emissions, as the process releases significant quantities of CO_2 , contributing to greenhouse gas emissions (International Energy Agency, 2019).

In response to growing environmental concerns, the 1970s oil crises spurred research into alternative hydrogen production methods that were less dependent on fossil fuels. During this period, electrolysis experienced renewed interest, especially when powered by renewable energy sources. This shift marked the early conceptualization of "green hydrogen," a term now used to describe hydrogen produced with renewable electricity, resulting in zero direct emissions (Bockris, 2002).

In the 2000s, advances in carbon capture and storage (CCS) technologies led to the emergence of "blue hydrogen," a cleaner variant of gray hydrogen. Blue hydrogen employs the same SMR process but incorporates CCS to trap and store up to 90% of the CO_2 emissions, making it a lower-carbon option. This method has gained popularity as a transitional solution, leveraging existing natural gas infrastructure while reducing the carbon footprint of hydrogen production (Staffell et al., 2019).

Simultaneously, new production methods for hydrogen were developed, including methane pyrolysis, which produces "turquoise hydrogen" by decomposing methane into solid carbon and hydrogen. This method eliminates CO_2 emissions but remains less commercially viable due to high costs and limited technological readiness (Tollefson, 2021). Other emerging types include "pink hydrogen," produced via electrolysis powered by nuclear energy, which offers a low-emission alternative but is constrained by the availability and public acceptance of nuclear power (Hydrogen Council, 2021).

Recent developments have also introduced "yellow hydrogen," which is derived from grid-based electricity, and "white hydrogen," a naturally occurring form of hydrogen found in certain geological formations. While yellow hydrogen's environmental impact depends on the grid's energy mix, white hydrogen remains a niche resource that has not yet been widely developed for commercial use (He, 2020).

The evolution of hydrogen production reflects the progress and challenges of balancing technological efficiency with environmental considerations. As new methods and technologies develop, hydrogen is increasingly viewed as a vital component of sustainable energy systems capable of reducing global carbon emissions.

3. Analysis of Hydrogen Technologies by Color

Hydrogen production methods are classified by color to represent different processes, each with unique environmental impacts, cost considerations, and applications. This section examines each “color” of hydrogen—gray, blue, green, turquoise, pink, yellow, and white—based on their production process, emissions, readiness, and usage potential in diverse sectors.

Gray Hydrogen

Gray hydrogen is derived from steam methane reforming (SMR), a mature, low-cost method that uses natural gas to produce hydrogen but releases substantial carbon dioxide (CO₂) emissions. Representing the most common form of hydrogen today, gray hydrogen is integral to industries needing affordable hydrogen, despite its high emissions profile (Rostrup-Nielsen, 2017; International Energy Agency, 2019).

Blue Hydrogen

Blue hydrogen also relies on SMR but incorporates carbon capture and storage (CCS) to reduce CO₂ emissions. By capturing up to 90% of these emissions, blue hydrogen presents a cleaner alternative to gray hydrogen while using existing SMR technology and infrastructure (Staffell et al., 2019). This makes blue hydrogen a feasible “bridge” technology for industrial applications, though challenges remain with the cost of CCS technology and long-term storage viability (Tollefson, 2021).

Green Hydrogen

Green hydrogen is produced through electrolysis powered by renewable energy sources, resulting in zero direct CO₂ emissions. This method offers the greatest potential for carbon neutrality but is currently limited by high production costs and the need for extensive renewable energy capacity (Bockris, 2002; U.S. Department of Energy, 2020). Green hydrogen’s applications are expanding into areas like transportation and power storage, where emissions reduction is prioritized, particularly as renewable energy prices continue to decline.

Turquoise Hydrogen

Turquoise hydrogen, produced via methane pyrolysis, decomposes methane into hydrogen and solid carbon, avoiding direct CO₂ emissions. This method is in the developmental stage and holds promise for regions with access to natural gas and solid carbon markets. However, its commercial viability remains constrained by cost and technological readiness (Abánades, 2020; Hydrogen Council, 2021).

Pink Hydrogen

Pink hydrogen production uses nuclear power to drive electrolysis, providing a low-emission solution without the variability of renewable energy sources. This consistent power supply can support continuous hydrogen production, which is particularly advantageous for heavy industrial applications. However, its scalability is limited by public

H₂	Gray Hydrogen: Fossil fuel-based production without carbon capture, low environmental benefit, and relatively low cost.
H₂	Blue Hydrogen: Derived from fossil fuels but incorporates carbon capture and storage (CCS) to reduce emissions significantly, balancing feasibility, impact, and cost.
H₂	Green Hydrogen: Renewable energy-driven electrolysis, high environmental impact, and moderate-to-high cost.
H₂	Turquoise Hydrogen: Methane pyrolysis, producing solid carbon byproducts, moderate environmental impact, and competitive cost.
H₂	Pink Hydrogen: Electrolysis using nuclear energy, moderate environmental impact, and moderate cost.
H₂	Yellow Hydrogen: Solar-powered electrolysis, significant environmental impact, and high cost.
H₂	White Hydrogen: Naturally occurring hydrogen extracted from underground, high environmental impact but rare and costly.

Figure-1: Hydrogen production technologies and its environmental impacts – categorized by color (Authors illustration)

perceptions of nuclear energy, regulatory challenges, and infrastructure costs (Glasstone & Sesonske, 2014; International Atomic Energy Agency, 2020).

Yellow Hydrogen

Yellow hydrogen, produced through grid-based electricity, varies in environmental impact depending on the grid's energy composition. In regions where renewable energy contributes significantly to the grid, yellow hydrogen can be a cleaner option. However, in fossil fuel-heavy grids, the carbon footprint of yellow hydrogen remains relatively high (Ball & Wietschel, 2009).

White Hydrogen

White hydrogen, or naturally occurring hydrogen, is found in specific geological formations. Although it has potential as a clean hydrogen source, it remains underdeveloped, with significant challenges in extraction and storage. White hydrogen offers a unique opportunity if technology advances allow for efficient and economical extraction from natural deposits (He, 2020).

Table 1: Summary of Hydrogen Production Methods by Color, Emissions, and Applicability.

Color	Production Method	Emissions	Applicability
Gray	Steam Methane Reforming (SMR)	High CO ₂ , emissions	Cost-effective; widely used in industry but high emissions
Blue	SMR with Carbon Capture and Storage (CCS)	Low CO ₂ , emissions due to CCS	Lower emissions option for industries with existing infrastructure
Green	Electrolysis powered by renewable energy	Zero direct emissions	Ideal for sectors requiring low carbon, e.g., transport, energy storage
Turquoise	Methane Pyrolysis	No direct CO ₂ , emissions; solid carbon byproduct	Potential for regions with natural gas and solid carbon markets
Pink	Electrolysis powered by nuclear energy	Minimal emissions; low-carbon	Suitable for continuous industrial applications with nuclear power
Yellow	Electrolysis powered by grid electricity	Varies based on grid energy mix	Flexible; emission levels dependent on regional grid composition
White	Naturally occurring hydrogen in geological formations	Zero direct emissions; extraction impact uncertain	Under research; future potential if extraction methods are developed

4. Feasibility in Developing and Least-Developed Countries

The feasibility of hydrogen technologies varies significantly across regions, particularly in developing and least-developed areas, where access to resources, infrastructure, and capital are often limited. In the MENA and Africa regions, the potential for deploying different types of hydrogen production technologies is influenced by each region's unique resource availability, economic priorities, and environmental policies. This section explores the feasibility of various hydrogen production methods in these regions, examining factors such as energy availability, technological readiness, and regional needs.

Gray Hydrogen

Gray hydrogen, produced through steam methane reforming (SMR), is the most widely accessible hydrogen technology, largely due to its low production cost and the MENA region's abundant natural gas resources (International Energy Agency, 2021). Countries like Algeria, Egypt, and Qatar, which have significant natural gas reserves, can feasibly produce gray hydrogen to support industries such as ammonia production and refining. However, the high CO₂ emissions associated with gray hydrogen make it an unsustainable option in the long term, particularly as many African and MENA countries face increasing pressure to meet emissions targets (Bensaid et al., 2021).

Blue Hydrogen

Blue hydrogen, which incorporates carbon capture and storage (CCS) with SMR, offers a cleaner alternative that leverages the existing natural gas infrastructure. The MENA region, with its geological formations suited for carbon storage, has high potential for blue hydrogen projects. For example, Saudi Arabia and the United Arab Emirates have already begun exploring blue hydrogen initiatives; Saudi Arabia's NEOM project, in collaboration with Air Products, plans to develop a large-scale blue hydrogen facility aimed at reducing emissions (Hydrogen Council, 2021). However, the high cost and technological requirements of CCS can be barriers for wider adoption in lower-income African nations, where capital constraints and limited access to CCS technology may hinder blue hydrogen's feasibility (Lutsey & Sperling, 2018).

Green Hydrogen

Green hydrogen, produced through electrolysis powered by renewable energy, is an appealing option for regions with substantial renewable resources, particularly solar and wind. The MENA region has significant potential for green hydrogen production due to its vast solar capacity. Morocco has been a leader in green hydrogen development; its Noor Ouarzazate Solar Complex, one of the largest solar farms globally, has been linked to green hydrogen initiatives with support from the European Union and Germany (IRENA, 2021). In Africa, Namibia has also initiated green hydrogen projects, capitalizing on its high solar irradiance and favorable climate for year-round renewable production (UNIDO, 2022). Despite this potential, the high initial costs and need for renewable infrastructure present challenges in regions with limited capital, where access to affordable finance is crucial for green hydrogen's scalability.

Turquoise Hydrogen

Turquoise hydrogen, generated by methane pyrolysis, produces hydrogen with a solid carbon byproduct, eliminating CO₂ emissions in the production process. This method could be viable in natural gas-rich countries like Algeria and Egypt if paired with solid carbon markets. However, methane pyrolysis is a relatively new technology and lacks widespread infrastructure, limiting its immediate feasibility in developing economies. The need for technological advancement and cost reductions makes turquoise hydrogen a longer-term option for the MENA and Africa regions (Hydrogen Council, 2021).

Pink Hydrogen

Pink hydrogen, produced via electrolysis powered by nuclear energy, is attractive for countries with nuclear infrastructure. The MENA region currently has limited nuclear capacity, with only the United Arab Emirates operating a commercial nuclear power plant (Barakah Nuclear Energy Plant). This restricts the feasibility of pink hydrogen in the region, though it could serve as a low-emission hydrogen source in countries that invest in nuclear power in the future (International Atomic Energy Agency, 2020). In Africa, pink hydrogen remains largely unfeasible due to minimal nuclear infrastructure and public resistance to nuclear development.

Yellow Hydrogen

Yellow hydrogen, derived from grid-based electricity, offers flexibility but carries a variable carbon footprint depending on the grid's composition. In MENA countries with high fossil fuel-based energy production, such as Egypt and Libya, yellow hydrogen's emissions profile remains high. However, in nations like Morocco, where the energy grid includes substantial renewable sources, yellow hydrogen production could offer a lower-carbon alternative (Ball & Wietschel, 2009). The feasibility of yellow hydrogen in Africa and the MENA region will depend heavily on national energy policies and the pace of renewable energy integration into regional grids.

White Hydrogen

White hydrogen, or naturally occurring hydrogen, is an emerging resource with potential in certain African countries, including Mali, where naturally occurring hydrogen deposits have been discovered (He, 2020). This form of hydrogen offers an intriguing opportunity for local, clean hydrogen production without the need for extensive infrastructure. However, the technology to safely and economically extract white hydrogen is still in its infancy, limiting its immediate feasibility.

The MENA and Africa regions exhibit varied potential for hydrogen production technologies. Countries with abundant natural gas resources, like Algeria and Saudi Arabia, may initially focus on gray and blue hydrogen, leveraging existing infrastructure while transitioning to lower-emission alternatives. Regions with substantial solar resources, such as Morocco and Namibia, have opportunities to scale up green hydrogen production, provided they can secure financing and develop necessary infrastructure. Table-2 provides a comparative overview of each hydrogen production method's feasibility in these regions.

For hydrogen technology adoption in Africa, several challenges are specific to the continent's context, including infrastructure, economic, and policy-related hurdles. Competing governmental priorities, such as healthcare and poverty alleviation, often divert resources away from large-scale hydrogen and renewable investments.

Policy and regulatory support for hydrogen is also limited, with few hydrogen-specific frameworks to guide development. The absence of carbon pricing and incentives for low-carbon technologies diminishes motivation for cleaner hydrogen options. Meanwhile, renewable energy infrastructure is underdeveloped, and inconsistent solar and wind availability can hinder green hydrogen production. Water and land constraints further complicate large-scale projects, especially in regions where these resources are scarce or essential for local communities.

The following graph serves as a framework to evaluate the trade-offs between technological viability, environmental benefits, and economic considerations for each hydrogen production pathway.

In the Gulf Corporation Council (GCC) countries and Iraq, hydrogen development faces both unique challenges and promising opportunities. These nations are heavily reliant on hydrocarbon industries, making them well-positioned to produce blue hydrogen due to their abundant natural gas reserves and established infrastructure for carbon capture and storage (CCS). For example, Saudi Arabia and the UAE have both initiated significant blue hydrogen projects, leveraging their geological formations suitable for CO₂ storage (Hydrogen Council, 2021; IRENA, 2022). However, high CCS costs and the need for substantial technological investments pose challenges for broader adoption across the region, particularly in Iraq, where infrastructure and political stability are more limited (World Bank, 2022).

Green hydrogen presents a major opportunity due to the GCC's substantial solar potential, offering a pathway to diversify energy portfolios and meet decarbonization goals. Countries like Saudi Arabia and the UAE are investing in large-scale solar plants, which could support green hydrogen production (IRENA, 2021). Oman is also emerging as a green hydrogen hub, with planned projects to serve both domestic and export markets (UNDP, 2021). Yet, green hydrogen development requires high initial investments and extensive renewable energy infrastructure, which may be challenging to implement rapidly in Iraq due to economic and infrastructure constraints (World Bank, 2022).

Despite these challenges, the GCC countries and Iraq can leverage their strategic geographic position to access both European and Asian hydrogen markets, creating a competitive export opportunity. With supportive policies and investment in infrastructure, these countries can potentially transition from traditional fossil fuel economies to global leaders in the hydrogen sector, balancing hydrocarbon production with sustainable energy development (Hydrogen Council, 2021).

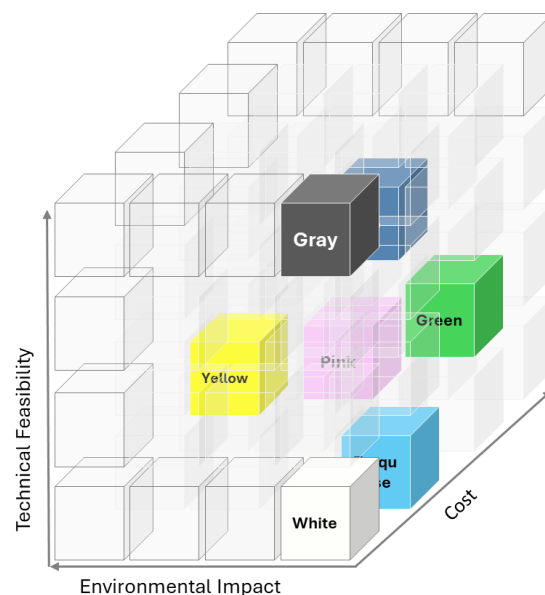


Figure 4: Hydrogen Technologies Categorization
(Source: Author's illustration)

Table 2: Assessment of Hydrogen Technology Feasibility in Different Economic Contexts.

Hydrogen Type	Economic Context	Technical Feasibility	Environmental Impact
Gray	Low-cost; feasible in natural gas-rich countries; high emissions limit long-term viability	High; mature technology with widespread adoption in industry	High CO ₂ emissions; unsustainable for low-carbon goals
Blue	Higher cost due to CCS; suitable for gas-rich areas with carbon storage infrastructure	Moderate; dependent on CCS technology and storage infrastructure	Reduced emissions; CCS mitigates CO ₂ , but relies on storage safety
Green	High initial costs; viable in regions with strong renewable infrastructure and financing	Moderate; requires renewable energy infrastructure and high capital investment	Zero direct emissions; ideal for low-carbon applications
Turquoise	Emerging technology; potential for natural gas regions with solid carbon markets	Low; experimental stage with limited infrastructure and market support	No CO ₂ emissions; carbon byproduct requires solid carbon markets
Pink	Limited to regions with nuclear power; high costs and regulatory challenges	Low; reliant on nuclear infrastructure and public acceptance	Minimal emissions; limited by nuclear energy availability
Yellow	Variable emissions based on grid mix; feasible where renewable share in grid is high	Moderate; dependent on grid energy mix, suitable where renewables are prevalent	Varies; emissions depend on grid composition and renewables share
White	Under research; potential in areas with natural hydrogen deposits if extraction develops	Low; early-stage research, limited knowledge on extraction feasibility	Zero direct emissions; environmental impacts uncertain due to limited extraction data

5. Hydrogen Technologies and Climate Change

Hydrogen technologies play a crucial role in mitigating climate change by providing a cleaner alternative to fossil fuels in sectors that are challenging to decarbonize, such as heavy industry, transportation, and energy storage. Each hydrogen production method has a distinct carbon footprint, influencing its potential impact on reducing greenhouse gas (GHG) emissions.

Gray hydrogen, the most widely produced type, contributes significantly to CO₂ emissions due to its reliance on natural gas and lack of carbon capture mechanisms. This high emissions profile makes gray hydrogen incompatible with climate goals focused on net-zero carbon targets (International Energy Agency, 2019). Transitioning from gray to cleaner hydrogen types is essential for minimizing the carbon impact of hydrogen use.

Blue hydrogen offers a lower-emission alternative by integrating carbon capture and storage (CCS) with steam methane reforming (SMR), capturing up to 90% of the CO₂ emissions generated. Although blue hydrogen's emissions are not zero, it provides a transitional solution for industries reliant on natural gas, particularly in regions with suitable CO₂ storage capacities (Staffell et al., 2019). However, CCS technology is still costly and energy-intensive, raising concerns about long-term sustainability.

Green hydrogen holds the highest potential for climate impact reduction, as it is produced through water electrolysis powered by renewable energy sources, resulting in zero direct emissions. Green hydrogen's role is particularly promising in areas with abundant renewable resources, as it could support the decarbonization of high-emission sectors like steel, cement, and transportation (U.S. Department of Energy, 2020). However, the scalability of green hydrogen is currently constrained by the availability and cost of renewable energy infrastructure.

Turquoise hydrogen is another low-emission option, using methane pyrolysis to split methane into hydrogen and solid carbon, avoiding CO₂ emissions in the process. While this method is still in its developmental phase, it shows promise in regions where solid carbon can be utilized in industrial applications, adding a new dimension to hydrogen's climate compatibility (Hydrogen Council, 2021).

Pink hydrogen, produced via nuclear-powered electrolysis, and **yellow hydrogen**, produced from grid-based electricity, offer additional flexibility. Pink hydrogen provides a steady, low-emission hydrogen source in regions with

nuclear energy, while yellow hydrogen's emissions vary depending on the grid mix. These technologies can contribute to climate goals, especially in countries with a low-carbon grid or nuclear capacity (IRENA, 2021).

White hydrogen, a naturally occurring form, could also offer a low-emission solution if viable extraction methods are developed. As of now, white hydrogen remains largely unexplored but could serve as a sustainable hydrogen source with minimal environmental impact if research advances (He, 2020).

By providing carbon-free or lower-carbon alternatives, hydrogen technologies have significant potential to aid global decarbonization efforts. However, realizing this potential will depend on continued advancements in renewable energy infrastructure, CCS technology, and the development of hydrogen storage and distribution systems.

Table 3: Emission Reduction Potential of Hydrogen Technologies by Color

Hydrogen Type	Emission Reduction Potential	Key Factors for Climate Impact
Gray	Low	High CO ₂ emissions; unsuitable for net-zero goals
Blue	Moderate	CO ₂ reduction with CCS; feasible transitional option
Green	High	Zero direct emissions; ideal for net-zero targets
Turquoise	Moderate	Solid carbon byproduct; low emissions but limited scaling
Pink	High (in nuclear-capable regions)	Minimal emissions; reliant on nuclear infrastructure
Yellow	Variable	Depends on grid energy mix; cleaner in renewable-heavy grids
White	High (if viable extraction)	Zero direct emissions; under research for sustainable use

6. Conclusion and Recommendations

Conclusion

Hydrogen technology represents a versatile and promising avenue for decarbonizing high-emission sectors and transitioning to a more sustainable global energy system. The spectrum of hydrogen production methods—gray, blue, green, turquoise, pink, yellow, and white—offers a variety of options, each with distinct technical and environmental characteristics. While gray hydrogen remains prevalent due to its low cost and widespread availability, its high emissions make it unsuitable for long-term climate goals. Blue hydrogen offers a transitional solution by incorporating carbon capture, although its sustainability depends on further CCS advancements. Green hydrogen, with its zero-emission production process, holds the greatest potential for aligning with net-zero targets, particularly in regions with abundant renewable resources. Emerging technologies like turquoise and white hydrogen offer additional opportunities but require further development for widespread adoption.

For countries in the MENA and Africa regions, where unique geographic and economic contexts present specific challenges, the successful adoption of hydrogen technologies will require a tailored approach. Gas-rich countries can leverage blue hydrogen as a transitional technology, while regions with high solar potential, such as North Africa and the GCC, are well-positioned to capitalize on green hydrogen. In developing regions, addressing infrastructural limitations, access to financing, and policy support will be essential to support the growth of a sustainable hydrogen economy.

General recommendations

1. **Policy and Regulatory Support:** Governments should establish clear regulatory frameworks and incentives for hydrogen technology development, including subsidies for green and blue hydrogen, carbon pricing mechanisms, and investment in renewable infrastructure to support green hydrogen production.
2. **Investment in Infrastructure and Innovation:** Large-scale hydrogen deployment requires robust infrastructure, including production facilities, storage, and distribution networks. Public and private sector collaboration should focus on building hydrogen hubs, developing CCS technology for blue hydrogen, and advancing solid carbon markets for turquoise hydrogen.
3. **Promoting International Collaboration:** Cross-border partnerships can help mobilize financial resources and share technical expertise, especially in developing countries. Collaborations with hydrogen leaders, such as

the EU, can facilitate knowledge transfer and create export opportunities for regions like the MENA, where hydrogen production potential is high.

4. **Supporting R&D for Emerging Technologies:** Investment in research and development (R&D) is essential for scaling emerging hydrogen technologies like turquoise and white hydrogen. R&D initiatives should focus on enhancing the efficiency and economic viability of these methods, enabling more regions to integrate low-carbon hydrogen into their energy portfolios.
5. **Public Education and Workforce Development:** Developing a skilled workforce is critical to supporting hydrogen technology infrastructure and operations. Educational programs, technical training, and community outreach will be important to build public support, particularly in regions considering the adoption of nuclear-based pink hydrogen.

By adopting a strategic, region-specific approach to hydrogen technologies, countries can capitalize on their natural resources, economic strengths, and existing infrastructure to create a cleaner, more resilient energy future. With sustained investment and regulatory support, hydrogen has the potential to become a cornerstone of the global transition toward sustainable energy and climate resilience.

7. References

1. Abánades, A. (2020). Hydrogen production from methane pyrolysis (turquoise hydrogen): A possible bridge between natural gas and renewable hydrogen. *International Journal of Hydrogen Energy*, 45(15), 8310-8320.
2. Ball, M., & Wietschel, M. (2009). *The Hydrogen Economy: Opportunities and Challenges*. Cambridge University Press.
3. Bensaid, S., et al. (2021). Hydrogen production from renewable resources in the MENA region: Challenges and perspectives. *Renewable and Sustainable Energy Reviews*, 144, 110-125.
4. Bockris, J. O. M. (2002). *The Hydrogen Economy: Its History and Prospects*. *Science and Public Policy*, 29(1), 31-37.
5. Carroll, B. W., & Ostlie, D. A. (2017). *An Introduction to Modern Astrophysics*. Cambridge University Press.
6. Glasstone, S. (2013). *Sourcebook on the Hydrogen Economy*. Elsevier.
7. Glasstone, S., & Sesonske, A. (2014). *Nuclear Reactor Engineering: Reactor Systems Engineering*. Springer.
8. He, Y. (2020). Naturally occurring hydrogen: A new frontier in energy. *Renewable Energy*, 145, 1625-1632.
9. Hydrogen Council. (2021). *Hydrogen Insights Report 2021*. Retrieved from [HydrogenCouncil.com](https://hydrogencouncil.com)
10. International Atomic Energy Agency. (2020). *Nuclear Power and the Clean Energy Transition*. Retrieved from IAEA.org
11. International Energy Agency. (2019). *The Future of Hydrogen*. Retrieved from IEA.org
12. International Energy Agency. (2021). *Hydrogen in North Africa*. Retrieved from IEA.org
13. International Renewable Energy Agency. (2021). *The Role of Hydrogen in the Energy Transition*. Retrieved from [IRENA.org](https://irena.org)
14. International Renewable Energy Agency. (2022). *Energy Transition in the Middle East and North Africa*. Retrieved from [IRENA.org](https://irena.org)
15. IRENA. (2021). *Green Hydrogen Supply Chain: Morocco as a case study*. Retrieved from [IRENA.org](https://irena.org)
16. Lutsey, N., & Sperling, D. (2018). The role of advanced transportation technology in developing countries. *Annual Review of Environment and Resources*, 43, 1-26.
17. National Aeronautics and Space Administration. (2021). *The Use of Hydrogen in Space Exploration*. Retrieved from [NASA.gov](https://nasa.gov)
18. Rostrup-Nielsen, J. R. (2017). Industrial Uses of Hydrogen. *Catalysis Today*, 103(1-4), 9-15.
19. Sharma, S., & Ghoshal, S. K. (2015). Hydrogen the future transportation fuel: From production to applications. *Renewable and Sustainable Energy Reviews*, 43, 1151-1158.
20. Smil, V. (2017). *Energy and Civilization: A History*. MIT Press.
21. Speight, J. G. (2020). *The Chemistry and Technology of Petroleum*. CRC Press.
22. Staffell, I., Scamman, D., Abad, A. V., et al. (2019). The role of hydrogen and fuel cells in the global energy system. *Energy & Environmental Science*, 12(2), 463-491.
23. Tollefson, J. (2021). How green is blue hydrogen? *Nature*, 597, 15-17.
24. Turner, J. A. (2004). Sustainable hydrogen production. *Science*, 305(5686), 972-974.
25. U.S. Department of Energy. (2020). *Hydrogen Program Plan*. Retrieved from DOE.gov
26. UNIDO. (2022). *Namibia's Green Hydrogen Future*. Retrieved from [UNIDO.org](https://unido.org)
27. United Nations Development Programme. (2021). *Oman's Pathway to Green Hydrogen Leadership*. Retrieved from [UNDP.org](https://undp.org)
28. World Bank. (2022). *Energy Infrastructure Challenges in the Middle East*. Retrieved from [WorldBank.org](https://worldbank.org)
29. Züttel, A. (2003). Materials for hydrogen storage. *Materials Today*, 6(9), 24-33.

Green Credit Lines in the Gulf Cooperation Council: Evaluating Opportunities, Overcoming Challenges, and Assessing the Impact on Energy Transition in Qatar

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Abstract

This study explores the potential of Green Credit Lines (GCLs) as a tool for accelerating the transition towards a sustainable economy in Qatar, a nation with ambitious renewable energy (RE) targets. Building upon existing literature, the study has developed a comparison method to assess the potentiality of establishing and operating GCLs within Gulf Cooperation Council (GCC), named GCL Implementation Potential Index (GCLPI). The GCLPI incorporates three factors. The first factor, *Financial Institutions Performance*, evaluates the readiness of financial institutions (FIs) based on their financial performance and commitment to sustainability. The second factor, the *Governmental Frameworks*, examines the supportive governmental frameworks, encompassing national green finance strategies, GCL regulations, and green building standards. The third factor, the *FIs Internal Capacities*, assesses the capacities of financial institutions to manage and execute GCL programs effectively, this includes, the FIs technical expertise in green project assessments, and the availability of a robust green projects pipeline. Factors including *Financial Institutions Performance*, *Governmental Frameworks*, and *FIs Internal Capacities* are qualitatively assessed to judge the potentiality of establishing GCLs by the FIs. Based on that, a GCL Potential Impact Indicator (GCLPII) is introduced to combining the GCLPI with the *RE Target Ambitiousness* of Qatar's to assess the potential energy transition impact of GCLs. The study considers the broader context of gulf region by conducting a comparison between Qatar, the United Arab Emirates (UAE) and Saudi Arabia (KSA). The study reveals that the UAE is the only country among the three with existing GCL products, providing them with a demonstrably earlier stage of implementation. All three countries share promising green projects pipelines and ambitious RE targets, with Qatar's target being the most ambitious. Analyzing GCL implementation potential alongside RE target ambition reveals Qatar could experience a significant impact from GCLs. However, limitations in Qatar's governmental frameworks compared to KSA require strategic development to unlock its full GCL potential. By prioritizing the development of supportive green finance frameworks and regulations, Qatar can bridge the gap with its competitors and leverage GCLs to achieve its ambitious clean energy goals.

Keywords: Green Credit Lines (GCLs), Renewable energy (RE), Energy Transition, GCL Implementation Potential Index (GCLPI), GCL Potential Impact Indicator (GCLPII), Financial sector, Qatar, GCC.

Introduction

The financial sector's mandate is to manage risk, allocate capital, mobilize savings and— as a financial intermediary— manage risk and maturity transformation and convenience denomination, all at the lowest possible transaction costs. It can have either an enabling role in the transition to a sustainable world or a hampering role depending on its actions[2].

The public sector does and will still play the leading role in commissioning green infrastructure projects and to guide and “jump start” investment when needed. But transformational change will ultimately require large-scale private sector engagement [3]. The transition requires a structural change in the financing patterns in terms of finance tenor and the result orientation to enable the private sector, including small and medium enterprises (SMEs) and Large Enterprises (LEs) to have a viable business case for green and RE projects. However, a clear distinction between 'green' and 'brown' investments is essential to ensuring transparency and effectiveness in green finance.

Green Credit Lines (GCLs) are specifically designed to finance environmentally sustainable projects, such as renewable energy, energy efficiency, and pollution control, while conventional or "brown" investments typically fund projects associated with high carbon emissions, like fossil fuel extraction or traditional infrastructure developments. The key difference lies in the purpose and criteria: GCLs prioritize environmental benefits, offering favorable terms (e.g., lower interest rates, longer repayment periods) to encourage green projects, whereas brown investments focus on profitability without necessarily considering environmental impacts. Additionally, GCLs often adhere to strict

sustainability standards and frameworks, ensuring alignment with global climate goals, while brown investments may contribute to environmental degradation through activities that increase carbon emissions and harm natural ecosystems.

The following table compare between the conventional credit line and the GCL.

Table 1: Difference between the conventional credit line and the green credit line

Feature	Credit Line	Green Credit Line
Definition	A pre-approved loan amount that a borrower can access as needed, up to a specified limit.	A specific type of credit line designed to finance environmentally friendly projects and initiatives.
Purpose	Can be used for various purposes, such as working capital, debt consolidation, or emergency expenses.	Exclusively used for projects that promote environmental sustainability, like renewable energy, energy efficiency, or pollution control.
Eligibility	Based on creditworthiness, income, and other financial factors.	Requires meeting specific criteria related to the project's environmental benefits and adherence to sustainability standards.
Terms and Conditions	Interest rates, fees, and repayment terms can vary depending on the borrower's credit profile and the lender's policies.	Often comes with more favorable terms than traditional credit lines, such as lower interest rates, longer repayment periods, or reduced fees, to incentivize green investments.
Examples	Personal credit lines, business credit lines, home equity lines of credit	Green credit lines for renewable energy projects, energy-efficient building upgrades, or sustainable agriculture practices.

The GCC economies have been heavily reliant on oil and gas export revenues, with a narrow export profile dominated by oil and gas. World Bank data ranks Qatar, Kuwait, the UAE and Bahrain in positions one-to-four for the highest per capita emissions, with Saudi Arabia 10th and Oman 12th [4]. While Qatar's per capita emissions are the highest in the world, its small population means that total emissions are lower than those of Saudi Arabia and the UAE [5].

The GCC countries have progressively joined global forces to addressing climate change. They signed and ratified the Paris Climate Agreement and have submitted nationally determined contributions (NDCs) to the United Nations Framework Convention on Climate Change (UNFCCC), outlining their goals for mitigating and adapting to climate change. Notably, all GCC countries have pledged to achieve a net-zero target by the middle of the century. As of 2021, comprehensive national climate strategies, including adaptation ambitions, have been publicly declared by only three GCC states: the UAE, Oman, and Saudi Arabia [6]. Among them, Qatar's NDCs were submitted to UNFCCC in November 2021, showing that the country is committed to reducing its Greenhouse Gases (GHG) emissions by 25% compared to a baseline scenario by 2030¹. However, this is a complex challenge given Qatar's heavy dependence on oil and gas hydrocarbons' production, transformation, and consumption, which contribute significantly to nation's gross domestic product and overall wealth [7].

Large-scale projects require significant capital to develop and implement innovative technologies that can reduce emissions on a national scale. Simultaneously, SMEs play a vital role in the energy transition by fostering innovation and adopting sustainable practices at a smaller scale [8]. Therefore, ensuring adequate funding for both large and small energy projects is essential. This financial support will enable the deployment of cutting-edge technologies and sustainable practices necessary for reducing emissions, allowing Qatar and other GCC countries to effectively achieve the energy transition and contribute to global climate goals [9].

Green finance instruments, such as green bonds, green loans, grants, and carbon markets, are designed to fund projects that promote environmental sustainability and address climate change. Among these, GCLs stand out as impactful tools, providing pre-approved financing with favorable terms for green projects like renewable energy and energy efficiency. By offering lower interest rates, longer repayment periods, and technical assistance, GCLs help overcome funding barriers, particularly for SMEs enabling faster adoption of sustainable practices.

¹ Qatar's baseline scenario is a projection of future GHG emissions assuming no additional mitigation measures are implemented beyond those already planned before the NDC submission. The 25% reduction target is measured against this projected emissions trajectory.

The study adopts a comprehensive approach that includes a literature review and a conceptual framework to define and explain the opportunities and challenges of establishing GCLs by the FIs. It also presents a methodology for assessing the impact of GCLs mechanisms on Qatar's energy transition goals. By examining publicly available and historical data, the study aims to evaluate the progress of establishing GCLs in the GCC, with a particular focus on Qatar.

To measure the potential and impact of GCLs establishment by the FIs, the study introduces two metrics: (1) an index to assess the potential of the country to successfully implement green credit programs based on its existing financial and regulatory infrastructure, and (2) an indicator which measures the expected influence of implementing GCL by the FIs within the country on advancing energy transition goal. These metrics will explore various aspects, including the performance of the FIs, Governmental Framework, Internal capacities, and the energy transition target which are assessed qualitatively to judge the potentiality and impact of establishing GCLs.

The study enriches its findings by comparing Qatar, the United Arab Emirates (UAE), and Saudi Arabia (KSA). This comparative analysis aims to draw lessons from the sustainable finance experiences of these three countries. This provides a comprehensive perspective on the factors that may hinder or facilitate the success of GCLs in the GCC region.

Literature Review

Energy Transition Challenges in Qatar and the GCC

Founded in 1981, the GCC council countries, Bahrain, Kuwait, Oman, Qatar, Saudi Arabia and United Arab Emirates, together, covers a total area of 2410 km², has population of about 47 million and a GDP of \$ 1.6 trillion or about \$ 33,300 per capita [10]. The GCC countries ranked among the highest countries in term of energy consumption per capita. Qatar ranked first, followed by Kuwait, United Arab Emirates, Bahrain, Oman and Saudi Arabia. In comparison, the per capita energy consumption in Qatar is 4 times that in the United Kingdom and 20 times the average Indians per capita energy consumption. This can be attributed to the harsh climate in the GCC region leading to high demand for air conditioning and cooling throughout the year. For instance, Qatar's per capita electricity consumption reaches approximately 17,500 kWh annually, with a significant portion dedicated to cooling, especially during peak summer months [11]. In contrast, colder countries like Germany exhibit lower per capita energy consumption for heating, averaging around 130-150 kWh/m² per year, thanks to better insulation, energy-efficient building designs, and a more diverse energy mix that includes renewables [12]. For example, the Passivhaus standard in Germany reduces heating energy needs to as low as 15 kWh/m² annually [13]. This difference in energy demand highlights the efficiency and sustainability challenges faced by the GCC, where the reliance on electricity for cooling contrasts sharply with the more varied and efficient energy strategies in colder regions [14]. Also, low fuel prices over the past decades led to higher energy demand and less efficient energy use in the GCC countries [15].

The energy consumption across the GCC region showed an average annual growth rate (AAGR) of 6 percent to 7 percent, largely driven by the residential sector (which accounts for about 44 percent of the region's total consumption) [16]. Although GCC countries are not the world's largest emitters in terms of total GHG emissions and total GHG emissions per GDP, several Gulf countries fall in the top 10 emitters in terms of GHG emissions per capita. Despite that GCC countries collectively contribute about 2.7% of global greenhouse gas (GHG) emissions due to its oil and gas production [17], they have embarked in wide-ranging reforms to achieve socio-economic transformation. As shown in Qatar National Vision 2030, Vision 2030 of Saudi Arabia, "We the UAE 2031", Oman Vision 2040. Main progress areas are social and business-friendly reforms, efforts to enhance sustainability and resilience, investments in strategic industries and digital and green infrastructure [18]. All the GCC countries are progressing with the SDGs' agenda implementation and four of the countries have already exceeded two-thirds of the overall SDG score, Qatar being one of them [19]. The Gulf Arab states have already taken steps and developed institutional architecture conducive to mitigating the effects of climate change. Before the announcement of net-zero commitments, each GCC state launched initiatives, regulations, and programs intended to mitigate different aspects of climate change impacts [20].

Qatar is taking a range of actions to reduce its greenhouse gas emissions; Qatar ratified the Kyoto Protocol in 2005 and submitted an initial National Communication to the UNFCCC in 2011. In 2012, Qatar demonstrated its leadership role and support for global action on climate change action by hosting the 18th Conference of Parties (COP18) to the

UNFCCC and submitted the Updated First NDC in 2021. In terms of the impact of the climate change, Qatar's ND-GAIN² Indicator rank is 38. It is the 139th most vulnerable country and the 44th most ready country. The low vulnerability score and high readiness score of Qatar places it in the lower-Bottom quadrant of the ND-GAIN Matrix, which means that adaptation challenges still exist, but Qatar is well positioned to adapt [21]. The next table presents an overview of Qatar's strategic initiatives aimed at sustainable development and climate action.

Table 2: Qatar's strategic plans for sustainable development and climate action

Category	Action/Plan	Description	Lead Institution(s)
National Strategy	Qatar National Vision 2030	Outlines a long-term vision for economic diversification, environmental sustainability, and human development. Emphasizes a balanced approach to economic growth and environmental protection.	Qatar Secretariat for General Development Planning (QGSDP)
Climate Change Strategy	Qatar National Environment and Climate Change Strategy (QNECS)	Establishes a national framework for mitigating climate change impacts and adapting to a changing climate. Sets national climate targets, including those outlined in Qatar's NDC, and outlines various measures across different sectors.	Ministry of Municipality and Environment (MME)
Nationally Determined Contribution (NDC)	Qatar's First NDC	Sets a national target to reduce greenhouse gas emissions by 25% by 2030 compared to a business-as-usual (BAU) scenario.	Ministry of Municipality and Environment (MME)
Energy Sector	QatarEnergy Sustainability Strategy	Aims to reduce greenhouse gas emissions throughout the natural gas value chain. Focuses on improving energy efficiency, reducing methane emissions, and exploring carbon capture and storage (CCS) technologies.	QatarEnergy
Renewable Energy (RE) Strategy	Qatar National Renewable Energy (RE) Strategy	Aims to increase the share of RE in the country's energy mix.	Ministry of Municipality and Environment (MME)
Carbon Capture and Storage (CCS)	QatarEnergy Research and Development	CCS Explores the potential of CCS technologies to capture and store carbon emissions from the natural gas industry.	QatarEnergy
Aero Emissions Reduction	The Qatar Airways' Fuel Optimization	Aims to reduce fuel consumption and carbon emissions from aircraft operations.	Qatar Airways
Zero Emission Targets	N/A	N/A	N/A

Source: prepared by the authors based on published data.

In their recent National RE strategy, Qatar focuses in three main objectives: (1) Reduce GHG emissions through sustainable policies and trends targeting 10% reduction in the annual GHG emissions, (2) Increase RE penetration while maintaining network reliability targeting to increase RE share in the power mix from its current 5% to 18% by 2030, and (3) Maximize the socio-economic contribution from its RE targeting reduction in the average cost of electricity generation by 15% by 2030 program.

In conclusion, the literature review highlights the pressing energy challenges facing Qatar and the broader GCC region, characterized by high energy consumption per capita, significant greenhouse gas emissions, and a growing imperative for sustainable development. Moreover, there exists a pronounced ambition towards RE targets, signifying a pivotal shift towards a more sustainable and resilient energy future.

² The ND-GAIN Country Indicator summarizes a country's vulnerability (1) to climate change and other global challenges in combination with its readiness (2) to improve resilience.

The Role of the Financial Sector and GCL

Several empirical studies have shown that the financial sector can play an important role in reducing GHG emissions. Financial firms' ability to secure and allocate funds, and manage financial risks, are recognized as an essential in response to climate change risks [22]. Financial firms are adapting their traditional financial purpose, function, and philosophy, to change financial decision activities and products to reflect climate change risks and sustainability aims [23]. However, several barriers currently inhibit green lending in developing countries. GCL is considered useful in addressing some of these barriers to green lending, for example, an unfavorable investment environment can hinder the competitiveness of green investments compared to traditional alternatives. This lack of competitiveness stems from several factors. One major issue is the high upfront capital required for green projects, such as renewable energy plants, which often exceeds the initial costs of conventional fossil fuel investments. These projects also tend to have longer payback periods, making them less attractive to investors looking for quicker returns [24].

Additionally, the fossil fuel industry in many developing countries benefits from significant subsidies, creating market distortions that make it difficult for green projects to compete on a level playing field [25]. Another challenge is the lack of financial instruments, such as risk guarantees, which increases the perceived financial risk of green investments, further reducing their appeal to potential investors [26].

Also, challenges include a lack of viable projects, insufficient demand for financing, and limited financial knowledge and products [27]. While the availability of finance through GCLs is crucial, it is not a standalone solution. If viable projects are lacking, the effectiveness of available financing is limited. However, GCLs can stimulate project development by offering technical assistance aimed at helping project developers structure bankable green projects. GCLs serve a dual purpose as a financial tool. Firstly, they aim to facilitate lending to projects with environmental benefits. Secondly, they seek to enhance the capacity of local FIs to expand the local green lending market³ [27]. Under a GCL arrangement, funds are typically provided by a public financial institution whether its national or international to participating local FIs, who then lend them to developers of eligible green projects (SMEs and Large Enterprises (LEs)). The eligibility of green projects is generally determined by either the public financial institution providing the funds or by adhering to internationally recognized frameworks such as the Green Bond Principles or standards set by the Climate Bonds Initiative. These frameworks ensure that projects align with global environmental goals, such as reducing carbon emissions or promoting renewable energy. This support helps build a pipeline of viable projects, thereby bridging the gap between available finance and project readiness [28]. GCLs play a pivotal role in overcoming supply-side barriers by providing dedicated financing for green projects and leveraging the intermediation function of FIs [27][29]. Additionally, GCLs may encompass technical assistance funded by the participated FIs, aimed at enhancing the capacity of local banks to finance green investment projects and/or assisting end-borrowers in structuring investment proposals.

Recent developments in the taxonomy of the Green Loan Principles (GLP) provide clear guidelines for distinguishing between green and brown projects. The GLP taxonomy, established by the Loan Market Association, outlines specific criteria that projects must meet to be considered environmentally sustainable. These criteria include improvements in energy efficiency, significant reductions in greenhouse gas emissions, and the promotion and development of renewable energy sources [30]. For instance, projects aimed at enhancing energy efficiency might involve upgrading to energy-efficient appliances or retrofitting buildings to reduce energy consumption. Renewable energy projects could include the development of solar farms, wind energy installations, or hydroelectric power stations [30]. Similarly, the EU Taxonomy for Sustainable Activities provides a comprehensive framework for classifying environmentally sustainable economic activities. It includes detailed criteria across six environmental objectives, ensuring a broad and rigorous approach to sustainability [31]. The Climate Bonds Standard and Certification Scheme also offers sector-specific criteria to ensure that projects deliver genuine environmental benefits, particularly in high-impact areas like energy and transport [32].

Conversely, brown projects are characterized by activities that significantly contribute to carbon emissions and environmental degradation. Examples of brown projects include fossil fuel extraction, coal mining, and the development of new oil and gas infrastructure. These activities are not aligned with the sustainability goals outlined by the GLP and other taxonomies and are typically excluded from green financing. The differentiation between green

³ A green lending market refers to the financial ecosystem specifically targeting environmentally sustainable projects, which includes products and services like green loans or bonds. It aims to channel funds into renewable energy, energy efficiency, and other green initiatives. By offering GCLs, local financial institutions can develop expertise in green financing and expand the market for environmental projects by providing specialized financial products ((World Bank, 2020).

and brown projects is critical to avoid "greenwashing," where projects are misleadingly marketed as environmentally friendly without meeting the necessary sustainability criteria. Greenwashing can undermine the integrity of green finance and erode investor and public trust. By adhering to established taxonomies such as the GLP, EU Taxonomy, and Climate Bonds Standard, financial institutions can ensure that only genuinely sustainable projects receive green financing, thereby maintaining the integrity of green credit lines. This adherence involves rigorous assessment and verification processes to confirm that projects meet the relevant criteria and deliver the intended environmental benefits [31] [32].

Figure 5 below illustrates the mechanism of the GCL establishment by FIs.

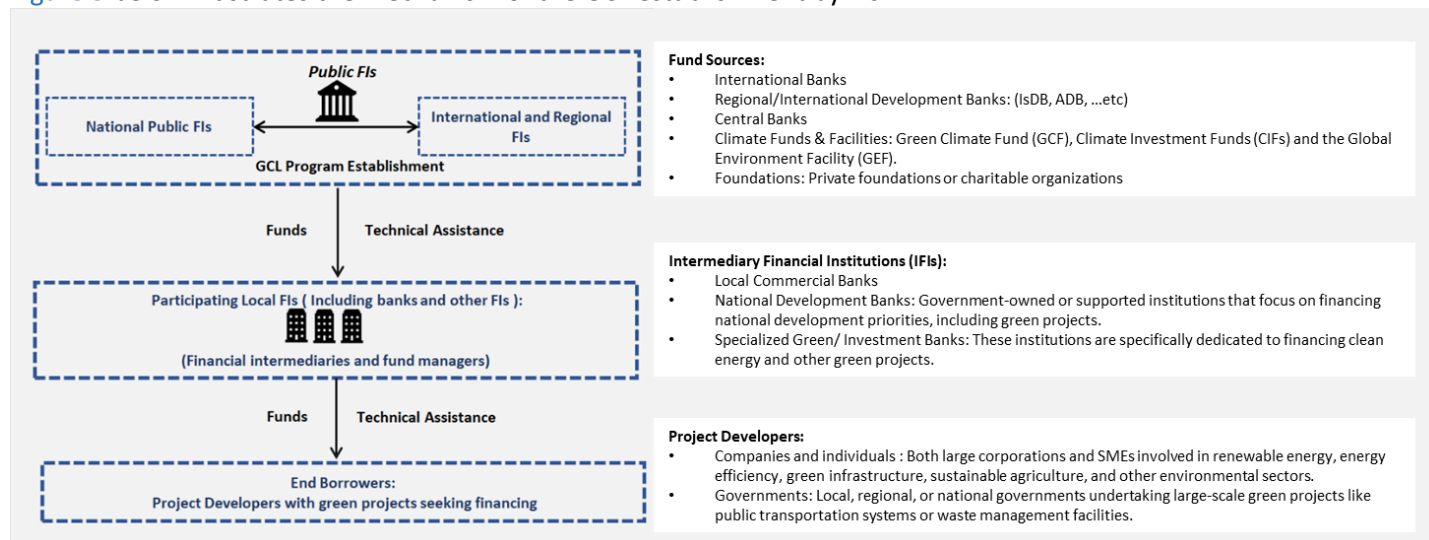


Figure 5: GCL Mechanism

Source: Authors illustration.

Impact Assessment

GCLs play a pivotal role in promoting green investments by addressing financial and non-financial barriers throughout the lending process both from the supply and demand sides. Financial incentives embedded in GCLs -typically provided by national or international public financial institutions, such as development banks, government agencies, or international climate funds- such as concessionally and favorable terms, help alleviate supply-side challenges like high costs and perceived risks. Lower interest rates, longer tenors, and grace periods align loan terms with the lifecycle of green projects, thereby enhancing their financial feasibility. Moreover, incentives like performance payments and credit enhancements encourage the adoption of green lending practices among financial intermediaries [34].

Public FIs (such as International Banks, Development Banks, Central Banks, Climate Funds & Facilities, Foundations or charitable organizations) facilitate GCLs by leveraging their access to stable financing and blending commercial finance with concessional contributions, making green lending more appealing. These institutions can mobilize external public finance and private co-financing, thereby expanding the pool of available funds for green projects. However, empirical evidence on the extent to which financial benefits are passed on to end-borrowers remains limited [35]. Non-financial benefits of GCLs are equally significant, with Technical Assistance programs addressing capacity gaps among financial intermediaries and end-borrowers [36].

The Qatari banking sector is increasingly recognizing the importance of environmental sustainability and is actively participating in the green finance movement. This shift aligns with Qatar's National Vision 2030. As a result, several Qatari banks have implemented green finance initiatives to promote environmentally friendly practices among businesses and consumers. According to secondary data published as of May, 2024, GCLs are not yet a green established within the Qatari financial sector.

The below table provides a snapshot of key green finance initiatives offered by leading Qatari banks. It highlights the diverse range of products and services available, serving various segments of the market:

Table 3: key green finance initiatives offered by leading Qatari banks

Bank	Green Initiatives
------	-------------------

Qatar National Bank (QNB)	National Issued USD 600 million in green bonds
Doha Bank	Green car loans, green mortgage program, green credit cards, green accounts
Qatar Development Bank (QDB)	Green Financing Program: Provides preferential financing for eco-friendly projects and products.
Qatar Central Bank (QCB)	Regulatory and Strategic Efforts: 1/Facilitates green bond issuance 2/Collaborates with QDB for green financing initiatives. 3/ Encourages environmental and social considerations in bank governance
Dukhan Bank	1/ ESG Strategy: Implements a comprehensive strategy with a focus on sustainable finance. 2/Green Products: Offers eco-friendly vehicle financing and partners with GORD for sustainable building practices.
International Bank of Qatar (IBQ)	N/A
Qatar Islamic Bank (QIB)	1/ Sustainable Deposit: Sharia-compliant product allocating funds to environmentally and socially responsible projects. 2/ Sustainability Strategy: Aims to be a leader in Sharia-compliant 1/ Sustainable finance. Aligns with UN SDGs and Qatar National Vision 2030. 2/Equator Principles: Adopted framework to manage environmental and social risks in financed projects.
Masraf Al Rayan	1/ Sustainable Finance Framework: Promotes environmental and social objectives through financing activities. 2/ Green Deposit: Qatar's first Sharia-compliant green deposit product. 3/Green Vehicle Finance Campaign: Offers competitive financing for electric and hybrid vehicles.
Ahli Bank	N/A
Qatar International Islamic Bank (QIIB)	1/ Sustainable Finance Framework: Promotes environmental and social benefits through financing activities aligned with Green Bond Principles (GBP), Social Bond Principles (SBP), Green Loan Principles (GLP), and Social Loan Principles (SLP). 2/Green Financing Products: Offers financing for various green projects like RE, clean transportation, energy efficiency, and sustainable water management.

Source: Compiled by the authors from multiple publicly accessible sources, including annual reports, official bank websites, and sustainability publications.

Research Methodology

Research Framework

The paper proposes a conceptual framework to qualitatively assess the potential for GCL establishment by the FIs and its impact on energy transition in Qatar. The framework highlights two metrics:

Green Credit Line Implementation Potential Index (GCLIP): this index assesses the potentiality of the GCL establishment by the FIs and considers three main factors namely; (1) Financial Institution Performance (based on common aspects specified by frameworks employed by many donors and development banks, such as those used by the World Bank, OECD DAC, IFC, GCF, and ADB), (2) Governmental Frameworks (based on WB, 2023, ADB, 2023), and (3) FIS Internal Capacities (derived from a review of various established GCL programs)⁴. While these factors reflect

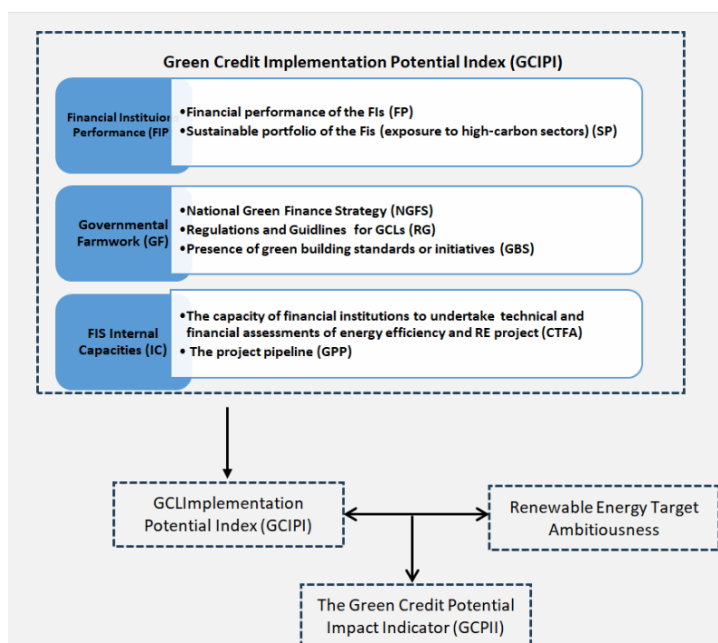


Figure 6: conceptual framework for evaluating the GCL potential in GCC and impact on energy transition

the financial sector's readiness to integrate GCLs into their operations, they represent a preliminary assessment towards a more comprehensive evaluation.

The Green Credit Potential Impact Indicator (GCPII): this indicator aims to provide a preliminary assessment of the potential impact of GCL establishment by the FIs on energy transition.

Figure 6 shows that the impact of GCL establishment for a country will be examined by three factors: the *performance of financial institutions* (FIs) as intermediaries, the overall *governmental frameworks*, and the *internal capacities* of the bank regarding the GCL-specific program. To further enrich the assessment, the framework proposes developing a *GCL Implementation Potential Index (GCLIP)* for Qatar. This index will aggregate scores based on the three identified factors. Additionally, the GCIPI score for Qatar will be compared with scores calculated for two other GCC countries namely; (UAE) and Saudi Arabia (KSA). This comparative analysis will offer valuable insights into the relative strengths and challenges of each country's GCL potential within the regional context. In addition, to assess the Potential Impact of GCL, the framework incorporates an additional concept: "*RE Target Ambitiousness*". This factor, independent of the GCL Potential index, represents a country's level of ambition for increasing its contribution from RE sources. A more ambitious target suggests a stronger commitment to RE, potentially leading to a higher GCL Potential Impact (GCPII) score.

GCL Implementation Potential Index (GCIPI)

The GCIPI assesses the potential of a country to successfully implement green credit programs based on its existing financial and regulatory infrastructure.

To assess the potential of GCLs in Qatar, the UAE, and KSA, a comparative analysis using GCL Potential Index is proposed. This index utilizes the publicly available information. Table 4 illustrates the GCLIP factors, sub-indices, measurements and data sources.

The index will be a composite score (0-100) derived from weight of the three factors:

1. Financial Institution Performance (30%)
2. Governmental Frameworks (40%)

⁴ Namely, Egyptian Pollution Abatement Project II, IFC's Commercializing Sustainable Energy Finance Program (CSEF) in Turkey – Credit Line for Leasing to three Turkish Leasing Companies, EBRD Green Economy Finance Facility Serbia - Green Credit Line to, UniCredit Bank Serbia.

3. Internal Capacities (30%)

1. **The performance of the FIs (30%):** Assesses the degree to which FIs are prepared to meet the Credit Line criteria. This factor is weighted at 30% and includes the following sub-indices:
 - **Financial Performance of the FIs - 50%:** this is measured by the "credit rating" of the FIs. Credit rating indicates the institution's ability to meet its financial obligations and reflects the overall financial health and stability of the institution. It is a critical factor in ensuring the trustworthiness and reliability of financial operations.
 - **Sustainability Portfolio - 50%:** refer to the efforts made to reduce exposure to high-carbon sectors. This involves assessing how banks contribute to or mitigate climate change, particularly through their financing decisions. One key metric for this is the ratio of loans to carbon-intensive sectors relative to total loans.
2. **Governmental Frameworks (40%):** this component evaluates the broader policy and regulatory context that supports green finance. This factor is weighted at 40% and includes the following sub-indices:
 - **National Green Finance Strategy - 33%:** measured by the presence of relevant information in published materials is used to assess this variable.
 - **Regulations for GCLs - Reporting - 33%:** similar to the national strategy, this is assessed by evaluating the presence of relevant information in published materials.
 - **Presence of Green Building Standards or Initiatives - 33%:** the assessment is based on the presence of relevant information in published materials.
3. **Internal Capacities (30%):** this component focus on the specific capabilities and activities related to specific GCL programs. This factor is weighted at 30% and includes the following sub-indices:
 - **Capacity of FIs to Undertake Technical and Financial Assessments of Energy Efficiency and RE Projects - 50%:** measured by examining the conduction of sustainable finance training programs.
 - **Green Projects Pipeline - 50%:** assessed by evaluating the presence of relevant information in published materials.

The rationale for assigning these weights is grounded in the recognition of the fundamental importance of the governmental frameworks in ensuring the success and sustainability of the GCLs⁵. Even if the performance and the internal capacities of the FIs fully met, the absence of a supportive governmental frameworks would significantly hinder the effectiveness of the GCLs. Given these insights, the governmental frameworks are prioritized with a 40% weight to reflect its critical role in supporting the GCL's success. The comprehensive assessment of the *national green finance strategy*, *GCL regulatory and guidelines*, and *green building standards* ensures that the foundational conditions necessary for the GCL are in place. The FIS performance is essential for defining the specific eligibility criteria for the FIs, focusing on the *financial performance* of the FIs and the commitment to the *sustainability (exposure to high-carbon sectors)*. Internal capacities focus on the specific design and implementation aspects of the GCL. This includes the *green finance expertise*, the availability of the *green projects pipeline* among other. The sub-indices for each factor are illustrated in Table 4 with relative importance percentage (out of 100%). A pre-defined scoring system (e.g., 0 for no evidence, 1 for limited evidence, 2 for moderate evidence, 3 for strong evidence) will be applied to each sub-indices. Scores for each factor will be calculated and multiplied by their respective weights. The final GCLPI score for each country will be the sum of the weighted factors scores. The GCLPI scores will serve as a starting point for assessing the potential impact on the energy transition. The below explanatory note shows the scale definitions, the model equations and the benchmark matrix.

⁵ According to the OECD, a clear national strategy for green finance is critical for mobilizing investments and ensuring that financial markets support sustainable development goals. The International Monetary Fund (IMF) emphasizes the importance of regulatory frameworks in ensuring financial stability and transparency, which are crucial for the success of financial instruments like GCLs (IMF, 2023). The World Bank highlights that the adoption of green building standards is a key indicator of a country's commitment to sustainable development and can drive significant environmental benefits (World Bank, 2023).

Table 4: GCL Potential Index, factors, sub-indices, measurements and data sources

Factors	1		2		3	
	Financial Institution Performance (30%)	Measurement	Governmental Frameworks (40%)	Measurement	Internal Capacities (30%)	Measurement
Weighted sub-indices	Financial performance of the FIs (50%)	measured by the credit rating	National Green Finance Strategy (33%)	assessed by evaluating the presence of relevant information in published materials.	The capacity of FIs to undertake technical and financial assessments of energy efficiency and RE project (50%)	measured by examining the conduction of the Sustainable finance training program.
	Sustainable portfolio of the FIs (exposure to high-carbon sectors) (50%)	measured by the efforts to reduce exposure to high-carbon sectors (Loans to carbon intensive sectors / Total loans)	Regulations for GCLs (33%) Presence of green building standards or initiatives (33%)	assessed by evaluating the presence of relevant information in published materials. assessed by evaluating the presence of relevant information in published materials.	The green projects pipeline (50%)	assessed by evaluating the presence of relevant information in published materials.
Data Sources	FIs websites, annual reports industry reports, central bank reports.		Regulatory body websites, government policy documents		government policy documents, national plans and related publications.	
References	World Bank's Financial Intermediary Funds (FIFs) Framework, https://www.worldbank.org/en/programs/financial-intermediary-funds		WB, 2023, ADB, 2023, WB, Financing Energy Efficiency, Part 2: Credit Lines, 2018.		Derived from a review of various established GCL programs.	

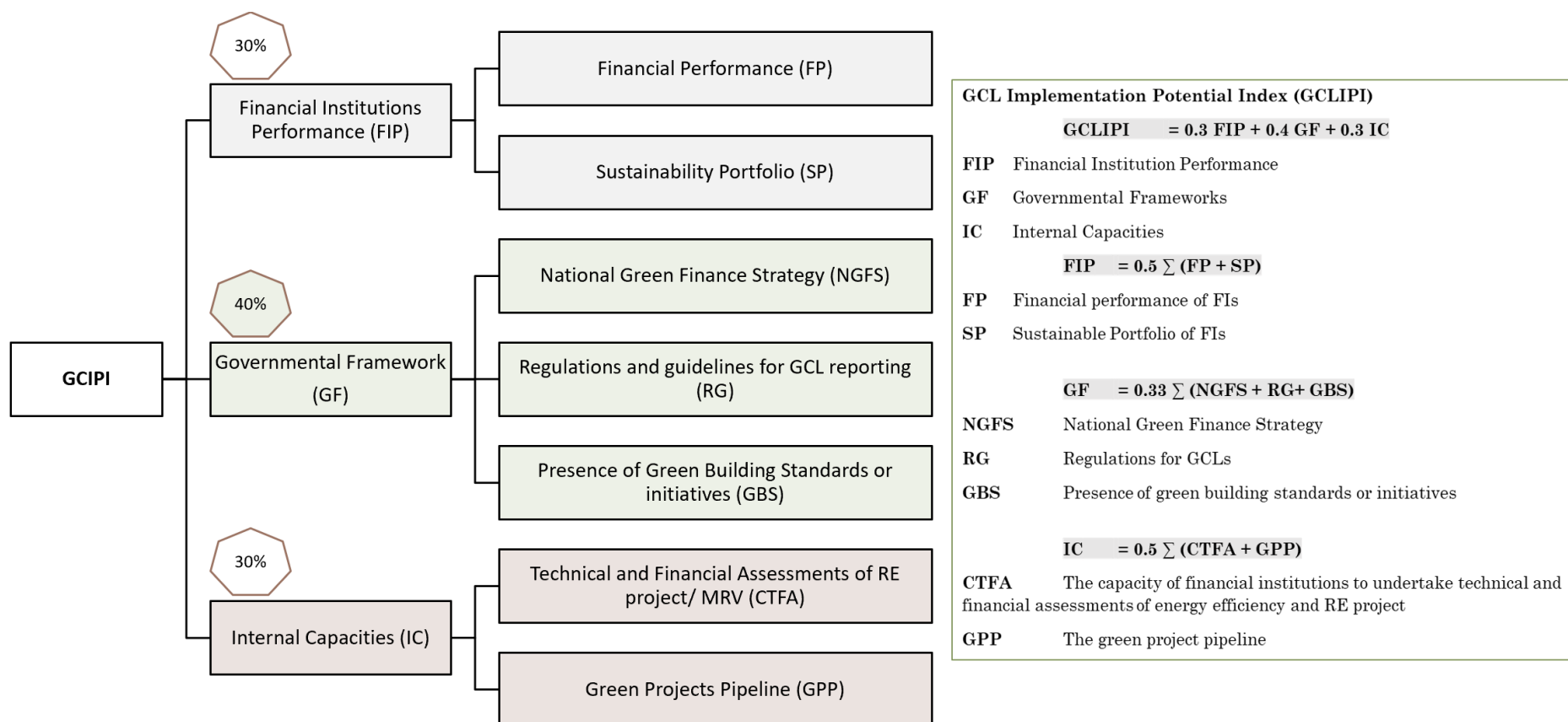


Figure 7: Breakdown of the factors that composing the Green Credit Implementation Potential Index (GCIPI)

Table 5: Scale Definition for GCL Implementation Potential Index (GCIPI): Criteria and Sub-Indicators

Factors	Sub-indices	Abbreviation	Scale definition			
			0	1	2	3
Financial Institution Performance	Financial performance of the FIs	FP ⁶	D and SD (In default)	Low: ranging from CCC+, CCC, CCC-, CC, C.	Med: ranging from BBB+, BBB, BBB-, BB+, BB, BB-, B+, B, B-.	High: ranging from AAA, AA+, AA, AA-, A+, A, A-
	Sustainable portfolio of the FIs (exposure to high-carbon sectors)	SP	No significant efforts to reduce exposure to high-carbon sectors	Some initial efforts towards sustainability, and minor reductions in high-carbon investments	Substantial commitment to sustainability, and moderate reductions in exposure to high-carbon sectors	Leader in sustainability, with aggressive targets and substantial reductions in high-carbon sector exposure
Governmental Frameworks	National Green Finance Strategy	NGFS	No evidence of a National Green Finance Strategy in publicly available data or reports	Indirect mention of a National Green Finance Strategy in publicly available data or reports, but no comprehensive details or dedicated strategy documents	Some documentation or references to a National Green Finance Strategy, possibly including high-level goals or intentions, but lacks detailed implementation plans or strategies	Clear evidence of a National Green Finance Strategy with documented goals, strategies, and initial implementation plans available in publicly published reports.
	Regulations and Guidelines for GCLs - Reporting	RG	No evidence of specific regulations or guidelines for Green Credit Lines (GCL) in publicly available data or reports	Direct mention of regulations or guidelines for Green Credit Lines (GCL), but no comprehensive details or dedicated regulatory framework.	Some documentation or references to regulations or guidelines for Green Credit Lines (GCL), possibly including broad principles or intentions, but lacks detailed implementation guidelines or	Clear evidence of specific regulations or guidelines for Green Credit Lines (GCL), with documented criteria, eligibility requirements, and implementation guidelines available in publicly published reports.

⁶ According to the S&P Global credit rating scale.

					regulatory frameworks	
	Presence of green building standards or initiatives	GBS	No evidence or acknowledgment of specific green building standards or initiatives in publicly available data or reports	Minimal or indirect mention of green building standards or initiatives, but no comprehensive details or dedicated framework	Some documentation or references to green building standards or initiatives, possibly including initial guidelines or principles, but lacks widespread adoption or comprehensive implementation	Clear evidence of specific green building standards or initiatives with documented criteria, certification processes, and adoption by stakeholders available in publicly published reports
Internal Capacities	The capacity of financial institutions to undertake technical and financial assessments of energy efficiency and RE project	CTFA	No evidence or acknowledgment of specific Sustainable Finance training programs or capacity building initiatives related to energy efficiency or renewable energy projects in publicly available data or reports	Indirect mention of Sustainable Finance training programs, but no comprehensive details or dedicated framework for energy efficiency or renewable energy projects	Some documentation or references to Sustainable Finance training programs, possibly including basic training or awareness sessions, but lacks comprehensive coverage of technical and financial assessments	Clear evidence of specific Sustainable Finance training programs with documented curriculum, training modules on technical and financial assessments for energy efficiency and renewable energy projects
	the green project pipeline	GPP	No evidence or acknowledgment of a specific green project pipeline or renewable energy targets in publicly available data or reports	indirect mention of green project pipeline or renewable energy targets, but no comprehensive details or dedicated framework for identifying and developing green project	Some documentation or references to green project pipeline or renewable energy targets, possibly including preliminary listings or intentions, but lacks a structured and detailed pipeline management system	Clear evidence of a specific green project pipeline with documented projects aligned with renewable energy targets, stages of development, and potential financing needs available in publicly published reports

GCL Potential Impact Indicator (GCLPII)

GCLPII measures the expected influence of implementing Green Credit Lines (GCL) by financial institutions (FIs) within a country on advancing energy transition goals.

As mentioned above, this indicator utilizes two factors namely; (1) the GCLPI, and (2) the RE target ambitiousness which measures the level of ambition of a country's RE targets (see explanatory note 2). Data sources include government RE plans and international reports tracking RE targets. The RE target ambitiousness is scored based on a pre-defined scale (e.g., Highly ambitious, moderate, Ambitious with challenges) reflecting the target's level of ambition compared to current energy production. The indicator is calculated as a simple average of the GCLPI score and the RE target ambitiousness score (after converting it to a numerical value).

GCLPII = (GCIPI + RETA) / 2					
GCIPI: Green Credit Implementation Potential Index					
RETA: Renewable Energy Target Ambitiousness					
Green Credit Potential Impact (GCIPI)		Green Credit Implementation Potential Index (GCIPI)			
		0	1	2	3
Renewable Energy Target Ambitiousness (RETA)	1	0.5	1	1.5	2
	2	1	1.5	2	2.5
	3	1.5	2	2.5	3
		GCLPII Score	Category	Expected Impact of Green Credit Lines	Description
			No Impact	No expected positive impact on renewable energy adoption or transition.	The country has low GCIPI and unambitious renewable energy targets, making green credit lines unlikely to significantly accelerate the energy transition.
			Limited Impact	Potential for marginal improvements in renewable energy adoption, but significant barriers remain.	The country has moderate GCIPI and/or limited renewable energy targets, suggesting that green credit lines might have a positive but constrained impact due to regulatory, financial, or technological limitations.
			Moderate Impact	Green credit lines are likely to contribute to a noticeable acceleration of renewable energy adoption and a gradual energy transition.	The country has good GCIPI and ambitious renewable energy targets, indicating a favorable environment for green credit lines to effectively support the transition to cleaner energy sources.
			High Impact	Green credit lines are expected to play a crucial role in rapid renewable energy adoption and a transformative energy transition.	The country has excellent GCIPI and highly ambitious renewable energy targets, creating a strong foundation for green credit lines to significantly catalyze the development and deployment of renewable energy technologies.

Figure 8: Benchmark matrix for the Green Credit Line Potential Impact Indicator (GCLPII)

Table 6: Scale Definition for GCL Potential Impact Indicator (GCIPI)

Highly ambitious (3 points):

The country has a target for RE exceeding a certain percentage of total energy generation by a specific year (e.g., 50% RE by 2040). This target should also be considered ambitious compared to the region's average or the country's current energy production level with suitable timeframe.

Moderate (2 points):

The country has a RE target but it might be less ambitious than the "High" category (e.g., 30% RE by 2050). Targets with long timelines or unclear definitions might also fall into this category.

Ambitious with challenges (1 point):

The country has ambitious target for high RE use, but significant efforts are required and challenges need to be addressed.

Table 7: RE Target Ambitiousness

Qatar		
RE Target Ambitiousness	Note	References
National RE Strategy aiming to increase the RE contribution from 5% to 18% by 2030	A significant increase in RE contribution within a defined timeframe (2030) indicates a highly ambitious target	Qatar National RE Strategy, May-2024.

		REN21 Renewables Global Status Report 2023.
UAE		
RE Target Ambitiousness	Note	References
The 50% target by 2050 represents a long-term vision but is still ambitious compared to the region's average	The 50% target by 2050 represents a long-term vision but is still ambitious compared to the region's average	- UAE Net Zero by 2050 Strategic Initiative. - IEA World Energy Outlook 2023.
KSA		
RE Target Ambitiousness	Note	References
Saudi Arabia's Vision 2030 targets 50% of its electricity generation from RE sources by 2030	The 50% target by 2050 represents a long-term vision but is still ambitious compared to the region's average	- Saudi Arabia Vision 2030. - IEA Executive Director statement.

Source: prepared by the authors based on published documents and national plans.

Analysis and Discussion

GCL Implementation Potential Index (GCLPII)

This study aimed to assess the GCL potential of Qatar compared to the UAE, and KSA by developing a GCL Implementation Potential Index. According to the first national survey on contributions of the FIs to Green Economy⁷, the UAE appears to be the only country among the three with existing GCL within their green finance offerings. This suggests a potential head start for the UAE in terms of GCL implementation. Regarding the degree to which the financial system within the three countries is prepared to establish the GCL, which measured by the financial performance, and the sustainable portfolio of FIs, Qatar and UAE scored highest in this sub-indicator, indicating both the health of the FIs and higher commitment by the FIs to sustainability and energy transition. Regarding the governmental frameworks in the three countries, the analysis revealed that while each country has implemented supportive policies and took steps toward the green finance, the KSA demonstrated a more developed and comprehensive green Financing Framework. Qatar and the UAE could benefit from exploring similar initiatives to enhance their GCL potential. For the internal capacities (measured by the green finance expertise, green projects pipeline), while all three countries showed similar results regarding the green projects pipeline, the UAE demonstrates a documented effort regarding the sustainable finance training. Based on the factors analyzed, UAE appears to be the most prepared for GCL establishment by the FIs and deployment. The KSA shows promise with a focus on building enabling governmental frameworks. While Qatar is possessing a strong financial sector, limitations in the enabling environment criteria hinder its immediate GCL readiness.

GCL Potential Impact Indicator (GCLPII)

The RE target ambitiousness is assessed through reviewing the RE target for the three countries within the specified timeframe (Table 7). Qatar's RE target stands out as highly ambitious, aiming for a significant increase in the share of RE in the power mix by 2030 (from 5% to 18%) as documented in Qatar National RE strategy [37]. By facilitating access to financing for green projects, GCLs can play a crucial role in enabling Qatar to achieve its ambitious RE targets. Many studies demonstrate how GCLs can accelerate clean energy development by providing tailored financing solutions for RE projects [38].

For KSA, the RE target highlights ambitious within the specified timeframe, however, there are many challenges the kingdom might encounter and thus the transition requires more effort. The result reveals that KSA may witness high impact from establishing the GCL by the FIs given its GCL potential and RE target. Hence, the GCL potential impact indicator is the highest for the KSA given the challenging target of RE, the GCL might act as catalyzer for enabling such transition.

⁷ UAE Green Agenda 2015-2030 State of Green Finance in the UAE The first national survey on contributions of financial institutions to Green Economy

While the UAE demonstrates strong green finance initiatives, the relatively longer timeframe for RE targets reduces the potential impact of GCLs in accelerating the transition.

To sum up, the study compared the GCL potential of Qatar, the UAE, and KSA. While the UAE has existingly limited GCL products, KSA possesses the strongest readiness of financial sector. All three countries share similar green projects pipelines and ambitious RE targets, with Qatar having a particularly new set high target. Analyzing GCL potential alongside RE target ambition, the study suggests KSA has the highest GCL potential impact on energy transition. Qatar also shows promise due to its RE target, while the UAE's more moderate RE target leads to a relatively lower impact of GCLs. Overall, GCLs hold promise for the GCC, but each country needs to address its limitations to unlock full potential of GCLs.

Table 8: The results of scoring the sub-indices of the GCLPI and references

Qatar	Sub-index weight	Score	weighted scores	References
Financial Institutions Performance - country scale (FIP)	30%			(1,2)
Financial performance of the FIs	0.5	3	1.5	(1) Credit rating, S&P Global Ratings 2022
Sustainable Portfolio of FIs	0.5	2	1	(2) Loans to carbon intensive sectors / Total loans, Banks — Gulf Cooperation Council: 2022 Outlook
Sub index score			2.5	
weighted sub index score			0.75	
Governmental Frameworks (GF)	40%			(3,4,)
National Green Finance Strategy	0.33	1	0.33	(3) Qatar Central Bank's Plan (2017-2022) with intention to implement strategic actions on green finance
Regulations for GCL	0.33	0	0	No regulations found yet.
Presence of green building standards or initiatives	0.33	3	0.99	(5) Qatar has 1,406 sustainability certified buildings, according to the Ministry of Environment and Climate Change
Sub index score			1.32	
weighted sub index score			0.528	
Internal Capacities (IC)	30%			(5,6,7)
The capacity of financial institutions to undertake technical and financial assessments of energy efficiency and RE project	0.5	0	0	(5) Opportunities for the GCC to strengthen the sustainable finance ecosystem, PwC research adapted from the Atlantic Council, 2022. Official public resources, 2023 https://www.strategyand.pwc.com/m1/en/strategic-foresight/sector-strategies/financial-services/middle-east-green-finance/middle-east-green-finance.pdf
the project pipeline	0.5	2	1	(6) Qatar National Vision 2030 - Renewable Energy Goals
Sub index score			1	
weighted sub index score			0.3	
Total GCL Implementation Potential Index	1.578			
UAE	Sub-index weight	Score	weighted scores	References
Financial Institutions Performance - country scale (FIP)	30%			(1,2)
Financial performance of the FIs	0.5	3	1.5	(1) Liquid assets/assets (%), S&P Global Ratings, 2022
Sustainable Portfolio of FIs	0.5	2	1	(2) Loans to carbon intensive sectors / Total loans, Banks — Gulf Cooperation Council: 2022 Outlook
Sub index score			2.5	
weighted sub index score			0.75	
Governmental Frameworks (GF)	40%			(8,9)
National Green Finance Strategy	0.33	2	0.66	(8) UAE Sustainable Finance Framework 2021 - 2031
Regulations for GCLs -Reporting	0.33	0	0	No regulation found yet
Presence of green building standards or initiatives	0.33	3	0.99	(9) Estidama Green Building Rating System (UAE)
Sub index score			1.65	

weighted sub index score			0.66	
Internal Capacities (IC)	30%			(5, 10, 11)
The capacity of financial institutions to undertake technical and financial assessments of energy efficiency and RE project	0.5	1	0.5	(5) Opportunities for the GCC to strengthen the sustainable finance ecosystem, PwC research adapted from the Atlantic Council, 2022. Official public resources, 2023 https://www.strategyand.pwc.com/m1/en/strategic-foresight/sector-strategies/financial-services/middle-east-green-finance/middle-east-green-finance.pdf
The project pipeline	0.5	2	1	(10) Emirates NBD Green Financing Initiatives, UAE Energy Strategy 2050, Dubai Green Fund.
Sub index score			1.5	
weighted sub index score			0.45	
Total GCL Potential Index	1.86			
KSA	Sub-index weight	Score	weighted scores	References
Financial Institutions Performance - country scale (FIP)	30%			(1,2)
Financial performance of the FIs (efficiency)	0.5	3	1.5	1) Liquid assets/assets (%), S&P Global Ratings
Sustainable Portfolio of FIs	0.5	1	0.5	(2) Loans to carbon intensive sectors / Total loans, Banks — Gulf Cooperation Council: 2022 Outlook
Sub index score			2	
weighted sub index score			0.6	
Governmental Frameworks (GF)	40%			(12, 13)
National Green Finance Strategy	0.33	3	0.99	(12) Green Financing Framework March 2024, https://ndmc.gov.sa/investorsrelations/Documents/Green-Financing-Framework-KSA-28March2024.pdf
Regulations for GCL	0.33	0	0	No regulation found yet
Presence of green building standards or initiatives	0.33	3	0.99	(13) MOSTADAM Certificate
Sub index score			1.98	
weighted sub index score			0.792	
Internal Capacities (IC)	30%			(5, 14, 15)
The capacity of financial institutions to undertake technical and financial assessments of energy efficiency and RE project	0.5	0	0	Opportunities for the GCC to strengthen the sustainable finance ecosystem, PwC research adapted from the Atlantic Council, 2022. Official public resources, 2023 https://www.strategyand.pwc.com/m1/en/strategic-foresight/sector-strategies/financial-services/middle-east-green-finance/middle-east-green-finance.pdf
the project pipeline	0.5	2	1	(14) Green Financing Framework March 2024 and Giga projects
Sub index score			1	
weighted sub index score			0.3	
Total GCL Implementation Potential Index	1.692			

Table 9: The results of Scoring the factors of the GCL Potential Impact Indicator

Qatar		
RE Target Ambitiousness	Score	
National RE Strategy aiming to increase the RE contribution from 5% to 18% by 2030	3	High
GCLPI	1.58	
GCL Potential Impact Indicator	2.29	
UAE		
RE Target Ambitiousness	Score	
The 50% target by 2050 represents a long-term vision but is still ambitious compared to the region's average	2.5	Medium
GCLPI	1.86	
GCL Potential Impact Indicator	2.18	
KSA		
RE Target Ambitiousness	Score	
Saudi Arabia's Vision 2030 targets 50% of its electricity generation from RE sources by 2030	3	High
GCLPI	1.69	
GCL Potential Impact Indicator	2.35	

As shown in the above table, the three countries have good GCLPI and ambitious renewable energy targets, indicating a favorable environment for green credit lines to effectively support the transition to cleaner energy sources.

Conclusion

This study adopted a qualitative conceptual framework to assess the GCLs potential in Qatar and the wider GCC region. It begins with a literature review that defined and explored the enablers, opportunities, and challenges associated with the potential and impact of GCLs. Building upon this foundation, the study presented a methodology to assess both the GCL establishment potential and the impact on achieving Qatar's energy transition goals compared with two GCC countries, namely, the UAE and KSA. This methodology involved examining publicly published and historical data to evaluate the progress of GCL establishment and sustainable finance frameworks.

To comprehensively measure GCL potential and impact, the study utilizes two metrics: the GCL Implementation Potential Index and the GCL Potential Impact Indicator. The study identified the GCL potential across all three Gulf nations. While Qatar possesses a strong financial sector, limitations in its governmental frameworks hinder its immediate GCL readiness. However, Qatar's highly ambitious RE target, aiming for a significant increase in the share of RE in the power mix, necessitates innovative financing solutions like GCLs. UAE emerged as the country with the strongest overall GCL potential based on the GCL Implementation Potential Index. This suggests a head start for the UAE in terms of practical GCL experience. This is attributed to its robust financial sector and the existence of current GCL products. The KSA demonstrated a more developed and comprehensive green Financing Framework. All three countries share ambitious RE targets and similar green projects pipelines, highlighting a strong commitment to clean energy development.

Policy Recommendations

This study emphasizes the need for Qatar to prioritize the development of *a comprehensive green financing and investment framework*. Such a framework can address the limitations identified by the GCL Implementation Potential Index within the country's enabling environment. A well-designed framework should aim to enhance the competitiveness of green investments compared to traditional brown alternatives [39]. The main objective is to reduce perceived high risks associated with the energy transition projects from the investor point of view. This will increase the investor base for this type of projects which, in turn, will have a positive effect on the green projects pipeline. Furthermore, sufficient financial regulations will address the uncertainty and risk aversion among potential investors.

In addition to the overall finance and investment framework, Qatar can introduce policies that incentivize or require potential borrowers to participate in training programs on green project development and financing. These programs, potentially co-developed with the Qatari Central Bank and FIs, should equip borrowers with the necessary skills to develop technically sound and bankable green projects. This also will accelerate the increase in the investor base and green projects pipeline.

Financial Sector Recommendations

Qatar and GCC region in general need to focus on capacity building within the banking sector, the FIs should prioritize building capacity within through targeted training programs. These programs should focus on equipping the finance-oriented team (bank staff) with the necessary expertise to assess green projects. Training should cover the technical, economic & financial, and socio-environmental attributes of the green loan and the underlying project. Bank staff must be proficient in technical terminologies such as energy efficiency measures, greenhouse gas emissions, CO2 savings, efficiency factors, and baseline energy sources. A deep understanding of these concepts will aid in client acquisition, product promotion, and effectively communicating the benefits of GCL. Additionally, to enhance the promotion of green loans and their benefits to the borrows (investors), bank officers should possess both theoretical and practical knowledge of common green measures, including energy efficiency and RE solutions enabling them to confidently discuss these topics during introductory or basic conversations with potential clients.

By addressing the identified limitations and implementing the recommended actions, Qatar can unlock the full potential of GCLs and significantly accelerate its transition towards a sustainable economy. GCLs can play a pivotal role in mobilizing financial resources towards RE projects, contributing meaningfully to achieving Qatar's ambitious climate change mitigation goals. Further research exploring the specific details of potential GCL programs and tracking the effectiveness of GCLs in promoting green investments would provide valuable insights for ongoing development and refinement of GCL implementation strategies in Qatar. Moreover, adopting clear taxonomies such as the GLP will help in differentiating between green and brown projects, ensuring the credibility and effectiveness of green credit lines. By realizing these recommendations, Qatar can bridge the gap with its competitors and leverage GCLs to achieve its ambitious clean energy goals.

Study Limitations

This study acknowledges several limitations that require further consideration. The quality and accessibility of publicly available data vary significantly across Gulf countries, affecting the accuracy of the GCL implementation Potential Index and GCL Potential Impact Indicator. Additionally, the GCL Potential Index and Impact Indicators, while useful for comparative analysis, could benefit from further refinement by integrating more data sources and expert insights. Incorporating specific features and eligibility criteria of existing GCL products and obtaining qualitative insights from green finance specialists would strengthen the assessment. Addressing these limitations through stakeholder engagement and additional data sources in the UAE and KSA would provide a more detailed understanding of the potential role of GCLs in promoting a sustainable Gulf economy.

References

- [1] Organization for Economic Co-operation and Development (OECD). *OECD Work on Green Finance and Investment*. OECD Environment Directorate, 2022. <https://www.oecd.org/cgfi/about/OECD-work-on-green-finance-and-investment.pdf>.
- [2] Battiston, Stefano, Irene Monasterolo, Keywan Riahi, and Bas J. van Ruijven. "Accounting for Finance is Key for Climate Mitigation Pathways." *Science* 372, no. 6545 (2021): 918-920.
- [3] Corfee-Morlot, Jan, Valérie Marchal, Céline Kauffmann, Christopher Kennedy, Fiona Stewart, Christopher Kaminker, and Graeme Ang. *Towards a Green Investment Policy Framework: The Case of Low-Carbon, Climate-Resilient Infrastructure*. 2012.
- [4] World Bank. 2020.
- [5] *Engineering Net Zero*, Gulf Cooperation Council Report. 2022.
- [6] Al-Sarhi, Aisha. *Climate Action in the GCC: Current Status and Future Prospects*. 2021.
- [7] Al-Noaimi, Fatima, Tareq Al-Ansari, and Yusuf Bicer. "Toward a Long-Term Low Emission Development Strategy: The Case of Energy Transition in Qatar." *Global Challenges* 7, no. 3 (2023): 2200229.
- [8] International Energy Agency (IEA). 2021. *Energy Policies of IEA Countries: GCC Report 2021*. Paris: International Energy Agency.
- [9] World Bank. 2021. *World Development Report 2021: Data for Better Lives*. Washington, DC: World Bank
- [10] Almulla, Yousef, and Mark Howells. *Gulf Cooperation Council (GCC) Countries 2040 Energy Scenario for Electricity Generation and Water Desalination*. Master of Science Thesis, KTH Industrial and Engineering Management, 2014.
- [11] World Bank, *Air Conditioning and Energy Demand in Hot Climates*.
- [12] European Commission, *Energy Efficiency in Buildings*.
- [13] Passivhaus Institute, *Passivhaus Standard*.
- [14] International Energy Agency (IEA), *Energy Efficiency 2021* and Lawrence Berkeley National Laboratory (LBNL), *The Future of Air Conditioning for Buildings*.
- [15] Sakhrich, Ahmad. "Energy Analysis for GCC Countries." In *Global Conference on Renewables and Energy Efficiency for Desert Regions*, 1-4. 2016.
- [16] Wogan, David, Shonali Pradhan, and Salman Albardi. *GCC Energy System Overview–2017*. Riyadh: King Abdullah Petroleum Studies and Research Center, 2017.
- [17] General Electric. (n.d.). *Pathways to faster decarbonization in the GCC's power sector: Accelerating the energy transition with a combination of renewables and gas power*. General Electric.
- [18] International Monetary Fund. *Gulf Cooperation Council Economic Prospects and Policy Priorities for the GCC Countries*. 2023.
- [19] International Monetary Fund. *Gulf Cooperation Council Economic Prospects and Policy Priorities for the GCC Countries*. 2023.

- [20] *The GCC and the Road to Net Zero*. Middle East Institute. <https://mei.edu>.
- [21] University of Notre Dame. 2023.
- [22] UNEP Finance Initiative (2015), *The Principles for Positive Impact Finance: A Common Framework to Finance- The Sustainable Development Goals*. <http://www.unepfi.org/positive-impact/principles-for-positive-impact-finance/>
- [23] GFANZ, (2021), *The Glasgow Financial Alliance for Net Zero*.
- [24] International Energy Agency (IEA). (2021). *World Energy Investment 2021*. International Energy Agency. <https://www.iea.org/reports/world-energy-investment-2021>.
- [25] United Nations Environment Program (UNEP). (2022). *Global Trends in Renewable Energy Investment 2022*. UNEP. <https://www.unep.org/resources/report/global-trends-renewable-energy-investment-2022>.
- [26] World Bank. (2020). *Financing Climate Action: Green Finance and the Role of Public Policy*. World Bank Group. <https://www.worldbank.org/en/topic/green-finance>.
- [27] Shishlov, Igor, Tina Bajohr, Mariia Deheza, and Ian Cochran. *Using Credit Lines to Foster Green Lending: Opportunities and Challenges*. Paris: Institute for Climate Economics, 2017.
- [28] Organization for Economic Co-operation and Development (OECD). (2020). *Green Financing: Challenges and Opportunities in Emerging Markets*. OECD.
- [29] Smith, J., et al. (2020). "Unlocking the potential of Green Credit Lines: Lessons from emerging markets." *Journal of Sustainable Finance & Investment*, 10(3), 257-273
- [30] Loan Market Association (LMA). (2021). *Green Loan Principles*. Available at: LMA Green Loan Principles.
- [31] Climate Bonds Initiative. (2020). *Green Bond Principles and Standards*. Available at: Climate Bonds Initiative.
- [32] European Commission. (2020). *EU Taxonomy for Sustainable Activities*.
- [33] International Capital Market Association (ICMA). (2021). *Green Bond Principles*.
- [34] Aravamuthan, Vijay, Miguel Ruete, and Andres Dominguez. "Enhancing Private Sector Investment in Green Growth and Climate Change Resilience." *Inter-American Development Bank*, 2015.
- [35] Organization for Economic Co-operation and Development. "Climate Finance in 2013-2014 and the USD 100 Billion Goal: A Report by the OECD in Collaboration with Climate Policy Initiative." OECD, 2016.
- [36] Enclude Limited. "Enhancing Private Sector Investment in Green Growth and Climate Change Resilience." Enclude, 2016.
- [37] Qatar National Renewable Energy Strategy. May 2024.
- [38] Brown, Jessica, and Michael Jacobs. "Leveraging Private Investment: The Role of Public Sector Climate Finance." *Overseas Development Institute*, 2011.
- [39] World Bank. "Climate Investment Funds." <https://fifitrustee.worldbank.org/en/about/unit/dfi/fifitrustee/funds>.

Supercritical Water Technology - A Sustainable Solution for Waste Management in Qatar

By: Dr. Muzamil Abdella

Introduction

Qatar's rapid industrialization and urbanization have resulted in escalating waste generation, placing enormous stress on its existing waste management infrastructure. With limited land availability and the environmental hazards posed by traditional waste management methods such as incineration and landfilling, there is an urgent need for innovative and sustainable approaches. Supercritical Water Technology (SCWT) emerges as a transformative solution, capable of efficiently treating waste, recovering valuable resources, and aligning with Qatar's vision for sustainable development under its National Vision 2030.

SCWT leverages advanced principles of thermal treatment, offering unparalleled efficacy in addressing a broad spectrum of waste management challenges. By integrating SCWT into its national strategy, Qatar can establish itself as a leader in environmental stewardship and circular economy practices.

SCWT Principles and Advantages

SCWT operates by exploiting the properties of water in its supercritical state, where it exhibits characteristics of both gas and liquid. Achieved at temperatures exceeding 374°C and pressures above 22.1 MPa, this state enables water to act as a powerful solvent and oxidant, breaking down organic waste into harmless byproducts such as water, carbon dioxide, and mineral salts.

One of the defining advantages of SCWT is its remarkable treatment efficiency. Unlike traditional waste treatment methods that often leave residual pollutants, SCWT ensures complete oxidation of even the most refractory organic compounds. This makes it highly effective for treating complex and hazardous waste streams, including industrial byproducts, medical waste, and municipal sludge.

In addition to its efficiency, SCWT minimizes waste generation by producing negligible sludge, reducing the volume of secondary waste that requires disposal. This makes it a cleaner alternative to conventional methods. Furthermore, the technology enables the recovery of valuable resources such as carbon dioxide, nutrients, and energy, turning waste into an economic asset while reducing the need for virgin materials.

SCWT also provides significant environmental benefits by drastically lowering greenhouse gas emissions and reducing reliance on landfills. By avoiding the harmful byproducts associated with incineration, such as dioxins and furans, SCWT ensures a cleaner and safer environment.

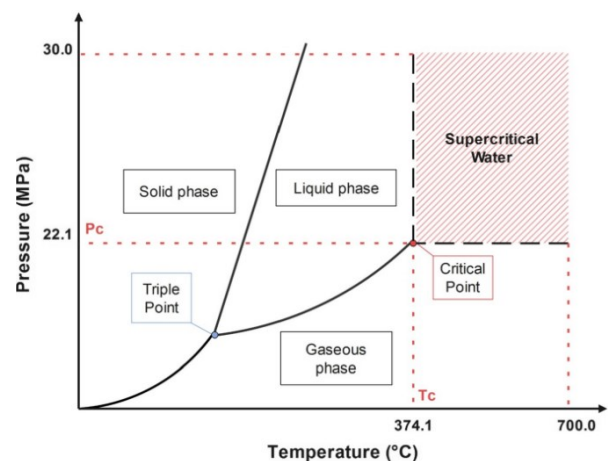


Figure-1: Generic phase diagram for pure water

SCWT and Resource Recovery

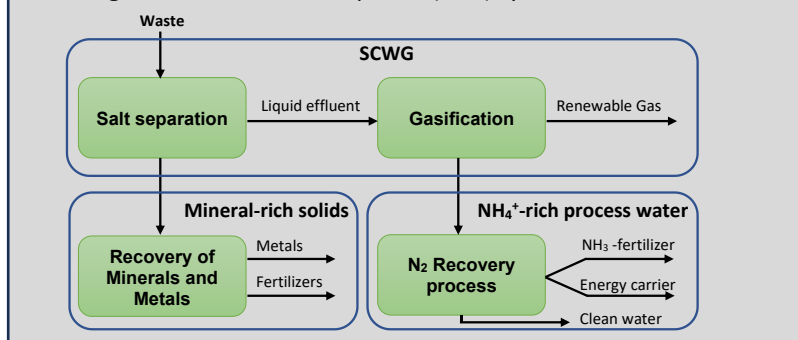
One of the most compelling aspects of SCWT is its ability to recover valuable resources from waste streams, contributing to economic and environmental sustainability. For instance, the process generates high-purity carbon dioxide under high pressure, which can be repurposed for industrial applications such as the production of methanol, urea, and other value-added chemicals. This not only provides an avenue for reducing greenhouse gas emissions but also creates economic opportunities.

SCWT also facilitates nutrient recovery, particularly from treated sewage effluent (TSE). By extracting essential elements such as phosphorus and nitrogen, the technology

supports the production of fertilizers, which can be used in agriculture to enhance soil fertility. This aligns with Qatar's goals of achieving greater resource efficiency and reducing its dependence on imported agricultural inputs.

Energy recovery is another significant benefit of SCWT. The high-temperature effluent produced during the process can be harnessed for electricity generation or used to preheat incoming waste streams. By integrating combined heat and power (CHP) systems, SCWT can further enhance energy efficiency and contribute to Qatar's renewable energy targets.

- **CO₂ capture and utilization:** SCWT generates high-pressure pure CO₂, which can be converted into value-added products like urea, methanol, and other chemicals.
- **Nutrient recovery:** Treating treated sewage effluent (TSE) can recover valuable nutrients like phosphorus and nitrogen for fertilizer production.
- **Energy recovery:** SCWT's high-temperature effluent can be utilized for generating electricity or preheating incoming waste streams through combined heat and power (CHP) systems.



SCWT and Circular Economy

SCWT is a vital enabler of Qatar's transition to a circular economy, where waste is minimized, and resources are continuously repurposed. By converting waste into usable products, SCWT reduces landfill dependency and promotes sustainable resource utilization. For example, the technology ensures that treated wastewater can be safely reused in industrial and agricultural applications, closing the loop on water use.

This approach significantly reduces the need for virgin materials, as recovered nutrients and carbon dioxide can replace raw inputs in various industries. By fostering resource efficiency, SCWT contributes to a more sustainable and resilient economy, aligning with global best practices in circularity.

SCWT and Water Scarcity

Water scarcity is a critical challenge for Qatar, exacerbated by its arid climate and growing demand for freshwater resources. SCWT offers innovative solutions to address this issue by enabling the treatment and reuse of wastewater. For example, treated sewage effluent (TSE) processed through SCWT can be repurposed for industrial cooling, irrigation, or other non-potable applications, reducing pressure on Qatar's freshwater reserves.

Additionally, SCWT enhances water conservation by enabling industries to reuse process water, lowering overall consumption. This dual benefit of water treatment and recovery makes SCWT a valuable tool in Qatar's efforts to secure its water future.

SCWT vs. Traditional Waste Management

Compared to conventional waste management methods, SCWT offers numerous advantages that make it a superior choice for Qatar. Traditional approaches like landfilling and incineration often result in significant environmental degradation. Landfilling consumes vast amounts of land, which is a scarce resource in Qatar, and poses risks of soil and groundwater contamination. Incineration, on the other hand, releases harmful pollutants and greenhouse gases.

In contrast, SCWT eliminates these concerns by providing a clean and efficient alternative. The technology minimizes emissions, ensuring compliance with stringent environmental regulations. Additionally, SCWT's ability to recover resources such as energy and nutrients not only offsets its operational costs but also contributes to long-term sustainability. By converting waste into valuable products, SCWT transforms waste management into an opportunity for economic growth.

SCWT Integration with Other Technologies

SCWT is most effective when integrated with existing waste management systems, creating a comprehensive and synergistic approach. For instance, SCWT can complement anaerobic digestion by treating the effluent generated during the process. This improves the quality of the treated output and enables the recovery of additional resources such as energy and nutrients.

Similarly, integrating SCWT with waste-to-energy (WtE) systems can enhance energy recovery by processing waste streams that are unsuitable for direct combustion. SCWT-treated water can also be reused in water treatment plants, supporting conservation efforts and reducing overall water demand.

Economic and Environmental Assessment

To ensure the successful implementation of SCWT in Qatar, a thorough economic and environmental assessment is essential. A cost-benefit analysis can evaluate the financial feasibility of the technology by comparing operational costs with the potential revenue from resource recovery. Life cycle assessment (LCA) can provide insights into the environmental impacts of SCWT, quantifying benefits such as reduced emissions and resource conservation. Additionally, market analysis can identify opportunities for commercializing recovered resources, further enhancing the technology's economic appeal.

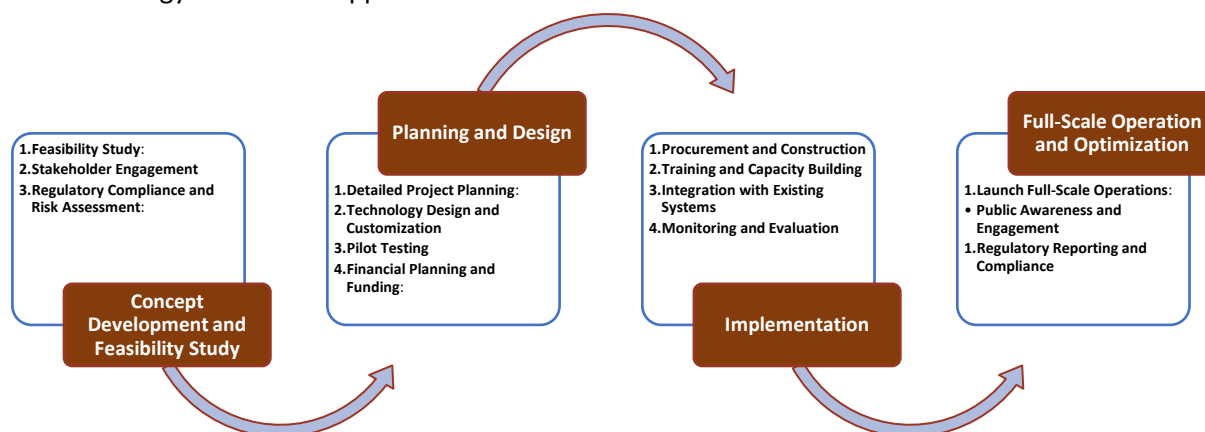


Figure 2: Supercritical Water Oxidation Innovation Road Map

Policy and Public Engagement

The adoption of SCWT requires supportive policies and regulatory frameworks. Financial incentives, such as subsidies or tax credits, can encourage investment in SCWT infrastructure. Streamlined permitting processes and dedicated research and development funding can further accelerate implementation.

Public awareness campaigns are equally important to build trust and acceptance. Transparent communication about SCWT's environmental and economic benefits can address concerns and foster collaboration among stakeholders.

Conclusion

Supercritical Water Technology (SCWT) offers a revolutionary approach to waste management, aligning seamlessly with Qatar's sustainability objectives. By efficiently treating waste, recovering valuable resources, and reducing environmental impact, SCWT addresses critical challenges while contributing to a circular economy.

To unlock its full potential, Qatar must invest in research, pilot projects, and supportive policies. Stakeholders such as Ashghal can play a pivotal role in championing SCWT adoption, showcasing Qatar's leadership in sustainable innovation. By embracing SCWT, Qatar can set a global benchmark for waste management excellence, fostering a cleaner, greener future.

References

1. enerQA Consultancy, 2024. *Concept Note on Supercritical Water Oxidation Technology for Sustainable Waste Management in Qatar*. Available from: SCWO.pdf.
2. U.S. Environmental Protection Agency, 2021. *SCWO: A Promising Technology for Hazardous Waste Management*. Available at: https://www.epa.gov/sites/default/files/2021-01/documents/pitt_research_brief_scwo_final_jan_25_2021_508.pdf.
3. SpringerLink, 2021. *Advances in Supercritical Water Oxidation Technology*. Available at: <https://link.springer.com/article/10.1007/s11157-021-09601-0>.
4. MDPI, 2023. *Resource Recovery from Waste via SCWO Processes*. *Energies*, [online] 16(4), p.2081. Available at: <https://www.mdpi.com/1996-1073/16/4/2081>.
5. Semspub, 2021. *Supercritical Water Oxidation for Effective Hazardous Waste Treatment*. U.S. Environmental Protection Agency. Available at: <https://semspub.epa.gov/work/06/9545963.pdf>.

Driving Climate Action Through Renewable Energy Finance: Insights from an Expert

By: Dr. Islam M. Awad and Giovanni Fabbio

The urgency of addressing climate change has never been more apparent. As global temperatures continue to rise and the impacts of extreme weather events become more severe, the need for sustainable solutions is critical. **Renewable energy finance** is at the heart of these solutions, providing the necessary capital to transition from fossil fuels to clean energy sources. Here to walk you through the pivotal role of finance in advancing climate action and to share insights from my experience in the field.

The Role of Finance in Climate Action

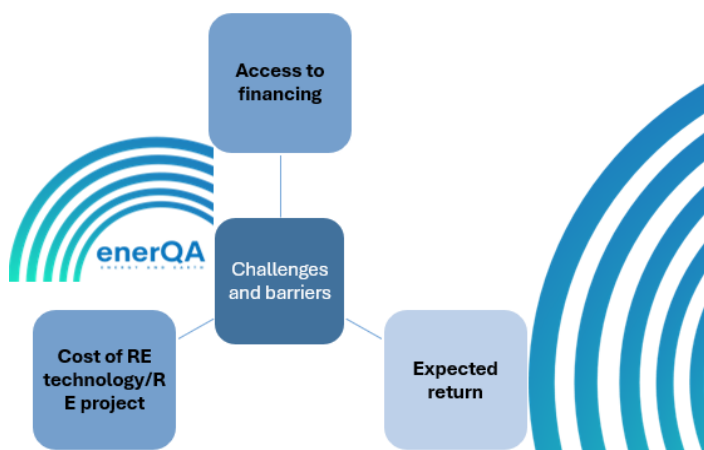
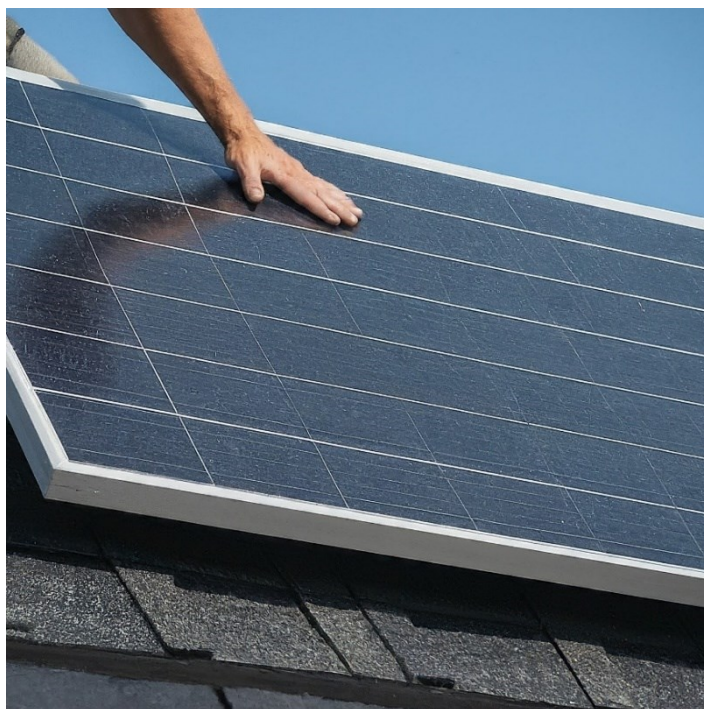
Overview

Finance is the lifeblood of climate action. It enables the development and deployment of technologies and infrastructures necessary for a sustainable future. From solar farms to wind turbines, the capital required to bring these projects to fruition is substantial. However, with the right financial mechanisms and investments, we can accelerate the transition to a low-carbon economy.

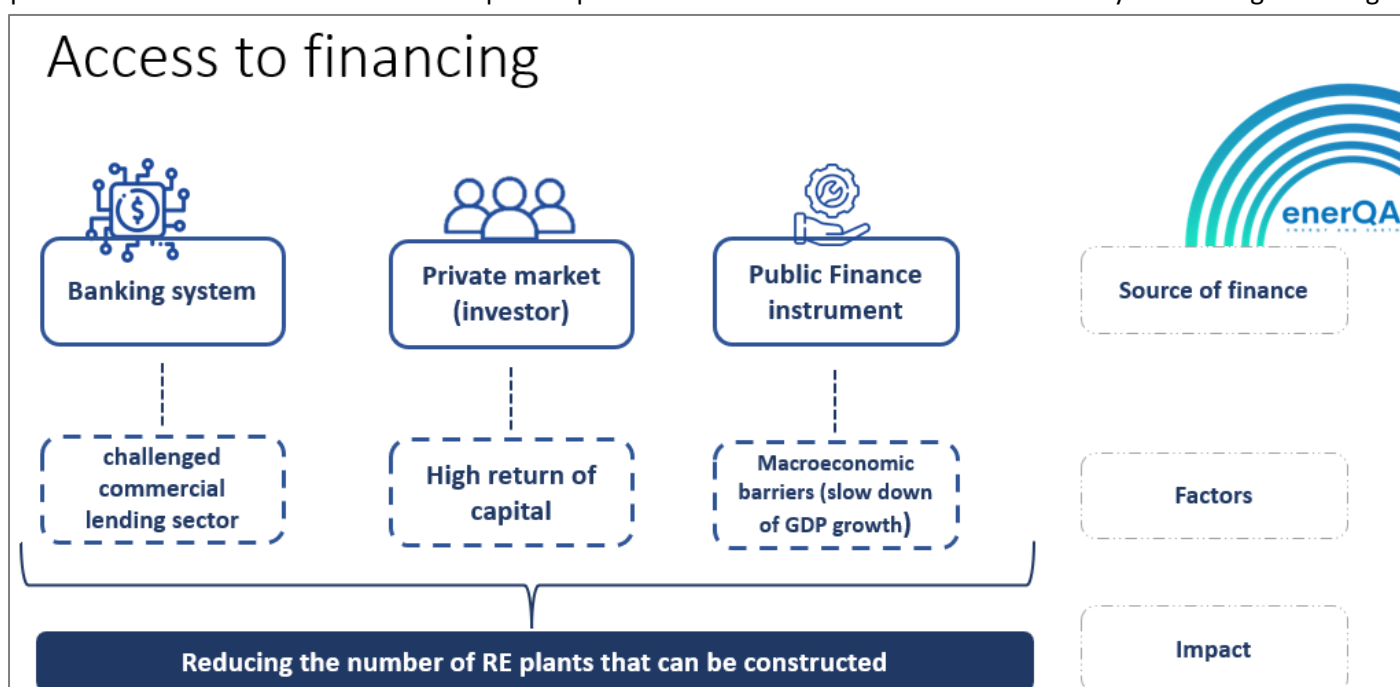
Challenges and barriers

In renewable energy projects, several challenges and barriers arise. Key factors include **access to financing**, often hindered by perceived risks and high initial costs. **Expected returns** are crucial for investors who require confidence in long-term profitability. Additionally, **the costs associated with renewable energy technologies and projects** can be significant, requiring substantial upfront capital. Addressing these challenges involves innovative financial mechanisms, effective risk assessments, and supportive policies. Overcoming these obstacles is essential for making renewable energy projects more appealing and feasible, ultimately promoting a transition to a sustainable and low-carbon economy.

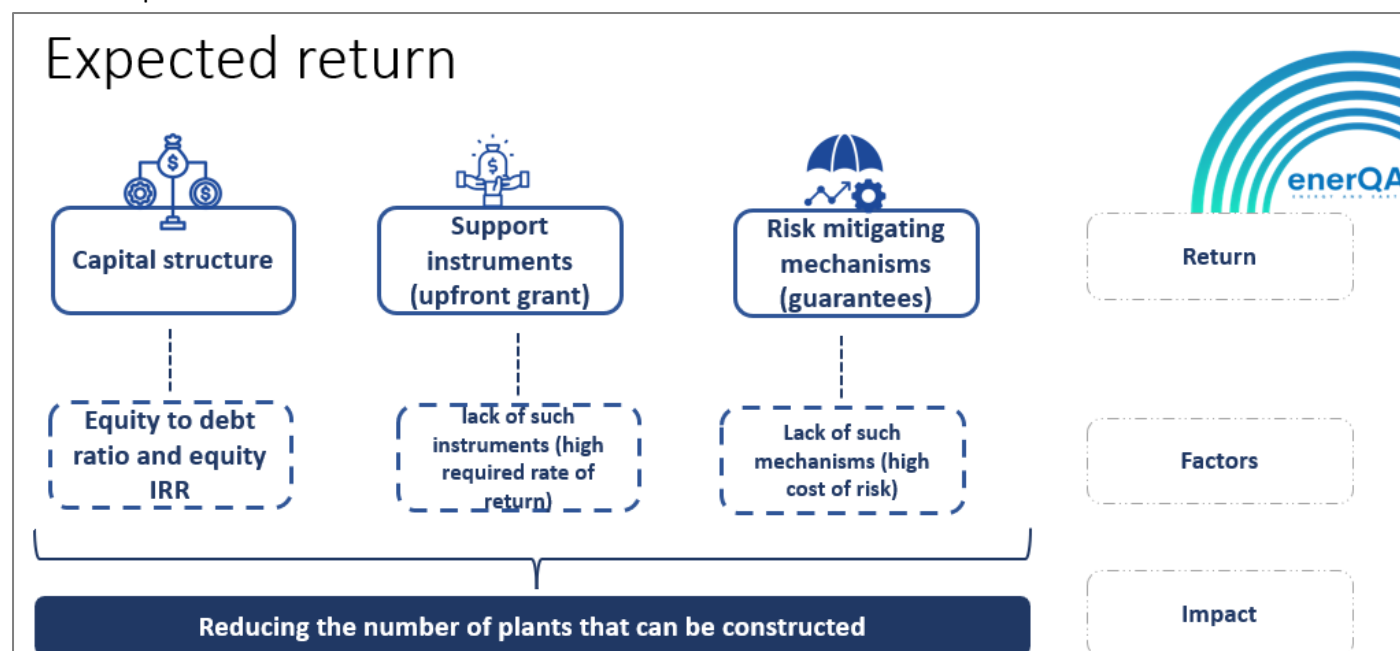
The challenges that project developers particularly in developing countries face in securing the necessary financing for their projects. There are three main sources of finance for renewable energy (RE) projects: **banking systems**, **private investors**, and **public finance instruments**. Each source faces obstacles, such as a challenged commercial lending sector, high capital requirements, and macroeconomic barriers. These limitations result in fewer RE plants being constructed, impacting the transition to renewable energy.



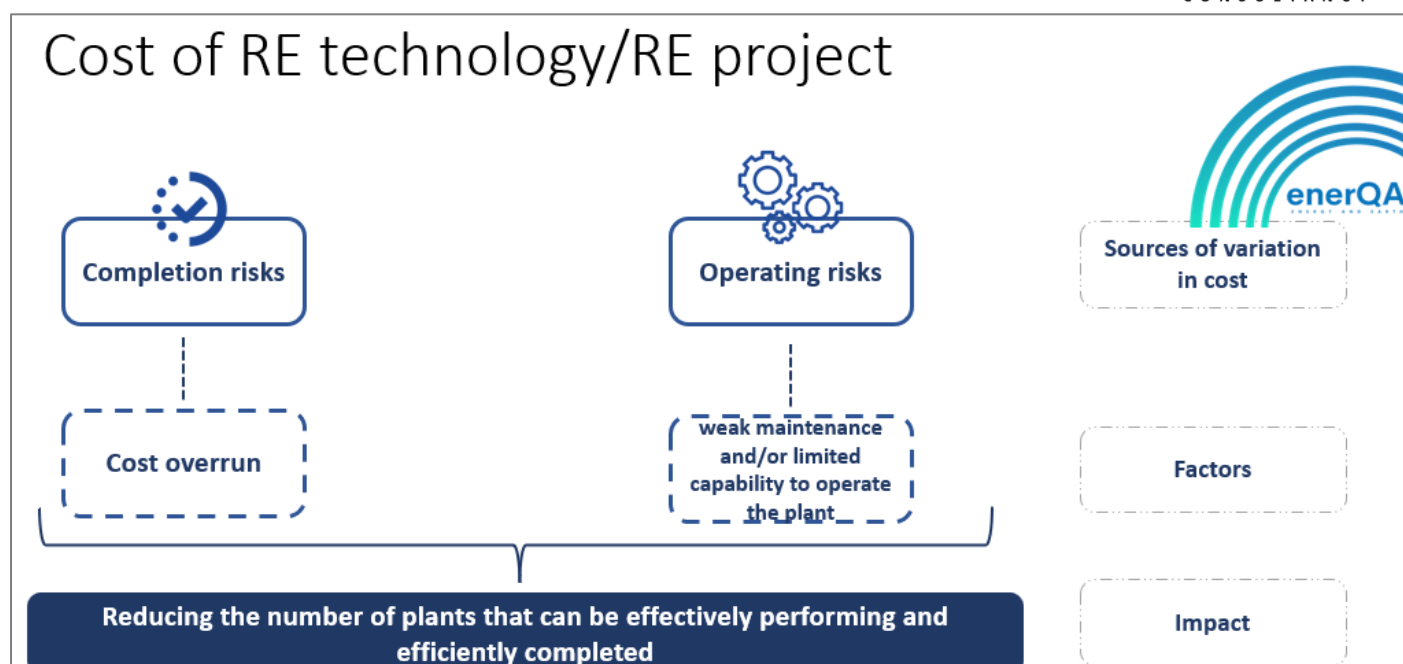
The lack of support instruments leads to higher project risks, deterring investors seeking stable returns. Developers must compensate for these risks by demanding higher expected returns, making projects less attractive to capital providers. This hurdle further limits the pool of potential investors and increases the difficulty of securing financing.



Furthermore, the absence of financial instruments like guarantees, which could significantly reduce the perceived risk of renewable energy projects. Without these risk mitigation mechanisms, investors face higher uncertainty, leading to higher required returns. This increased cost of risk further limits the pool of potential investors and reduces the number of plants that can be constructed.



The third barrier refers to the substantial upfront capital investment required for renewable energy projects, particularly pronounced in developing countries. Coupled with often limited access to affordable financing, developing countries face significant challenges in bridging the financial gap for renewable energy projects. Additionally, the complexities of project development, including securing land rights, permits, and grid connections, can further escalate costs. These factors, combined with the potential for cost overruns and higher operational expenses due to limited technical expertise.



Renewable Energy Finance: Key Considerations

Investment Trends

In recent years, there has been a notable shift in investment trends towards renewable energy. Solar and wind projects have seen substantial growth, driven by technological advancements and favourable policies. Energy storage is another area gaining traction, as it addresses the intermittency issues associated with renewables. These trends indicate a broader movement towards sustainable investments, reflecting the increasing importance of Environmental, Social, and Governance (ESG) criteria in decision-making.

Risk Assessment

While the renewable energy sector offers promising returns, it is not without risks. Investors must carefully assess factors such as regulatory changes, technological risks, and market dynamics. Due diligence is crucial, as it helps identify potential pitfalls and mitigate risks. Structured finance solutions, such as green bonds and public-private partnerships, can also play a significant role in de-risking projects and attracting capital.

Future Outlook and Recommendations

Looking ahead, we expect continued growth in the renewable energy sector. Technological innovations, such as advanced energy storage and smart grid technologies, will further enhance the viability of renewables. Additionally, increasing pressure from regulators and consumers for sustainable practices will drive more businesses to adopt green energy solutions.

Recommendations

For investors, the key to success in this sector lies in understanding the unique dynamics of renewable energy finance. It is essential to stay informed about policy developments, technological advancements, and market trends. Collaboration between public and private sectors is also critical, as it can unlock new financing mechanisms and expand



access to capital. Finally, transparency and accountability are vital in building trust with stakeholders and ensuring the long-term success of renewable energy projects.

Way Forward

As we move towards a sustainable future, the role of finance in supporting renewable energy cannot be overstated. By investing in clean energy projects, we are not only addressing climate change but also fostering economic growth and social well-being. I invite you to join me and enerQA in exploring the vast opportunities in this field. Together, we can drive meaningful change and create a more sustainable world.

References

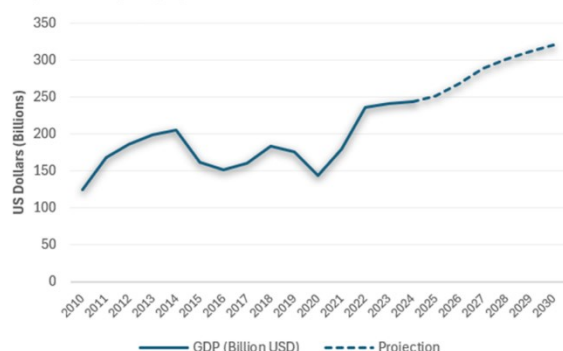
1. International Renewable Energy Agency (IRENA), 2022. *Renewable Energy Finance: Scaling Up Investment in Emerging Markets*. Available at: <https://www.irena.org/publications> [Accessed 28 November 2024].
2. United Nations Framework Convention on Climate Change (UNFCCC), 2021. *Financial Mechanisms and Climate Action: The Role of Innovative Finance*. Available at: <https://unfccc.int/topics/climate-finance/the-big-picture> [Accessed 28 November 2024].
3. BloombergNEF, 2023. *Clean Energy Investment Trends 2023*. Available at: <https://about.bnef.com/clean-energy-investment/> [Accessed 28 November 2024].
4. International Energy Agency (IEA), 2022. *World Energy Investment 2022*. Available at: <https://www.iea.org/reports/world-energy-investment-2022> [Accessed 28 November 2024].
5. World Bank, 2021. *Scaling Up Renewable Energy in Developing Countries: Overcoming Financial Barriers*. Available at: <https://www.worldbank.org/en/topic/energy/publication> [Accessed 28 November 2024].
6. OECD, 2022. *Aligning Private Investment with Climate Goals: Policy Guidance and Best Practices*. Available at: <https://www.oecd.org/climate-action/> [Accessed 28 November 2024].
7. REN21, 2023. *Global Status Report: Renewable Energy Finance and Policy*. Available at: <https://www.ren21.net/reports/global-status-report/> [Accessed 28 November 2024].
8. Gatzert, N. and Kosub, T., 2021. *Risks and Risk Management of Renewable Energy Projects: The Role of Financial Instruments for De-risking*. *Renewable Energy*, 153, pp. 1306-1316. Available at: <https://doi.org/10.1016/j.renene.2020.12.038> [Accessed 28 November 2024].
9. Global Infrastructure Facility (GIF), 2021. *Public-Private Partnerships in Renewable Energy: Enhancing Bankability and Attracting Investment*. Available at: <https://www.globalinfrastructure.org/> [Accessed 28 November 2024].
10. Lazard, 2022. *Levelized Cost of Energy Analysis 2022*. Available at: <https://www.lazard.com/perspective/levelized-cost-of-energy-and-storage-2022/> [Accessed 28 November 2024].

I-RECs - A catalyst for renewable energy investment in Qatar

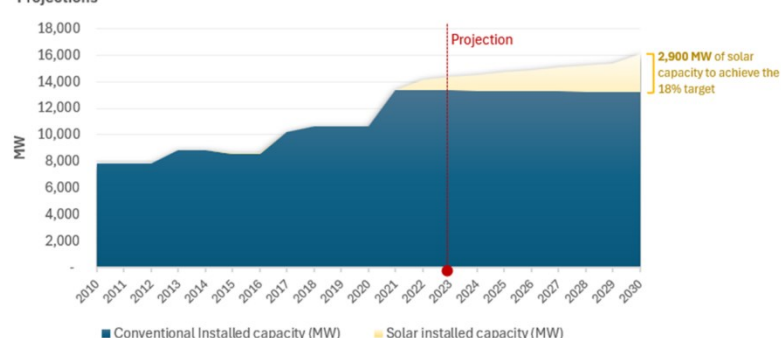
By: Amr Nasradin and Giovanni Fabbio

As a signatory to the Paris Agreement, the State of Qatar has reaffirmed its commitment to combating climate change by setting ambitious targets. The State aims to reduce greenhouse gas (GHG) emissions by 25% from 2010 levels and achieve an 18% share of renewable energy, primarily solar photovoltaics (PV), in its electricity generation mix by 2030. This is a significant challenge, especially given Qatar's status as having the highest GHG emissions per capita globally. Qatar's economy, heavily reliant on its abundant natural gas reserves, reached a record Gross Domestic Product (GDP) of USD 236 billion in 2022, with projections suggesting growth to USD 320 billion by 2030. To accommodate this economic expansion, the Qatar National Renewable Energy Strategy (QNRES) anticipates energy demand to rise to 11,500 Megawatts (MW) by 2030, up from 9,600 MW in 2021. Consequently, the supply capacity from conventional sources is planned at 13,200 MW, necessitating approximately 2,900 MW of solar PV capacity to meet the 18% renewable energy target.

GDP (Billion USD) and projection to 2030



Installed electricity generation capacity in MegaWatts (MW), Historical and Projections



Based on IRENA's Energy Profile for Qatar, the average solar PV potential is 1.7975 MWh/KWp annually. This means that the 2,900 MW of solar PV capacity could generate around 5.2 TWh of electricity each year from 2030 onwards, equivalent to 5.2 million International Renewable Energy Certificates (I-RECs).

I-RECs are Environmental Attribute Certificates (EACs) that represent the environmental attributes of electricity generated from renewable resources. They are issued by certified issuers accredited by the International Tracking Standard Foundation (ITS), ensuring transparency and credibility. Each I-REC represents 1 megawatt-hour (MWh) of renewable electricity, providing detailed information about its origin, technology, region, and vintage.

Qatar's anticipated economic growth over the next five years is expected to attract significant investments, with many international companies eager to contribute to and benefit from this growth. Aligning this economic expansion with global corporate initiatives like RE100, which aims for 100% renewable electricity procurement by 2050, could further boost investments in solar PV in Qatar. RE100 has attracted a significant number of leading businesses, including industry giants like Apple, Google, Intel, 3M, and many others.

As the global push for net-zero emissions and renewable energy adoption intensifies, the demand for I-RECs has surged. Since 2020, the annual growth rate of I-REC issuance has been an impressive 100%, underscoring the increasing importance of this tool in the transition to a sustainable energy future.

REC Market Development in Qatar

To fully leverage the benefits of Renewable Energy Certificates (RECs), Qatar can implement several strategic steps to develop a robust REC market:

- **Establish a Clear Regulatory Framework:** A transparent and well-defined regulatory framework is crucial for the successful development of a REC market. This framework should outline the rules and guidelines for REC issuance, trading, and compliance.
- **Promote REC Trading:** The government can foster REC trading by setting up a REC registry and facilitating transactions. This can be achieved through various initiatives, such as creating an online platform for REC trading and providing incentives for market participants.
- **Incentivize REC Purchases:** Government agencies and businesses can be encouraged to purchase RECs through financial incentives, tax benefits, or recognition programs. These incentives can drive demand for RECs and support the growth of renewable energy projects.
- **Educate the Public:** Raising awareness about the benefits of RECs and how they contribute to renewable energy development is essential. Public education campaigns can help increase participation in the REC market and promote a culture of sustainability.

RECs present a valuable tool for Qatar to accelerate its progress towards its renewable energy targets and ambitious climate goals. By developing a robust REC market, Qatar can stimulate investment in renewable energy, encourage consumer adoption, and effectively track progress towards its sustainability objectives. This strategic approach will not only help Qatar meet its environmental commitments but also position the country as a leader in the global transition to a sustainable energy future.

Data Sources:

1. *Qatar National Renewable Energy Strategy (QNRES)*, Qatar General Electricity and Water Corporation (KAHRAMAA).
2. <https://tradingeconomics.com/qatar/gdp#:~:text=GDP%20in%20Qatar%20averaged%2055.93,statistics%2C%20economic%20calendar%20and%20news>.
3. <https://www.statista.com/statistics/379978/gross-domestic-product-gdp-in-qatar/>
4. International Renewable Energy Agency (IRENA): *Energy Profile, Qatar* (https://www.irena.org/-/media/Files/IRENA/Agency/Statistics/Statistical_Profiles/Middle%20East/Qatar_Middle%20East_RE_SP.pdf).

The Role of Artificial Intelligence in Environmental Sustainability

By: Yasen Hatim

Abstract

Artificial Intelligence (AI) has emerged as a transformative technology to address critical global environmental challenges, including climate change, resource depletion, and biodiversity loss. This paper explores AI's diverse applications in environmental sustainability, such as predictive climate modeling, resource optimization, and biodiversity conservation. Case studies, including Google's DeepMind and Tesla's renewable energy solutions, highlight AI's practical contributions. Challenges, including data biases, energy consumption, and ethical concerns, are discussed, emphasizing the importance of interdisciplinary collaboration and responsible implementation. By aligning AI with global sustainability goals, this paper underscores its potential to advance technological and environmental harmony.

1. Introduction

The convergence of Artificial Intelligence (AI) and environmental sustainability has garnered significant global attention. The planet faces unprecedented challenges, including rising temperatures, biodiversity loss, water scarcity, and pollution. Advances in machine learning and deep learning have enabled AI to process vast datasets, make accurate predictions, and optimize complex systems, offering innovative solutions to these problems. This paper examines the transformative potential of AI, presenting real-world case studies, challenges, and future directions in its application for environmental sustainability.

2. Examples of AI Applications in Environmental Sustainability

2.1 Climate Change Mitigation

AI plays a critical role in monitoring and reducing greenhouse gas emissions:

- **Predictive Models:** AI algorithms analyze climate data to predict weather patterns and extreme climate events.
- **Carbon Footprint Reduction:** Smart grids and energy systems, powered by AI, improve energy efficiency, reducing carbon footprints.
- **Case Study:**
 - *Google's DeepMind* used AI to reduce energy consumption in its data centers by **40%**. [3]
 - Tesla's AI-powered energy solutions, such as Powerwall and grid balancing systems, enhance renewable energy adoption. [4]

CO₂ Emission Reduction Potential with AI-Driven Technologies

AI technologies could contribute to reducing global CO₂ emissions by up to 10%, showcasing their transformative potential in the fight against climate change. [5]

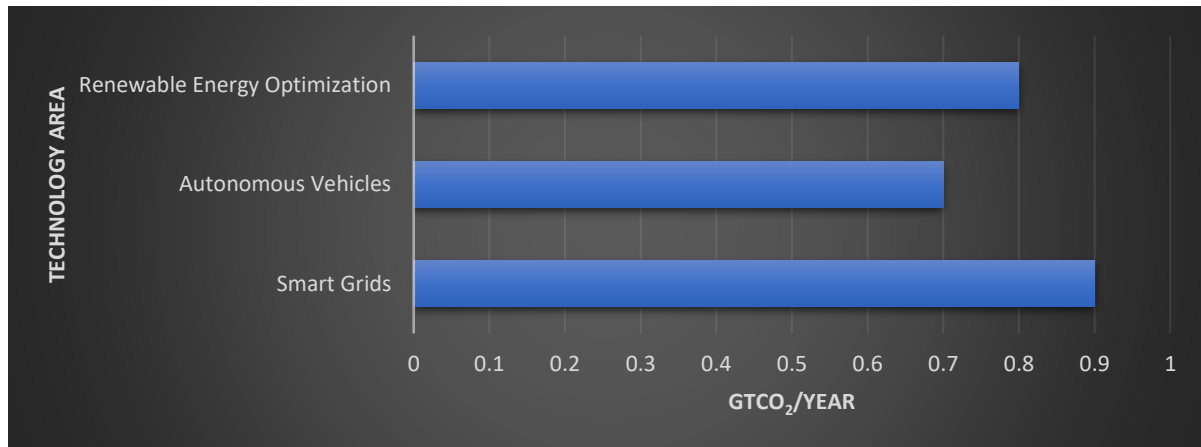


Figure 9: CO₂ Emissions Reduction With AI-Driven Technologies (Data Source: McKinsey, 2022)

2.2 Resource Optimization and Waste Management

AI systems optimize the use of natural resources and improve waste management strategies:

- **Water Management:** AI monitors and predicts water usage patterns, reducing wastage. For instance, IBM's AI tools saved **700 billion gallons** of water globally. [6]
- **Smart Recycling:** Image recognition and robotics powered by AI enhance sorting accuracy in recycling facilities.
- **Case Study:** The city of Seoul uses AI-powered waste sorting systems, achieving over 95% recycling efficiency for plastics. [7]

Key Achievements:

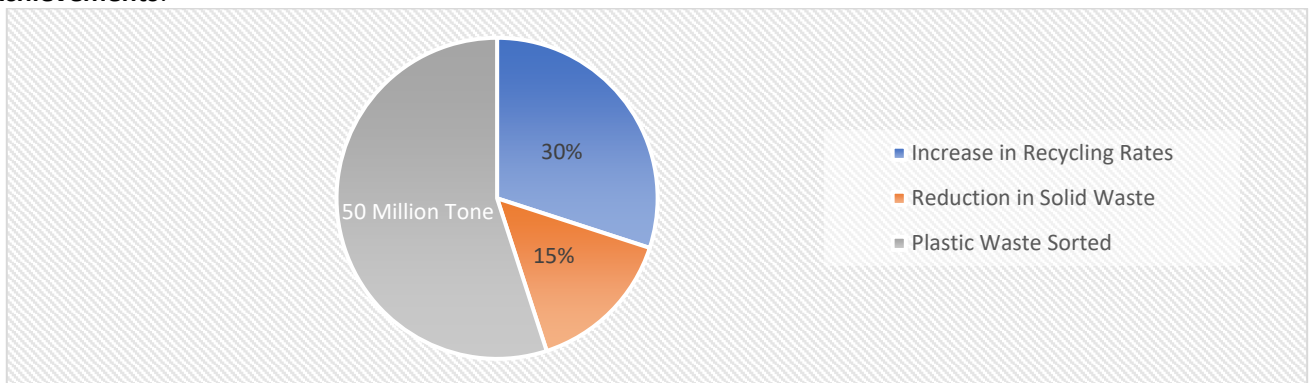


Figure 10: AI Impact on Global Waste Management

2.3 Biodiversity and Ecosystem Monitoring

Conservation efforts are greatly enhanced by AI:

- **Wildlife Monitoring:** AI-driven drones and camera traps track animal populations and prevent poaching.
- **Habitat Analysis:** Machine learning identifies and predicts threats to critical habitats.
- **Case Study:** Microsoft's AI for Earth initiative has mapped **10%** of Earth's land cover, aiding biodiversity conservation. [8]
- AI tools like TensorFlow and PyTorch are increasingly being adapted for conservation purposes, offering scalable and efficient solutions.

3. Challenges and Ethical Concerns

Despite its potential, AI faces challenges in environmental applications:

1. **Data Limitations:** Incomplete or biased datasets can lead to inaccurate predictions and flawed solutions.
2. **Energy Consumption:** Training AI models requires significant computational resources, paradoxically increasing carbon emissions. Green AI initiatives focus on addressing this paradox by designing energy-efficient AI models.

3. **Ethical Concerns:** Balancing technological advancements with local ecological and social impacts is a persistent challenge. For instance, deploying AI systems in developing regions must consider equitable access and cultural sensitivity.

4. The Future of AI in Sustainability

AI's capabilities are expected to grow with advancements in quantum computing and IoT integration. Future applications include precision agriculture, enhanced carbon capture methods, and AI-driven urban planning to create smart, sustainable cities.

According to Allied Market Research (2023), the green AI market is projected to grow at a CAGR of 25% over the next decade, underscoring its importance in achieving sustainability goals. [9]

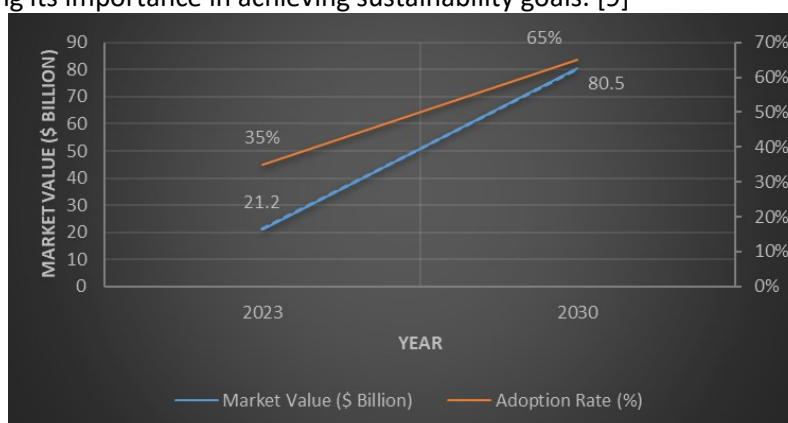


Figure 11: AI Market Growth Over Time (Data Source: Allied Market Research, 2023)

5. Conclusion

AI is a powerful tool for driving environmental sustainability, offering innovative solutions to some of the world's most pressing ecological challenges. However, collaboration across industries, policymakers, and environmental groups is essential to harness AI's potential responsibly. By integrating AI into global efforts aligned with the UN's Sustainable Development Goals (SDGs), we can create a harmonious balance between technology and the environment. Addressing ethical concerns and reducing AI's environmental footprint are key steps in ensuring its role as a force for good.

6. References

- [1] "Scoping Out: Tracking Nature Across The Supply Chain," CDP, 2023.
- [2] E. Thomson, "Droughts are creating new supply chain problems. This is what you need to know," World Economic Forum, 25 October 2023. [Online]. Available: <https://www.weforum.org/agenda/2023/10/drought-trade-rivers-supply-chain/>. [Accessed 25 September 2024].
- [3] Google, "Google," [Online]. Available: (DeepMind AI reduces energy used for cooling Google data centers by 40%).
- [4] Tesla, "Tesla," [Online]. Available: Best Tesla Powerwall in 2024: Here.
- [5] BCG, [Online]. Available: <https://www.bcg.com/publications/2023/how-ai-can-speedup-climate-action?utm>.

- [6] IBM, [Online]. Available: <https://newsroom.ibm.com/2023-03-14-IBM-launches-RFP-to-help-accelerate-global-water-management-solutions-for-vulnerable-populations?utm>.
- [7] Seoul. [Online]. Available: <https://english.seoul.go.kr/seoul-to-increase-recycling-rate-to-79-by-2026/?utm>.
- [8] Microsoft. [Online]. Available: <https://news.microsoft.com/apac/2021/10/11/microsoft-ai-for-earth-named-as-one-of-100-outstanding-biodiversity-positive-practices-and-actions-around-the-world/#:~:text=Launched%20in%202017%2C%20Microsoft%E2%80%99s%20%E2%80%9CAI%20for%20Earth%E2%80%9D%20is>.
- [9] A. M. Research. [Online]. Available: Artificial Intelligence (AI) Market Size, Share, Trends – 2033).
- [1] I. o. S. Studies, 14 October 2024. [Online]. Available:
 0] <https://instituteofsustainabilitystudies.com/insights/lexicon/what-are-scope-4-emissions/>.
- [1] W. B. C. f. S. D. (WBCSD), Guidance on Avoided Emissions: Helping business drive innovations and
 1] scale solutions toward Net Zero, Geneva, Amsterdam, New Delhi, London, New York, Singapore: World Business Council for Sustainable Development (WBCSD), 2023.
- [1] W. R. INSTITUTE, "WORLD RESOURCED INSTITUTE," 05 November 2013. [Online]. Available:
 2] <https://www.wri.org/insights/do-we-need-standard-calculate-avoided-emissions>.
- [1] M. D. G. G. C. L. M. A. Lauren Shwisberg, "How to Build Clean Energy Portfolios A Practical Guide to
 3] Next-Generation Procurement Practices," Rocky Mountain Institute, Basalt, CO, 81621 USA, 2020.
- [1] P. A. [Online]. Available: <https://plana.earth/glossary/scope-4-emissions>.
 4]
- [1] P. R. Attack, "Panama Canal Authority blames queues on drought and draft cut," Ship-Technology.com ,
 5] 11 August 2023. [Online]. Available: <https://www.ship-technology.com/news/panama-canal-authority-blames-queues-drought-draft-cut/>. [Accessed 25 September 2024].
- [1] . J. Verschuur, E. E. Koks and J. W. Hall, "Systemic risks from climate-related disruptions at ports,"
 6] *Nature Climate Change*, vol. 13, no. 8, p. 804–806, 2023.
- [1] L. Woetzel, D. Pinner, . H. Samandari, H. Engel, M. Krishnan, C. Kampel and J. Graabak, "Could climate
 7] become the weak link in your supply chain?," McKinsey Global Institute, 6 August 2020. [Online]. Available: <https://www.mckinsey.com/capabilities/sustainability/our-insights/could-climate-become-the-weak-link-in-your-supply-chain>. [Accessed 25 September 2024].

- [1] "Climate-Related Supply Chain Risk: Are You Prepared? - FTI Strategic Communications," FTI Consulting, 15 December 2022. [Online]. Available: <https://fticommunications.com/climate-related-supply-chain-risk-are-you-prepared/>. [Accessed 25 September 2024].
- [1] G. G. P. (. Protocol). [Online]. Available: <https://ghgprotocol.org/standards-guidance>.
- 9]

Environment and Society

1. The Hidden Costs of Your Burger and Pizza: What's Really at Stake?
2. Practical Tips for Reducing Food Waste at Home: A Step Towards Sustainability
3. Sustainable Tourism
4. The Hidden Link Between Cigarette Smoking and Climate Change
5. Empowering Communities Through Social Sustainability: A Call to Action
6. The Origins of Urban Greening
7. Environmental Impacts of Mercury Use in Artisanal Gold Mining in Africa (Sudan-Case)



The Hidden Costs of Your Burger and Pizza: What's Really at Stake?

By: Dr. Islam M. Awad

Introduction

When we buy a burger or pizza, we assume the price reflects all associated costs. However, many costs, especially environmental ones, remain hidden from consumers. Food production impacts the planet through carbon emissions, water consumption, and pollution—costs not reflected in prices. These are known as externalities, which contribute to market failures and environmental challenges like the tragedy of the commons (Pigou, 1932). This article delves into the concept of externalities and their impact on food choices.

Understanding Externalities

An externality is a cost or benefit affecting those not directly involved in a transaction (Coase, 1960). For instance, the price of a burger includes production and profit costs, but not the pollution from cattle farming, the carbon emissions from transportation, or water use for feed crops. These are **negative externalities**, which harm society without being accounted for in pricing. On the other hand, **positive externalities**, such as cleaner air from tree planting, provide societal benefits without cost. Ignoring externalities results in distorted pricing, misinforming consumer choices (Hardin, 1968).

Internalizing Externalities: The Solution

Internalizing externalities ensures goods' prices reflect true environmental and social costs. Governments achieve this through mechanisms like taxes and subsidies (Pigou, 1932). For example, a carbon tax increases the cost of high-emission products, while subsidies encourage sustainable practices like eco-friendly farming. If the environmental cost of a burger were included, its price might rise from \$5 to \$7, aligning with its true cost and encouraging sustainable choices (Nordhaus, 1994).

Conventional Burger



Price

\$5

Total Cost (with Externalities)

\$7

Environmental Impact:

7 Pounds

of CO2 emissions, deforestation
and water overuse

Climate Friendly Burger



Price

\$7

Total Cost (with Externalities)

\$7

Environmental Impact:

1 Pound

of CO2 emissions, deforestation
and water overuse

The Tragedy of the Commons

The tragedy of the commons describes how shared resources are depleted when individuals act in their immediate self-interest (Hardin, 1968). In food production, common resources like air, water, and land are exploited, leading to pollution, deforestation, and biodiversity loss. Overgrazing in shared fields, for instance, exemplifies this principle, demonstrating the need for regulation to prevent overexploitation.

Market Failure and Its Importance

Market failure occurs when prices fail to reflect true costs, leading to inefficient resource allocation. In conventional food production, unaccounted environmental costs keep prices low, promoting harmful consumption patterns. This exacerbates broader issues like climate change and soil degradation (Stiglitz, 2000).

Private Goods vs. Common Goods

While private goods like burgers or pizzas are individually consumed and paid for, the resources used to produce them—air, water, and fertile land—are common goods (Ostrom, 1990). These shared resources often face overuse and depletion due to a lack of accountability. Distinguishing between private and common goods is key to addressing market failures and promoting sustainable resource management.

Why Internalizing Externalities Matters

Correcting prices to reflect environmental costs addresses market failures. Products with high negative externalities become costlier, encouraging consumers to choose sustainable alternatives. This shift fosters responsible consumption, protects shared resources, and reduces environmental harm (Nordhaus, 1994).

Conclusion

Every choice between conventional and sustainable options impacts the environment. By internalizing hidden costs, market failures can be corrected, avoiding the tragedy of the commons and fostering a sustainable future. Spending a little more today on eco-friendly alternatives, like plant-based burgers or pizzas, contributes to a healthier planet tomorrow.

References

1. Coase, R. H. (1960). *The Problem of Social Cost*. *The Journal of Law and Economics*, 3, 1-44.
2. Hardin, G. (1968). *The Tragedy of the Commons*. *Science*, 162(3859), 1243–1248.
3. Nordhaus, W. D. (1994). *Managing the Global Commons: The Economics of Climate Change*. MIT Press.
4. Ostrom, E. (1990). *Governing the Commons: The Evolution of Institutions for Collective Action*. Cambridge University Press.
5. Pigou, A. C. (1932). *The Economics of Welfare*. Macmillan.
6. Stiglitz, J. E. (2000). *Economics of the Public Sector*. W.W. Norton & Company.

Practical Tips for Reducing Food Waste at Home: A Step Towards Sustainability

By: Yousif Ibrahim Hamza



Reducing domestic food waste is not just a trend—it's a necessity. Every year, humanity generates between 2.1 billion and 2.3 billion tonnes of municipal solid waste, a significant portion of which is food waste. Alarmingly, some 2.7 billion people lack access to waste collection, with 2 billion of them living in rural areas. Waste pollution is a growing threat to human well-being, economic prosperity, and the environment, contributing to the triple planetary crisis of climate change, nature and biodiversity loss, and pollution. Without urgent action, annual municipal solid waste generation is projected to reach 3.8 billion tones by 2050 (source: United Nations Environment Programme, International Day of Zero Waste 2024).

Food waste is a critical global issue, with approximately one-third of all food produced—1.3 billion tonnes—wasted annually. The drivers of food waste vary significantly between high- and low-income countries. In wealthier nations, food waste occurs predominantly at the consumer and retail levels due to over-purchasing, aesthetic standards, and poor understanding of food labeling. Conversely, in lower-income countries, inadequate infrastructure, storage facilities, and market access lead to substantial post-harvest losses. Social behaviors, such as cultural attitudes and consumer habits, also play a key role in influencing waste patterns.

Reducing food waste is essential for mitigating its environmental, social, and economic impacts. Food waste contributes significantly to climate change, with decaying waste in landfills releasing potent greenhouse gases like methane. It also accelerates biodiversity loss and pollution, as resources like land and water used to produce wasted food often result in habitat destruction and contamination. Alarmingly, food waste is a major component of the 2.1–2.3 billion tonnes of municipal solid waste generated annually, and this figure is projected to reach 3.8 billion tonnes by 2050 if urgent action is not taken.

Addressing food waste requires a combination of behavioral changes, policy interventions, and technological advancements. Individuals can reduce waste through better meal planning, understanding food labeling, and proper storage. Governments play a critical role in implementing food redistribution policies, enforcing sustainable packaging laws, and investing in infrastructure to minimize losses in supply chains. Additionally, waste-to-energy technologies and improved storage facilities can help repurpose unavoidable waste and reduce post-harvest losses in low-income regions.

The global response to food waste must be unified and proactive, leveraging platforms like the International Day of Zero Waste to promote awareness and collaboration. Reducing food waste is not just a matter of resource efficiency but also a crucial step toward addressing the triple planetary crisis of climate change, biodiversity loss, and pollution. Through combined efforts, food waste reduction can support sustainable development goals, improve food security, and mitigate its harmful environmental consequences.

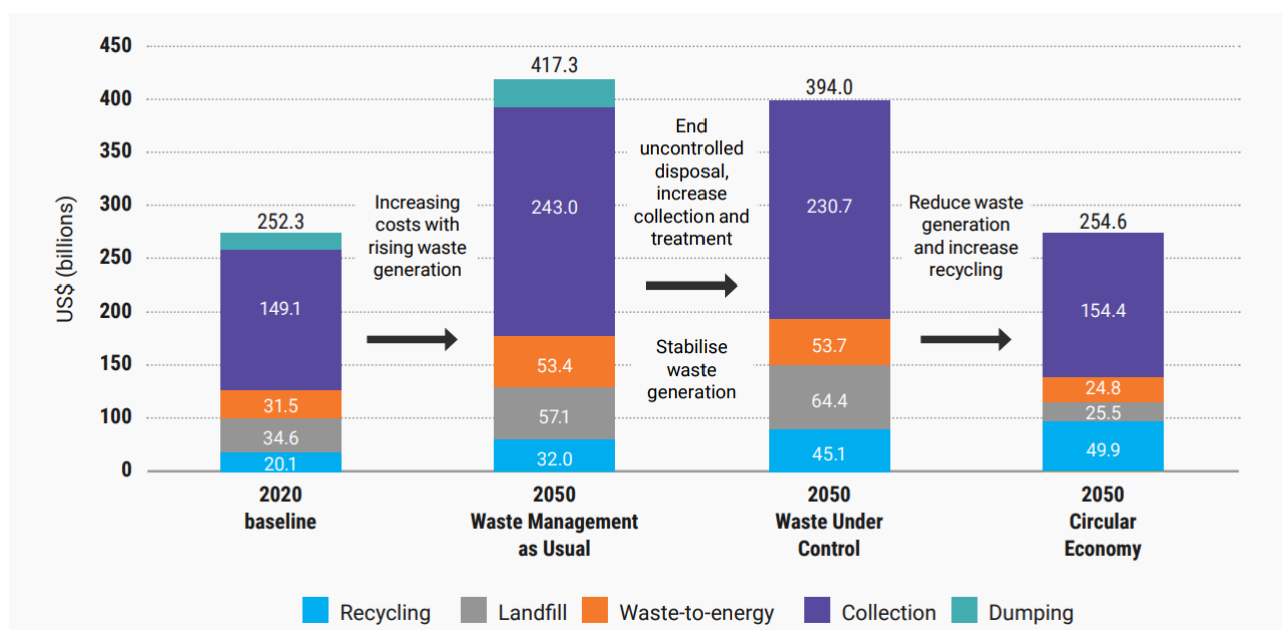


Figure-1: Global Direct Costs of Municipal Solid Waste Management in 2050 under Three Scenarios (Global Waste Management Outlook 2024)

The above figure illustrates the global direct costs of municipal solid waste management are expected to increase significantly by 2050 if waste generation continues to rise unchecked. By taking action now to reduce food waste at home, we can contribute to a circular economy and mitigate these costs.

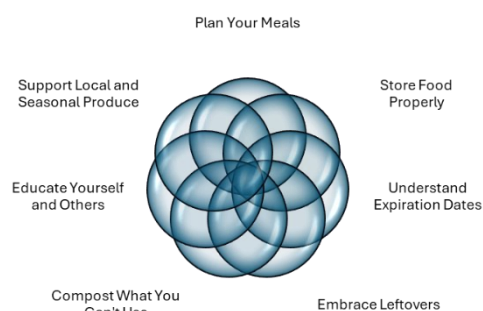
As we strive for a more sustainable future, here are some practical tips to help you develop good habits for food consumption and minimize waste:

1. Plan Your Meals

- **Why It Matters:** Planning your meals for the week ensures that you buy only what you need, reducing the risk of food going unused.
- **How to Do It:** Create a shopping list based on your meal plan and stick to it. Consider portion sizes to avoid over-purchasing.

2. Store Food Properly

- **Why It Matters:** Proper storage can extend the life of your food, reducing the chances of spoilage.



- **How to Do It:** Learn the best ways to store fruits, vegetables, and leftovers. For example, keep potatoes in a cool, dark place and store herbs in a glass of water in the fridge.
- 3. **Understand Expiration Dates**
 - **Why It Matters:** Misinterpreting "best before" and "use by" dates can lead to unnecessary waste.
 - **How to Do It:** "Best before" dates indicate quality, not safety. Use your senses to determine if food is still good past this date. For "use by" dates, adhere to them strictly, especially for perishable items.
- 4. **Embrace Leftovers**
 - **Why It Matters:** Leftovers are a great way to reduce food waste and save time on cooking.
 - **How to Do It:** Get creative with your leftovers. Turn yesterday's dinner into today's lunch or freeze portions for a quick meal later.
- 5. **Compost What You Can't Use**
 - **Why It Matters:** Composting turns food scraps into nutrient-rich soil, reducing the amount of waste sent to landfills.
 - **How to Do It:** Set up a compost bin in your backyard or use a kitchen compost container for smaller spaces. Compost fruit and vegetable peels, coffee grounds, eggshells, and more.
- 6. **Educate Yourself and Others**
 - **Why It Matters:** Awareness is the first step towards change. The more you know, the better you can prevent waste.
 - **How to Do It:** Stay informed about sustainable practices and share what you learn with family and friends. Encourage others to adopt similar habits.
- 7. **Support Local and Seasonal Produce**
 - **Why It Matters:** Buying local and seasonal produce reduces the carbon footprint associated with transportation and storage.
 - **How to Do It:** Visit farmers' markets or subscribe to a local produce delivery service. Plan meals around what's in season to enjoy fresher, tastier food.

Reducing food waste is a collective responsibility that starts at home. By implementing these practical tips, you can make a significant impact on your household's sustainability. Let's commit to making mindful food choices that benefit not only our planet but also our communities.

Without urgent action, the environmental and social costs of waste generation will continue to escalate, threatening our efforts to combat climate change and protect biodiversity.

References

1. FAO (2011). *Global Food Losses and Food Waste – Extent, Causes, and Prevention*. Food and Agriculture Organization.
2. Parfitt, J., Barthel, M., & Macnaughton, S. (2010). Food waste within food supply chains: Quantification and potential for change to 2050. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 365(1554), 3065–3081.
3. Aschemann-Witzel, J., De Hooge, I., Amani, P., Bech-Larsen, T., & Oostindjer, M. (2015). Consumer-Related Food Waste: Causes and Potential for Action. *Sustainability*, 7(6), 6457-6477.
4. IPCC (2019). *Climate Change and Land: An IPCC Special Report on Climate Change, Desertification, Land Degradation, Sustainable Land Management, Food Security, and Greenhouse Gas Fluxes in Terrestrial Ecosystems*.
5. UNEP (2024). *International Day of Zero Waste*. United Nations Environment Programme.

Sustainable Tourism

By: Yousif Ibrahim Hamza



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Introduction

Tourism is a powerful driver of economic growth and cultural exchange, offering people the chance to explore new places and immerse themselves in different cultures. However, unregulated tourism can harm the environment, disrupt local communities, and undermine the destinations it seeks to celebrate. Sustainable tourism addresses these challenges by promoting practices that minimize negative impacts while maximizing benefits for all stakeholders involved (UNWTO, 2023). This guide explores the principles of sustainable tourism and its importance in ensuring that destinations are preserved for future generations.

What is Sustainable Tourism?

Sustainable tourism involves responsible travel practices that prioritize environmental conservation, cultural preservation, and economic stability (UNEP & WTO, 2005). It ensures that tourism activities contribute positively to the destinations visited, fostering long-term benefits for local communities and ecosystems. Essentially, sustainable tourism is about making conscious choices to protect and respect the places and people encountered during travel.

The Three Pillars of Sustainable Tourism

Sustainable tourism is guided by three pillars:

1. **Environmental Sustainability:** Protecting ecosystems, conserving resources, and reducing pollution caused by tourism activities.
2. **Social Sustainability:** Preserving cultural heritage, respecting local traditions, and fostering community engagement.
3. **Economic Sustainability:** Ensuring that tourism generates equitable economic benefits for local communities, promoting long-term prosperity (UNWTO, 2017).

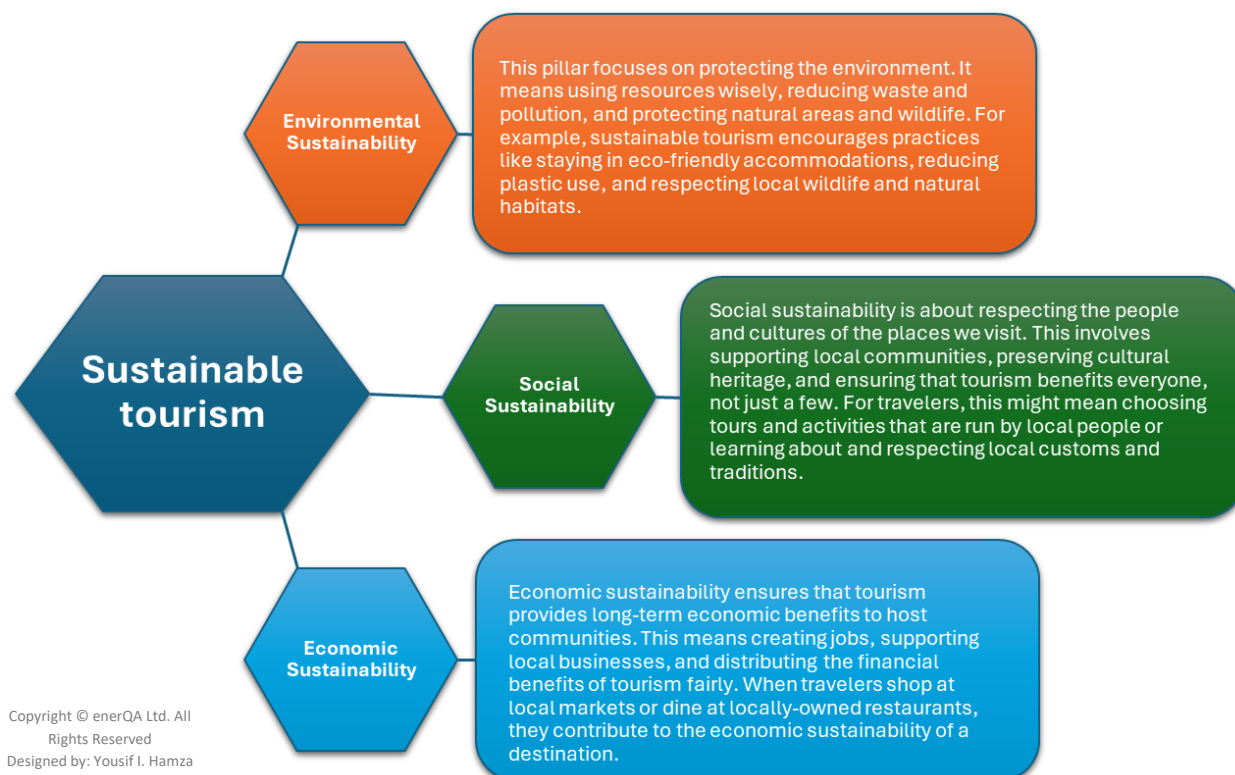
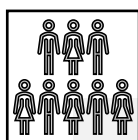


Figure-1: Pillar of sustainable tourism

The Triple Bottom Line: People, Planet, and Profit

Sustainable tourism aligns with the “triple bottom line” framework, which emphasizes the interconnectedness of environmental health (planet), social equity (people), and economic growth (profit). By balancing these three factors, sustainable tourism aims to create a holistic approach to development that benefits both current and future generations (Elkington, 1998).



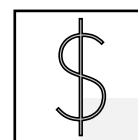
People

- Tourism should benefit local communities. This includes creating jobs, supporting cultural preservation, and ensuring that local people have a say in how tourism develops in their area



Planet

- Tourism should minimize its environmental impact. This means reducing carbon emissions, protecting natural habitats, and encouraging travelers and businesses to adopt eco-friendly practices



Profit

- Tourism should be economically viable, providing income for local businesses and governments. However, it's important that these profits are used to support sustainable development and are not generated at the expense of the environment or local communities

Why is Sustainable Tourism Important?

1. Protecting Natural Resources

Popular tourist destinations often house fragile ecosystems vulnerable to over-tourism. Sustainable tourism mitigates these risks by adopting measures such as limiting visitor numbers, promoting eco-friendly practices, and supporting conservation efforts (Buckley, 2012).

2. Preserving Cultures and Heritage

Tourism can lead to the erosion of cultural identity and traditions. Sustainable tourism fosters cultural respect and helps preserve unique heritages, ensuring that local communities retain their traditions while benefiting from tourism (Richards, 2018).

3. Supporting Local Economies

Properly managed tourism creates jobs, supports local businesses, and improves community infrastructure. Sustainable tourism ensures these economic benefits are distributed fairly, reducing inequalities and fostering long-term development (UNWTO, 2017).

How Can You Contribute to Sustainable Tourism?

- **As a Traveler:** Choose eco-friendly accommodations, reduce plastic use, and support local businesses.
- **As a Tour Operator:** Design environmentally sensitive tours, engage with local communities, and educate clients about sustainable practices.
- **As a Community Member:** Advocate for sustainability, participate in tourism planning, and educate visitors on preserving cultural and natural heritage.

Summary

Sustainable tourism transcends traditional travel by emphasizing responsibility and mindfulness in every journey. By protecting natural resources, preserving cultural heritage, and supporting local economies, sustainable tourism ensures that the benefits of travel are shared equitably while minimizing its adverse effects. As travelers, tour operators, or community members, we all have a role to play in fostering tourism that respects and sustains the world's diverse landscapes and cultures.

References

1. Buckley, R. (2012). *Sustainable Tourism: Research and Reality*. *Annals of Tourism Research*, 39(2), 528–546.
2. Elkington, J. (1998). *Cannibals with Forks: The Triple Bottom Line of 21st Century Business*. New Society Publishers.
3. Richards, G. (2018). *Cultural Tourism: A Review of Recent Research and Trends*. *Journal of Tourism and Cultural Change*, 16(2), 95–107.
4. UNEP & WTO (2005). *Making Tourism More Sustainable – A Guide for Policy Makers*. United Nations Environment Programme and World Tourism Organization.
5. UNWTO (2017). *Tourism and the Sustainable Development Goals – Journey to 2030*. United Nations World Tourism Organization.
6. UNWTO (2023). *What is Sustainable Tourism?* United Nations World Tourism Organization. Available at: <https://www.unwto.org>.

The Hidden Link Between Cigarette Smoking and Climate Change

By: Yasen Hatim

When we think about climate change, the usual suspects that come to mind are industries, automobiles, and deforestation. However, there is an often-overlooked contributor to environmental degradation and climate change: cigarette smoking. While smoking is primarily recognized for its health impacts, its environmental footprint is significant and alarming.

Cigarette Production and Deforestation

The journey of a cigarette begins with tobacco farming, a process that heavily relies on deforestation. Vast areas of forests are cleared to make way for tobacco plantations. According to the World Health Organization (WHO), tobacco farming is responsible for the loss of 200,000 hectares of forests each year. These forests, which act as carbon sinks, are vital in absorbing CO₂ from the atmosphere. When they are destroyed, the carbon stored in trees is released back into the atmosphere, contributing to global warming.

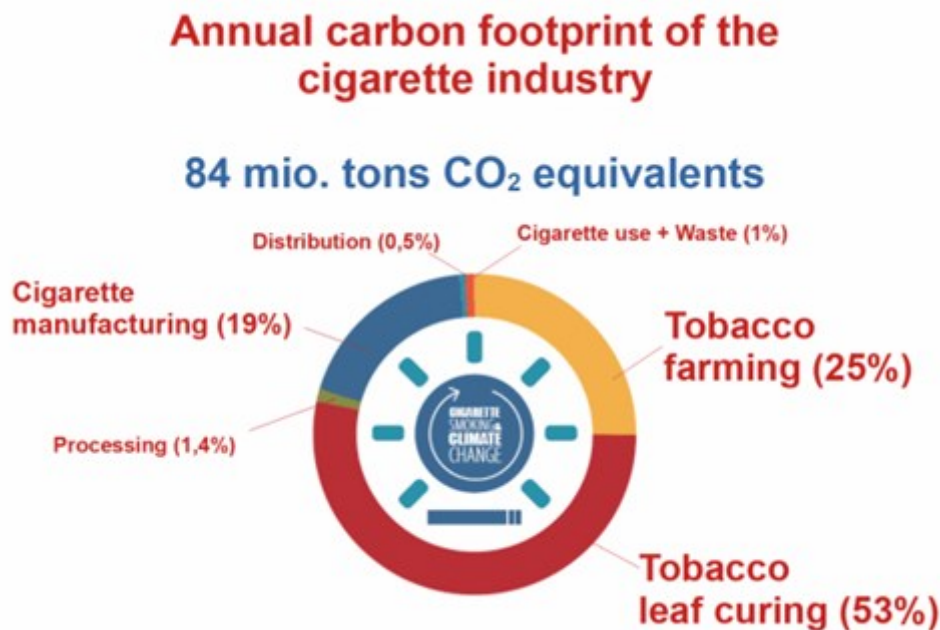


Figure-1: Annual carbon footprint of the cigarette industry. Source: <https://unfairtobacco.org/en/tobacco-harms-environment-climate>

Carbon Emissions from Tobacco Manufacturing

The manufacturing process of cigarettes is another significant source of carbon emissions. It's estimated that the production of just 300 cigarettes (around 15 packs) results in the emission of about 1 kilogram of CO₂. With nearly 6 trillion cigarettes produced annually worldwide, the carbon footprint is staggering.

Moreover, the energy consumption involved in curing tobacco, manufacturing cigarettes, and packaging is considerable. Factories rely on fossil fuels for energy, further increasing the carbon emissions associated with cigarette production.

Environmental Impact of Cigarette Waste

Cigarette butts are the most discarded waste item worldwide, with about 4.5 trillion butts littered each year. These butts are not just unsightly; they are also toxic. They leach harmful chemicals, such as nicotine, heavy metals, and carcinogens, into the soil and waterways, harming wildlife and ecosystems.

But beyond the direct pollution, the decomposition of cigarette butt's releases greenhouse gases. Though each individual butt contributes a small amount, the cumulative impact of trillions of discarded butts is significant.

Cigarettes contain over 4,000 chemicals, including...

Benzo(a)pyrene

Arsenic

Acetone

Lead

Formaldehyde

Toluene

Butane

Cadmium

Ammonia

Benzene



Source: <https://greenteampower.org/>

The figure showcases a selection of the over 4,000 chemicals found in cigarettes. Source:

Fires Caused by Smoking

Another link between smoking and climate change is the role of cigarettes in starting wildfires. Carelessly discarded cigarette butts are a common cause of forest fires, which can result in the loss of vast areas of forest and significant carbon emissions. Wildfires not only release carbon stored in trees but also reduce the planet's capacity to absorb CO₂, creating a vicious cycle.

The Bigger Picture

While smoking is often framed as a personal choice with personal health consequences, it is also an environmental issue with global implications. The entire lifecycle of a cigarette — from production to disposal — contributes to climate change. Reducing smoking rates worldwide could have a meaningful impact not only on public health but also on the health of our planet.

This intersection between personal behavior and global climate change highlights the importance of addressing all contributors to environmental degradation, no matter how seemingly small. By raising awareness of the environmental impacts of smoking, we can encourage more people to quit, benefiting both their health and the planet.

References

1. World Health Organization (WHO), 2017. Tobacco and its Environmental Impact: An Overview. Available at: <https://www.who.int/publications/tobacco-and-environment> [Accessed 28 November 2024].
2. Unfairtobacco.org, 2024. Tobacco Harms the Environment and Climate. Available at: <https://unfairtobacco.org/en/tobacco-harms-environment-climate> [Accessed 28 November 2024].
3. Novotny, T. E. and Zhao, F., 1999. Consumption and Production Waste: Another Externality of Tobacco Use. *Tobacco Control*, [online] 8(1), pp. 75-80. Available at: <https://tobaccocontrol.bmj.com/content/8/1/75> [Accessed 28 November 2024].
4. Truth Initiative, 2021. The Environmental Impact of Cigarette Smoking. Available at: <https://truthinitiative.org/research-resources/environment/environmental-impact-cigarette-smoking> [Accessed 28 November 2024].
5. Novotny, T. E. et al., 2021. The Environmental and Health Impacts of Cigarette Waste. *International Journal of Environmental Research and Public Health*, 18(6), p. 3243. Available at: <https://doi.org/10.3390/ijerph18063243> [Accessed 28 November 2024].
6. Campaign for Tobacco-Free Kids, 2020. Tobacco's Hidden Environmental Costs. Available at: <https://www.tobaccofreekids.org/assets/factsheets/0385.pdf> [Accessed 28 November 2024].
7. Heaton, C. G., 2022. Cigarette Smoking and its Role in Forest Fires. *Environmental Hazards Review*, 14(3), pp. 110-115. Available at: <https://doi.org/10.1080/1223456> [Accessed 28 November 2024].
8. Proctor, R. N., 2011. *Golden Holocaust: Origins of the Cigarette Catastrophe and the Case for Abolition*. University of California Press. [Covers tobacco's extensive environmental and societal harms].

Empowering Communities Through Social Sustainability: A Call to Action

By: Reem Elmalik

Are you ready to create a world where everyone thrives? where communities are inclusive and supportive, and where future generations inherit a healthy planet. This vision of a sustainable future is achievable through social sustainability.

Social sustainability is more than just a buzzword; it is a fundamental principle that guides our efforts to create a just, fair, and resilient society. It encompasses everything from ensuring access to basic needs like food and shelter to promoting social justice and equality.

In this article, we will explore the key dimensions of social sustainability, examine its importance in building thriving communities, and discuss practical steps you can take to contribute to a more sustainable world.

"We need to change the way we think about the Earth. It's not just a resource; it's our home." - David Attenborough

Understanding Social Sustainability

Imagine a world where everyone has access to food, shelter, and a good education. A world where everyone is treated fairly, regardless of their background. This is the idea behind social sustainability. It is about creating a society where everyone can thrive and live a happy, fulfilling life.

Social sustainability refers to the ability of a society to function in a way that promotes well-being, equality, and justice for all individuals over the long term. It involves creating systems, policies, and institutions that support the needs of current and future generations, while fostering inclusion, fairness, and cohesion within communities.

Caring about social sustainability in a country with a diverse range of cultures and ethnicities is vital because it ensures that all groups have equitable access to opportunities, resources, and a high quality of life, fostering a more inclusive and resilient society.

"The future belongs to those who believe in the beauty of their dreams." - Eleanor Roosevelt

The Role of the Sustainable Development Goals (SDGs)



United Nations Sustainable Development Goals (SDGs)

The United Nations has set out 17 goals for a sustainable future. These goals, called Sustainable Development Goals (SDGs), cover everything from ending poverty to protecting our planet. Many of these goals are about social sustainability.

- SDG 1 Ending Poverty: No one should go to bed hungry or live without a roof over their head. We need to work together to create a world where everyone has the necessities of life.
- SDG 3 Good Health and Well-being: Everyone deserves to be healthy and happy. This means having access to quality healthcare and a clean environment.
- SDG 4 Quality Education: Education is the key to unlocking your potential. Everyone should have the chance to learn and grow.
- SDG 5 Gender Equality: Boys and girls should have the same opportunities. We need to end discrimination and violence against women and girls.
- SDG 8 Decent Work and Economic Growth: Everyone deserves a fair job that pays a living wage. We need to create a strong economy that benefits everyone.
- SDG 10 Reduced Inequalities: No one should be left behind. We need to work to reduce the gap between the rich and the poor.
- SDG 11 Sustainable Cities and Communities: Our cities should be safe, clean, and affordable places to live. We need to build communities that are sustainable for future generations.
- SDG 16 Peace, Justice, and Strong Institutions: A peaceful and just society is essential for social sustainability. We need to work to prevent conflict and ensure that everyone has access to justice.

Knowing about the SDGs fosters global awareness and understanding of the interconnected challenges facing our planet. It encourages individuals to recognize their role in addressing these challenges and contributing to a more sustainable future.

Awareness of the SDGs can inspire individuals to advocate for sustainable solutions and take action in their communities. It facilitates collaboration and partnerships between governments, businesses, and civil society organizations to achieve common goals

Cultural Diversity & Migration

Migration is a complex and multifaceted phenomenon driven by a variety of reasons, both push and pull factors. Push factors are those that compel people to leave their home countries, such as economic hardship, political instability, conflict, natural disasters, or lack of opportunities. Pull factors, on the other hand, are those that attract people to a new country, such as better economic prospects, better quality of life, educational opportunities, or family reunification.

The decision to migrate is often a difficult one, filled with both hope and uncertainty. People who migrate may experience a range of emotions, including excitement, fear, anxiety, and homesickness. They may also feel a sense of loss and detachment from their home culture.

Despite the challenges, migration can also be a source of significant opportunity and personal growth. Migrants often bring new skills, perspectives, and cultural richness to their new homes. They can contribute to economic development, cultural diversity, and social innovation.

A Path to Social Sustainability

Social sustainability is essential in countries with diverse populations. It ensures that all individuals, regardless of their background, have equal access to opportunities, resources, and a high quality of life. This fosters social inclusion, promotes cultural understanding, and reduces inequality.

Locals have a responsibility to their country to create a welcoming and inclusive environment for newcomers. They can share their culture with migrants while also learning about and respecting their cultures. This exchange of cultures can enrich the lives of both locals and migrants and create a more vibrant and dynamic society.

As more people move to countries they have never lived in before, it's important to preserve and carry forward their own cultural identity. However, it's equally essential to learn about and respect the culture of the host country. Our values are what allow us to connect with others, share experiences, and embrace dissimilar cultures, ultimately enriching our quality of life.

By fostering social sustainability and promoting cultural exchange, we can create a more just, fair, and harmonious world for all.

Why Social Sustainability Matters

A More Just and Equitable World. Social sustainability is about creating a world where everyone has the same opportunities. It means ending discrimination and ensuring that everyone is treated with respect.

A Stronger Community. When everyone feels like they belong, our communities become stronger. Social sustainability helps to build a sense of belonging and community.

A Better Future. By working together to achieve social sustainability, we can create a better future for ourselves and for generations to come.

Pioneer in Social Sustainability

A remarkable example of social sustainability in action can be found in the work of Reena Naser, an entrepreneur in Qatar passionate about building inclusive and vibrant communities. Through her innovative startup, Language Agora, she has created a platform that fosters community engagement and introduces newcomers to the diverse cultural tapestry of Qatar.

Reena organises events that showcase the unique cultures of different countries, providing a space for people to connect, share their traditions, and learn from one another. These gatherings not only offer a chance for individuals from the same cultural backgrounds to meet and socialise but also create opportunities for cross-cultural exchange and understanding.

Recognizing the importance of communication in building bridges between communities, Reena emphasizes the sharing of languages. By encouraging participants to speak in their native tongues, she facilitates meaningful interactions and fosters a sense of belonging for all.

Through her efforts, Reena is not only contributing to the social fabric of Qatar but also inspiring others to embrace diversity and work towards a more sustainable and fair future.

Taking Action for a Sustainable Future

Get Involved. There are many ways to get involved in social sustainability. You can volunteer with local organisations, support sustainable businesses, or advocate for policies that promote social justice.

Educate Yourself. Learn more about social sustainability and the issues that affect your community. The more you know, the more you can do.

Inspire Others. Share your knowledge and passion with others. Together, we can create a more sustainable and equitable world.

Remember. Social sustainability is about creating a world where everyone can thrive. It is a challenge, but it is a challenge worth fighting for.

Conclusion Building a Sustainable Future Together

Social sustainability is the cornerstone of a thriving and equitable society. By fostering inclusivity, promoting cultural exchange, and addressing social challenges, we can create a world where everyone has the opportunity to flourish.

By embracing diversity, supporting sustainable initiatives, and advocating for social justice, we can collectively build a better future for ourselves and generations to come.

Let us commit to working together towards a more sustainable and fairer world. Together, we can create a legacy of hope, compassion, and resilience for all.

References

1. United Nations. (2020, September 25). The Sustainable Development Goals. <https://sdqs.un.org/goals>
2. European Commission. (n.d.). The EU and the United Nations Sustainable Development Goals. https://international-partnerships.ec.europa.eu/policies/sustainable-development-goals_en
3. Clean the World. (n.d.). What is Sustainability? <https://cleantheworld.org/about-us/about-us-overview/>
4. Agyeman, B. N. A., & Turnock, D. (2018). Sustainable development goals: Their interactions with governance, institutions, and social cohesion. *Sustainability* (Switzerland), 10(7), 3839. https://www.researchgate.net/publication/337760181_Governance_for_achieving_the_Sustainable_Development_Goals_How_important_are_participation_policy_coherence_reflexivity_adaptation_and_democratic_institutions
5. Organisation for Security and Co-operation in Europe. (2021). Local Authorities' Migrant Integration Guide: Fostering Social Cohesion and Inclusive Communities. <https://www.osce.org/migration>
6. International Organization for Migration. (2018). Integration and Social Cohesion: A Thematic Paper. https://www.iom.int/sites/default/files/our_work/ODG/GCM/IOM-Thematic-Paper-Integration-and-Social-Cohesion.pdf

The Origins of Urban Greening

By: Reem Almalik

Introduction

Urban greening, as a modern concept, is often traced back to the development of public parks in the 19th century. Notable early examples include New York City's Central Park, designed by Frederick Law Olmsted and Calvert Vaux in the 1850s, which aimed to provide green space for city dwellers amidst the rapid industrialization of the time. In Europe, similar ideas emerged in cities like London and Paris, where city planners sought to balance urban development with green spaces. Today, urban greening is recognized as a global priority, with cities worldwide working to integrate nature into their infrastructure.

Sustainable Urban Greening: Leveraging Existing Infrastructure

Integrating urban greening with existing infrastructure allows cities to enhance their sustainability without requiring extensive land or resources. Key strategies include:



Green Roofs and Walls: Installing vegetation on the roofs and walls of buildings can significantly reduce heat in cities, improve air quality, and add aesthetic appeal. Cities like Toronto and Singapore have implemented green roof policies to encourage such installations.

The photo is from Germany



Repurposing Abandoned Spaces: Old industrial sites and vacant lots can be converted into parks or community gardens. For instance, New York's High Line, a former railway line, is now a renowned elevated greenway that combines recreation with biodiversity.

The photo is for the High Line Park- New York



Street Tree Planting and Green Corridors:

Planting trees along roads and creating green corridors encourages biodiversity and reduces pollution. Melbourne, Australia, has planted thousands of trees and established "urban forests" that contribute to improved air quality and reduced heat.

The photo is for Melbourne city

Successful Examples of Urban Greening



Singapore: Often considered a "City in a Garden" Singapore is a leading example of urban greening. Initiatives like Gardens by the Bay, vertical gardens on high-rise buildings, and extensive tree-planting programs have transformed Singapore into a green oasis that balances urbanisation with nature.



by Robert GLOD (Bob) / 07.05.2018



Copenhagen, Denmark: Copenhagen aims to become the world's first carbon-neutral city by 2025, partly through urban greening. The city has invested in green rooftops, parks, and cycling infrastructure to reduce pollution and promote sustainable living.



Bogotá, Colombia: Bogotá has successfully created a network of green corridors, known as “Alamedas.” These corridors help regulate temperature, reduce flooding, and provide safe pathways for pedestrians and cyclists.



Advantages

Urban greening offers a multitude of benefits for both the environment and human well-being. **Environmentally**, it plays a crucial role in mitigating climate change by absorbing carbon dioxide and reducing air pollution. For instance, a study by the Environmental Protection Agency (EPA) found that trees can remove up to 60 pounds of air pollutants per year. Additionally, green spaces help to combat the urban heat island effect, cooling cities and reducing energy consumption.

From a **health perspective**, access to green spaces has been linked to numerous positive outcomes. Research by the World Health Organisation (WHO) suggests that spending time in nature can reduce stress, improve mental health, and lower the risk of chronic diseases. Furthermore, green spaces encourage physical activity, as they provide opportunities for walking, jogging, and cycling.

Economically, urban greening can contribute to a city's vitality. Studies have shown that green spaces can increase property values and attract businesses and tourists. For example, the High Line in New York City, a former elevated railway transformed into a public park, has revitalised the surrounding neighbourhoods and generated significant economic benefits.

Finally, green spaces foster **social cohesion** by providing places for people to gather, relax, and interact. Community gardens, parks, and urban forests can create a sense of belonging and strengthen social bonds.

Disadvantages

While urban greening has numerous advantages, it also presents certain challenges. **Initial costs** can be substantial, particularly for large-scale projects that require significant infrastructure investments. Moreover, **ongoing maintenance** is essential to ensure the health and vitality of green spaces. This includes tasks such as watering, pruning, and removing litter, which can be resource intensive.

In some cases, urban greening projects may inadvertently contribute to **gentrification**. As property values rise in areas with new green spaces, long-time residents may be displaced due to increased housing costs. This is a complex issue that requires careful planning and consideration of social equity.

In regions with limited water resources, **water usage** for urban greening can be a concern. However, innovative water management strategies, such as rainwater harvesting and drought-tolerant plant selection, can help to mitigate this challenge.

Implementing Urban Greening in the GCC

The Gulf Cooperation Council (GCC) countries, characterised by their arid climates and rapid urbanisation, face unique challenges in implementing urban greening. However, innovative strategies are appearing to address these challenges and create sustainable, liveable cities.

One such strategy is xeriscaping, which involves landscaping with native plants that require minimal water. These drought-tolerant plants, such as desert grasses and succulents, can thrive in the harsh GCC climate. By incorporating xeriscaping into urban design, GCC cities can reduce water consumption and create visually appealing, low-maintenance landscapes.

Another promising approach is the integration of green roofs and walls into urban architecture. Cities like Dubai and Abu Dhabi have already embraced this concept, using drought-resistant vegetation to cover building rooftops and facades. Green roofs and walls offer many benefits, including reducing energy consumption, mitigating the urban heat island effect, and improving air quality.

In addition to these strategies, GCC countries are investing in large-scale urban cooling projects to combat extreme heat. Qatar's national cooling strategy, for example, aims to create a more comfortable outdoor environment through a combination of technological solutions and urban planning initiatives. Saudi Arabia's ambitious NEOM project also incorporates green infrastructure, such as green corridors and urban forests, to mitigate the impact of heat and create a sustainable, liveable city.

By adopting these innovative approaches, GCC countries can transform their urban landscapes, improve the quality of life for their residents, and contribute to a more sustainable future.

Expo 2023 Doha: A Green Oasis in the Desert

Expo 2023 Doha, themed "Green Desert, Better Environment," offers a compelling example of urban greening in an extreme climate. This international exposition displays innovative solutions for creating sustainable and livable cities in arid regions.

By focusing on sustainable technologies, such as hydroponics and solar power, the expo how to expand green spaces in desert environments. This approach not only enhances the aesthetic appeal of urban areas but also contributes to improved air quality, reduced heat island effects, and increased biodiversity.

Moreover, Expo 2023 Doha emphasises community engagement, encouraging public participation in green initiatives. By fostering a sense of ownership and responsibility, the expo aims to inspire long-lasting positive impacts on urban sustainability.

As a global platform, Expo 2023 Doha serves as a beacon of hope, showing that even in the harshest conditions, it is possible to create thriving green cities. By sharing knowledge and best practices, the expo contributes to the global effort to address climate change and build a more sustainable future.

References

Urban Greening Success Stories

- National Parks Board, Singapore. (n.d.). Garden City. [Website: National Parks Board Singapore]
- Comune di Milano. (n.d.). Urban Forestry Plan. [Website: Comune di Milano]
- Friends of the High Line. (n.d.). High Line. [Website: Friends of the High Line]
- City of Vancouver. (n.d.). Greenest City 2020 Action Plan. [Website: City of Vancouver]

Urban Greening in the GCC

- Expo 2023 Doha. (2023). Green Desert, Better Environment. Retrieved from <https://www.dohaexpo2023.gov.qa/>
- QNA. (2023, September 29). Expo 2023 Doha: Qatar's cutting-edge initiative to elevate greening, battle climate change - report. Retrieved from <https://www.qna.org.qa/en/News-Area/News/2023-09/29/0009-expo-2023-doha-qatars-cutting-edge-initiative-to-elevate-greening,-battle-climate-change--report>

Urban Greening in Saudi Arabia

- Saudi Green Initiative
- NEOM: <https://www.neom.com/en-us>
- The Line: <https://www.neom.com/en-us/regions/theline>

Environmental Impacts of Mercury Use in Artisanal Gold Mining in Africa (Sudan- case): The Amplifying Effect of Torrential Rains and Floods

By: Yousif Ibrahim Hamza



Artisanal and small-scale gold mining (ASGM) is a crucial livelihood activity for many in Sudan, contributing significantly to the local economy (Ibrahim, 2015). However, the prevalent use of mercury in this sector introduces severe environmental and health hazards. The risks posed by mercury have been exacerbated by recent heavy rains and floods, particularly in mining markets where both milling and burning of mercury-amalgamated gold occur (Carsten, 2016). Figure-1 shows the Geographical distribution of artisanal mining activities

This article examines the environmental impacts of mercury in ASGM in Sudan, emphasizing the intensification of these effects due to extreme weather events.

ASGM is a significant economic activity across Africa, providing livelihoods for millions. However, the sector's extensive use of mercury poses severe environmental and health risks. Globally, ASGM is the largest source of anthropogenic mercury emissions, releasing over 2,000 tonnes annually.

In Africa, ASGM is prevalent in countries such as Ghana, Mali, Burkina Faso, Tanzania, and Zimbabwe. The

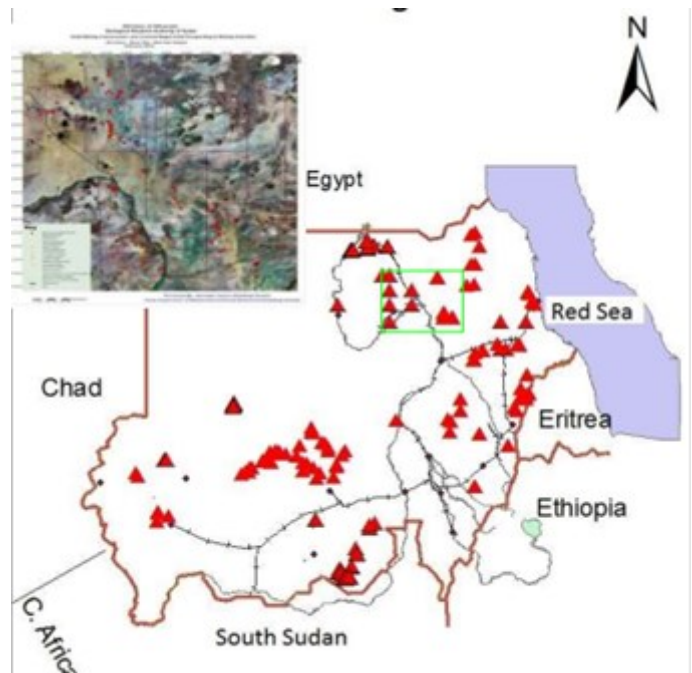


Figure 1: spotted areas for artisanal gold mining in Sudan (Source: Ibrahim, Artisanal Mining in Sudan - Opportunities, Challenges and Impacts, 2015)

mercury-to-gold ratio used in these operations varies, often ranging from 1.32 to 3.23 grams of mercury per gram of gold produced.

This indicates that for every gram of gold, up to 3.23 grams of mercury may be released into the environment. The lack of regulation and awareness exacerbates mercury pollution, leading to significant environmental degradation and health issues among mining communities.

Sudan has emerged as one of Africa's leading gold producers, with ASGM playing a crucial role in its economy. However, the sector's reliance on mercury has led to substantial environmental and health concerns. Figure 2 shows the Estimated mercury consumption for ASGM and estimated number of ASGM miners by country t/y. Carsten, 2016)

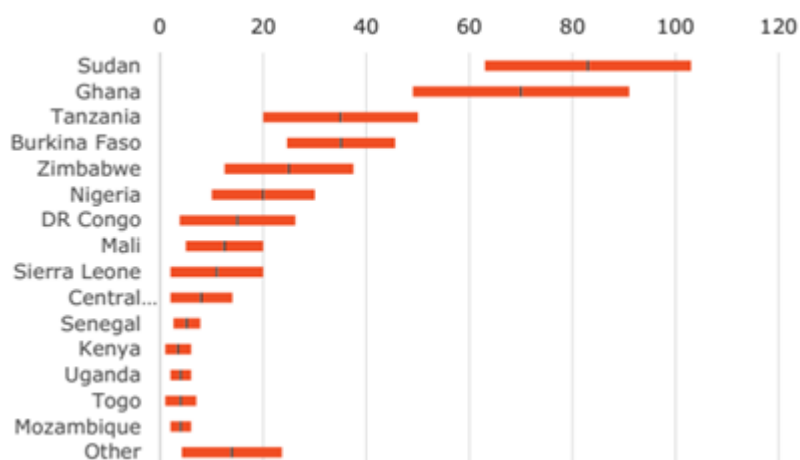
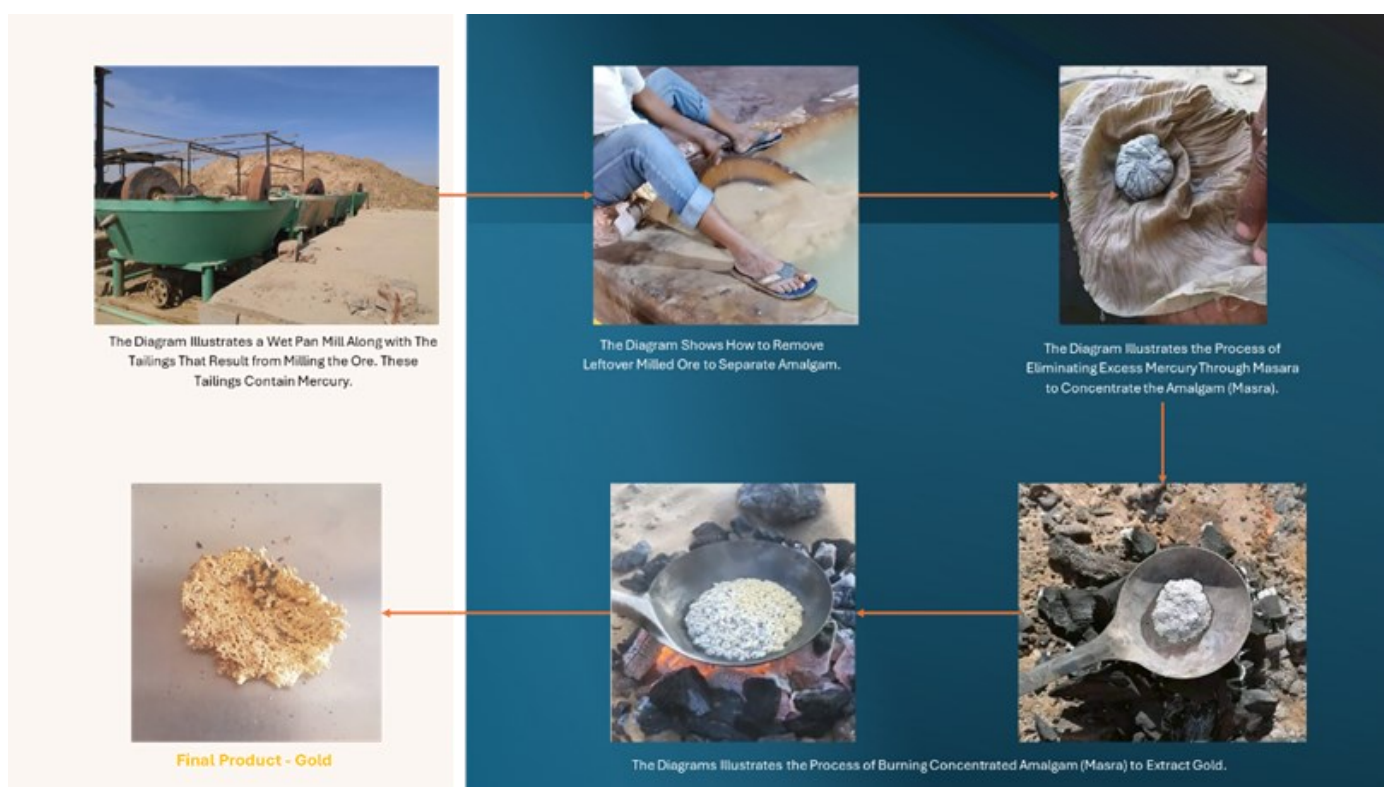


Figure 2: Estimated mercury consumption for ASGM and estimated number of ASGM miners by country t/y. Carsten, 2016)



The Process of Gold Extraction Using Mercury

1. Crushing and Grinding
 - Gold-bearing ore is crushed and ground into fine particles using rudimentary tools or mechanical mills.
2. Amalgamation
 - Mercury is mixed with the ground ore to form an amalgam. The mercury binds to the gold particles, separating them from other materials.
 - The mercury-to-gold ratio can range from 1.3:1 to 3.5:1, with substantial mercury losses into the environment.
3. Squeezing and Filtration

- The gold-mercury amalgam is collected and squeezed through a cloth to remove excess mercury. This step often exposes workers to direct skin contact with mercury.

4. Burning the Amalgam

- The amalgam is heated, often using open flames or rudimentary furnaces, to vaporize the mercury, leaving behind purified gold.
- Mercury vapors released during this process are highly toxic and contaminate the air, soil, and water.

A field study by the author was conducted in one of the artisanal mining hub in Northern Sudan. The study included investigating a lot of samples from different miners and integrating statistical analysis to obtain range of results. The following figure shows the Mercury to Gold Ratio loss during crushing operation.

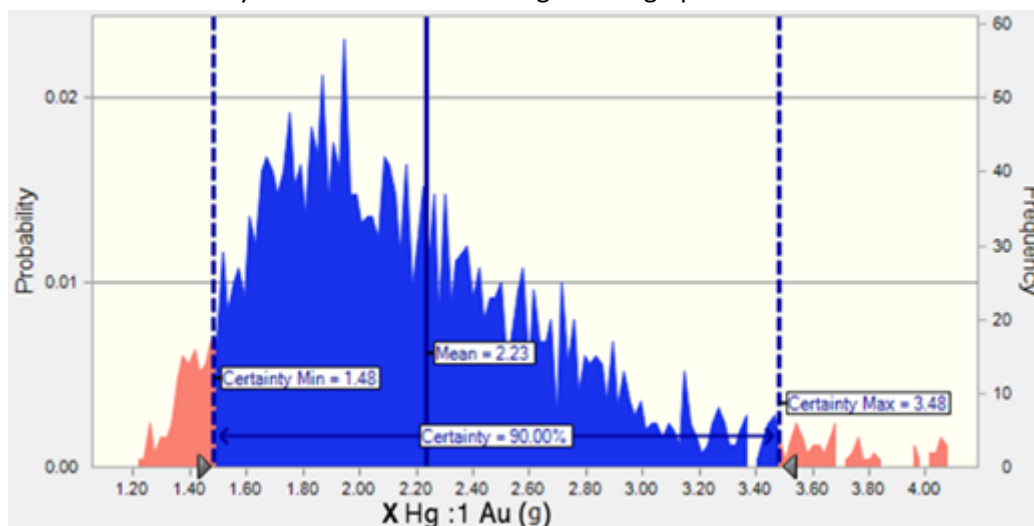


Figure3: Mercury-to-Gold Ratio (X Hg:1 Au) loss during the Milling Operation

While figure- 4 shows Mercury-to-Gold Ratio (X Hg:1 Au) Loss During the Burning of Concentrated Amalgam (Marsa) Operation

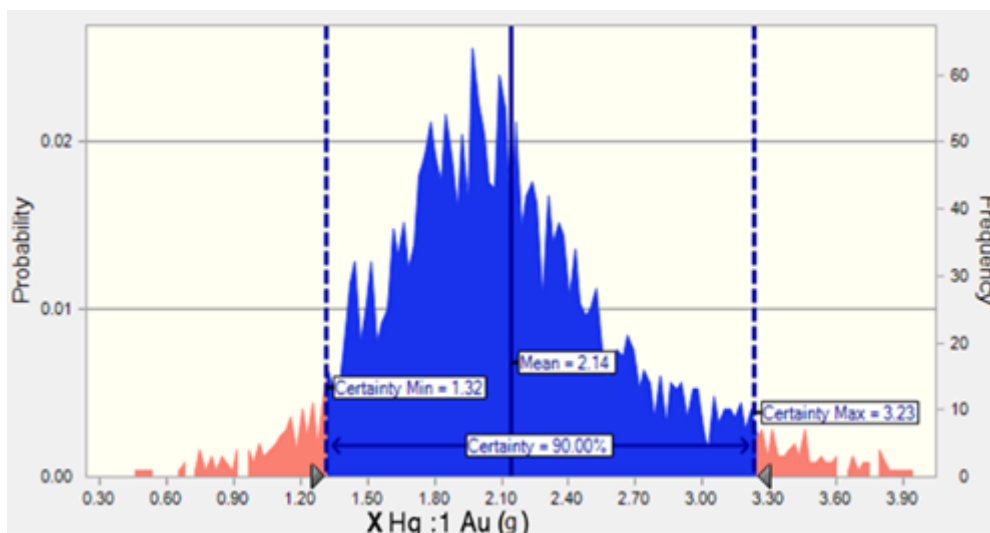


Figure 4: Mercury-to-Gold Ratio (X Hg:1 Au) Loss During the Burning of Concentrated Amalgam Operation

The extensive use of mercury in ASGM across Africa, and particularly in Sudan, presents critical environmental and public health challenges.

1. Soil Contamination

Mercury contamination of soil is a critical issue in ASGM. During gold extraction, mercury can spill during milling or be deposited atmospherically during amalgam combustion, leading to widespread soil pollution (Ibrahim, 2015). The mercury-to-gold ratio observed in milling operations often ranges from 1.48 to 3.48 grams, indicating significant

mercury loss into the environment (Carsten, 2016). The recent floods have further worsened this contamination by dispersing mercury-laden soil into agricultural and residential areas, rendering land infertile and threatening microbial biodiversity (Ibrahim, 2015).

2. Water Pollution

Nearby water bodies are particularly vulnerable to mercury pollution from ASGM. Mercury spills and runoff contaminate surface and groundwater, transforming into methylmercury—a highly toxic compound that bioaccumulates in aquatic organisms (Carsten, 2016). Flooding exacerbates this by washing mercury from mining sites into rivers, streams, and groundwater systems, significantly increasing the contamination zone and associated health risks (Ibrahim, 2015).

3. Air Quality Degradation

The burning of mercury-amalgamated gold releases toxic mercury vapors into the atmosphere, posing health risks through inhalation (Carsten, 2016). The mercury-to-gold ratio during this phase ranges from 1.32 to 3.23 grams, with an average of 2.14 grams of mercury released per gram of gold burned (Carsten, 2016). Flood-related turbulence spreads mercury vapors further afield, increasing the risk of atmospheric deposition in previously unaffected regions.

4. Impact on Vegetation

Mercury contamination adversely impacts vegetation, stunting growth and reducing agricultural productivity (Ibrahim, 2015). Plants absorb mercury through their roots, entering the food chain and posing risks to herbivores and predators alike. The spread of mercury-contaminated soil and water by floodwaters threatens crop safety and ecosystem stability, with severe implications for local food security (Carsten, 2016).

5. Threats to Wildlife

Mercury bioaccumulates in aquatic life, disrupting ecosystems and causing reproductive and behavioral issues in wildlife (Carsten, 2016). Floodwaters exacerbate these risks by concentrating mercury in aquatic habitats and creating contamination hotspots as waters recede. This has cascading effects on biodiversity, threatening species survival and ecosystem balance.

6. Human Health Hazards

The health impacts of mercury exposure in ASGM are severe, particularly for miners and nearby communities. Chronic mercury poisoning can cause neurological and cognitive impairments, respiratory problems, and developmental issues, especially in vulnerable populations such as children and pregnant women (Ibrahim, 2015). Flooding increases exposure risks by spreading mercury into residential areas, contaminating food and water sources.

The following photos show the recent (August, 2024) heavy rains and floods in the areas of artisanal mining Northern Sudan.



Conclusion

The environmental and health consequences of mercury use in Sudan's ASGM sector are amplified by extreme weather events, such as heavy rains and floods. Addressing these challenges necessitates a combination of reducing mercury use, adopting safer alternatives, and implementing robust environmental and health protection measures. Balancing the economic benefits of ASGM with the urgent need for sustainability is essential for mitigating its long-term impacts in a changing climate (Carsten, 2016; Ibrahim, 2015).

References

1. Carsten, S. (2016). *Mercury Use and Impacts in Artisanal and Small-Scale Gold Mining: Global Assessment and Implications*. United Nations Environment Programme.
2. Ibrahim, M. E. (2015). *Artisanal Mining in Sudan - Opportunities, Challenges and Impacts*. Sudan University of Science and Technology.
3. UNEP Global Mercury Partnership (n.d.). *Artisanal and Small-Scale Gold Mining (ASGM)*.
4. MDPI (2023). *Mercury Contamination in Artisanal Mining Areas in Sudan*.
5. Sudan Transparency Initiative (2023). *Mercury Use and Environmental Challenges in Sudan's Artisanal Mining Sector*.
6. Enerqa Consultancy (n.d.). *Artisanal Gold Mining and Environmental Impacts of Mercury Use in Sudan*.

Frameworks and Methodologies

1. **Understanding the DPSIR, DPCER, DPSWR, and DPSEIR Frameworks in Analyzing the Interactions Between Human Activities and the Environment**
2. **DPSIR Framework for Sustainable Aquaculture in the Red Sea Region (KSA): A Climate Adaptation and Mitigation Perspective**
3. **The Imperative for ESG Readiness Tools: Unlocking Benefits for Companies, Institutions, and Society**
4. **Scope 4 Emissions: The Concept of Avoided Emissions**



Understanding the DPSIR, DPCER, DPSWR, and DPSEER Frameworks in analyzing the interactions between human activities and the environment

By: Yasen Hatim

Environmental challenges such as climate change, pollution, and biodiversity loss require comprehensive frameworks for analysis and management. The **International Environmental Assessment and Strategy Centre (IEASC)**, which is a research institute associated with the **United Nations Environment Programme (UNEP)** develop framework called DPSIR in the late 1990s and has since become a widely used tool in environmental assessment and policy analysis. The DPSIR framework provides a structured approach to understanding the causal relationship between human activities and environmental impacts. This article discusses the original DPSIR framework and its derivatives—DPCER, DPSWR, and DPSEER—highlighting their unique features and applications.

Overview of the Frameworks

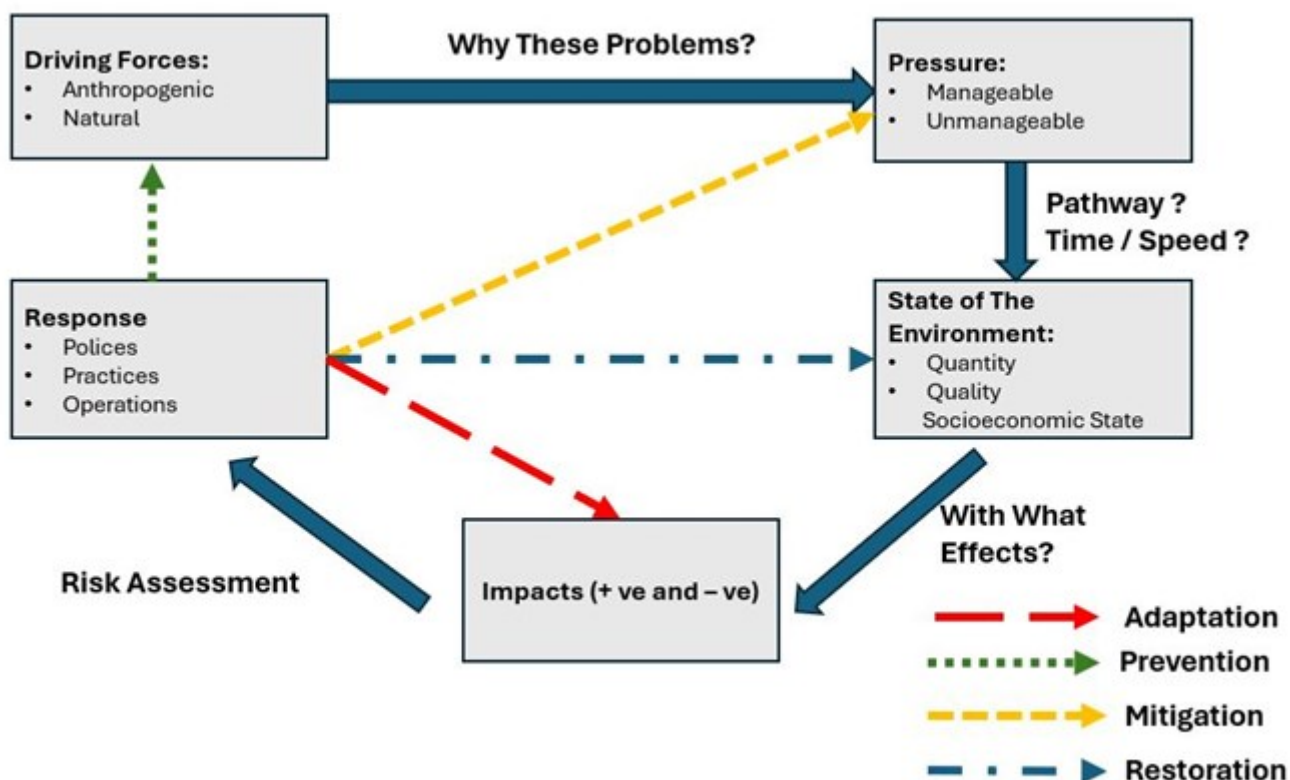


Figure-1: the relationship between the DPSIR elements

1. DPSIR Framework

The DPSIR framework consists of five components:

- **Drivers:** Socio-economic factors that lead to environmental changes.
- **Pressures:** Human activities that exert stress on the environment.
- **State:** The condition of the environment resulting from pressures.

- **Impact:** The effects of changes in the state on ecosystems and human health.
- **Response:** Actions taken to mitigate or adapt to environmental issues.

2. DPCER Framework

The DPCER framework focuses specifically on chemical pollution:

- **Drivers:** Socio-economic activities leading to chemical emissions.
- **Pressures:** Release of chemicals into the environment.
- **Chemical State:** Concentrations of chemicals in various media.
- **Ecological State:** Effects of chemical pollution on ecosystems.
- **Response:** Measures to control or mitigate chemical pollution.

3. DPSWR Framework

The DPSWR framework emphasizes human welfare:

- **Drivers:** Socio-economic factors influencing environmental change.
- **Pressures:** Activities that exert pressure on the environment.
- **State:** Environmental conditions resulting from pressures.
- **Welfare:** Impacts on human well-being and quality of life.
- **Response:** Actions taken to address impacts on welfare.

4. DPSEER Framework

The DPSEER framework highlights ecosystem services:

- **Drivers:** Human activities leading to environmental changes.
- **Pressures:** Actions that cause stress on ecosystems.
- **State:** Condition of the environment and its services.
- **Ecosystem Services:** Benefits provided by ecosystems to humans.
- **Response:** Strategies to maintain or restore ecosystem services.

Uses of the Frameworks

Each framework serves distinct purposes as shown below:



Figure-2: Uses of the Frameworks

Differences Between the Frameworks

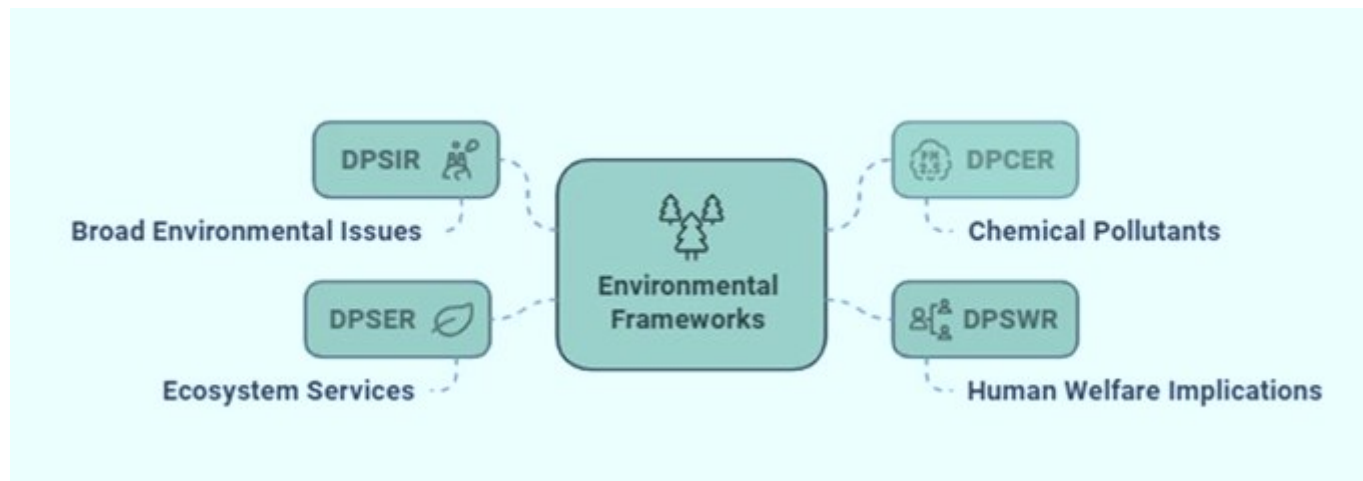


Figure-3: Differences Between the Frameworks

While all four frameworks share a common structure based on causal relationships, they differ in focus:

- **DPSIR** addresses a broad range of environmental issues without emphasizing specific aspects like chemicals or welfare.
- **DPCER** narrows down on chemical pollutants and their ecological impacts.
- **DPSWR** shifts attention to human welfare implications rather than purely ecological factors.
- **DPSEIR** integrates ecosystem services into the analysis, highlighting their role in human well-being.

Interrelationship Among the Frameworks

The frameworks are interconnected:

- The DPSIR framework serves as a foundation for both DPCER and DPSWR by providing a general structure for analysis.
- DPCER can be seen as a specialized application of DPSIR focused on chemical impacts, while DPSWR extends the discussion to human welfare outcomes.
- Similarly, DPSEIR builds upon DPSIR by incorporating ecosystem services into its analysis, reflecting the growing recognition of their importance in sustainability discussions.

Summary :

The DPSIR, DPCER, DPSWR, and DPSEIR frameworks provide valuable tools for understanding complex environmental issues. By recognizing their differences and interrelations, policymakers and researchers can choose appropriate frameworks for specific contexts, ultimately enhancing efforts toward sustainable development.

References

1. Gari et al. (2016). "DPSIR—Two Decades of Trying to Develop a Unifying Framework for Marine Environmental Management." *Frontiers in Marine Science*.
2. Allen, W. (2022). "Extending the DPSIR Framework." *i2Insights*.
3. GRID-Arendal (2005). "The DPSIR Framework."
4. US EPA (n.d.). "Systems Thinking Tutorials: Module 2: DPSIR Overview."
5. Antonescu, Ralu (2018). "Applying DPSIR Model to Sustainable Territorial Development." *Journal of Urban and Landscape Planning*.
6. Environmental Systems Research (2022). "Application of DPSIR Model to Identify Drivers and Impacts."
7. FAO (n.d.). "Driver-Pressure-State-Impact Response Framework."

DPSIR Framework for Sustainable Aquaculture in the Red Sea Region (KSA): A Climate Adaptation and Mitigation Perspective

By: Dr. Quosay A. Ahmed

This paper is a contribution to the 6th International Workshop on Applied Research for Enhancing the Aquaculture Sector in Saudi Arabia. It provides an overview of the DPSIR framework and its relevance to sustainable aquaculture development in the Red Sea, with a specific focus on aligning with Saudi Vision 2030 and global climate adaptation and mitigation goals.

Introduction

The Red Sea region, specifically along Saudi Arabia's coastline, represents a critical nexus of ecological, economic, and social significance. As one of the planet's most biodiverse and ecologically unique marine environments, the Red Sea is home to extensive coral reefs, seagrass meadows, and a wide variety of endemic marine species. These natural assets provide a foundation for economic diversification initiatives in Saudi Arabia, particularly in the aquaculture sector, which is identified as a key contributor to achieving the nation's Vision 2030 goals.

With increasing global emphasis on sustainable food systems, the aquaculture industry in the Red Sea holds potential not only for enhancing food security but also for addressing climate change challenges. However, this potential comes with substantial responsibilities, including mitigating environmental pressures such as habitat alteration, nutrient pollution, and overexploitation of marine resources. The development of aquaculture in such a sensitive ecosystem demands a nuanced, science-driven approach that integrates sustainability principles with innovative technological solutions.

The Drivers, Pressures, State of environment, Impacts, and Responses (DPSIR) framework provides a comprehensive method for analyzing and addressing the complex dynamics of aquaculture in the Red Sea. This article explores the application of the DPSIR framework in the Red Sea context, emphasizing its role in fostering a sustainable, resilient aquaculture sector that aligns with global climate goals and supports regional economic transformation.

DPSIR Framework for Aquaculture in the Red Sea Region

DPSIR (drivers, pressures, state, impact, and response model of intervention) is a causal framework used to describe the interactions between society and the environment. It seeks to analyze and assess environmental problems by bringing together various scientific disciplines, environmental managers, and stakeholders, and solve them by incorporating sustainable development.

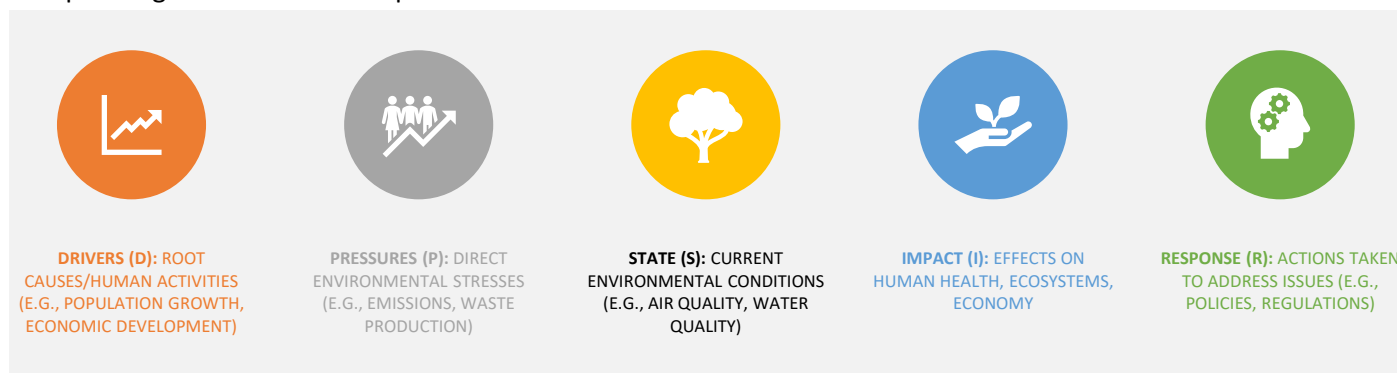


Figure 12: DPSIR framework

1. Drivers

Drivers are the underlying social, economic, and environmental factors that motivate aquaculture development in the Red Sea region.

- **Economic Growth and Diversification:** Saudi Arabia's Vision 2030 emphasizes economic diversification, with aquaculture identified as a key sector for reducing dependence on oil revenues.
- **Food Security:** The growing demand for sustainable protein sources drives the expansion of aquaculture.
- **Technological Advancements:** Innovations in aquaculture practices, such as recirculating aquaculture systems (RAS), facilitate development in arid and sensitive ecosystems.
- **Global Climate Commitments:** Saudi Arabia's pledge to reduce greenhouse gas emissions encourages the adoption of low-impact aquaculture practices.

The DPSIR framework provides a systematic approach to developing sustainable aquaculture in the Red Sea region, aligning with Saudi Arabia's Vision 2030 and global climate goals. By addressing the drivers, pressures, and impacts of aquaculture, and implementing targeted responses, Saudi Arabia can achieve a balance between economic growth, environmental conservation, and climate resilience.

2. Pressures

Pressures are the direct consequences of aquaculture activities that affect the marine environment.

- **Habitat Alteration:** Expansion of aquaculture infrastructure can disturb critical marine habitats such as coral reefs and seagrass beds.
- **Pollution:** Nutrient runoff, excess feed, and waste discharge contribute to eutrophication and reduced water quality.
- **Overexploitation:** Overharvesting of wild stocks for feed and broodstock can strain local ecosystems.
- **Climate Stressors:** Rising sea temperatures and increased salinity levels exacerbate pressures on aquaculture systems.

3. State of Environment

The state refers to the current conditions of the marine environment in the Red Sea region.

- **Marine Biodiversity:** The Red Sea hosts a unique and diverse array of marine species, many of which are sensitive to environmental changes.
- **Water Quality:** Increasing nutrient loads and pollutants from aquaculture operations threaten the pristine conditions of the Red Sea.
- **Climate Change Impacts:** Ocean acidification, warming seas, and sea level rise pose risks to aquaculture viability and marine ecosystem health.

4. Impacts

Impacts are the effects of changes in the state of the environment on ecosystems, human health, and socio-economic systems.

- **Ecosystem Degradation:** Loss of coral reefs and seagrass beds affects biodiversity and ecosystem services.
- **Economic Costs:** Degraded marine environments increase the costs of aquaculture operations, such as water treatment and disease management.
- **Community Livelihoods:** Disruptions to traditional fishing and tourism sectors can undermine local economies and social cohesion.
- **Contribution to Climate Change:** Unsustainable practices can exacerbate greenhouse gas emissions, counteracting mitigation efforts.

5. Responses

Responses are the actions taken to address the drivers, pressures, and impacts of aquaculture activities.

- | | |
|---------------------------------|---|
| a) Policy and Regulation | Implementing stringent environmental standards for aquaculture operations, including site selection and waste management. |
|---------------------------------|---|

b) Technological Innovations	Adoption of low-impact practices such as integrated multi-trophic aquaculture (IMTA) and renewable energy-powered systems.
c) Capacity Building	Training programs for local communities and stakeholders to promote sustainable aquaculture practices.
d) Climate Adaptation Strategies	Incorporating climate-resilient species and enhancing ecosystem-based approaches.
e) Monitoring and Research	Establishing robust systems to monitor environmental parameters and assess aquaculture's impacts.

Linking DPSIR to Climate Adaptation and Mitigation

Climate Change Impacts and Opportunities

Aquaculture in the Kingdom of Saudi Arabia (KSA) and the Gulf Cooperation Council (GCC) region presents both challenges and opportunities when examined through the lens of climate change, considering the ambitious nationally determined contributions (NDCs) and net-zero targets.

Table 10: Climate Impacts and Opportunities on the Aquaculture

Impacts	Opportunities
Temperature Rise: Increasing sea temperatures threaten aquaculture operations by affecting fish health, growth rates, and species viability.	Low-Carbon Protein Production: Aquaculture offers a sustainable alternative to land-based protein sources, aligning with GCC nations' efforts to decarbonize food systems.
Extreme Weather Events: Frequent storms and flooding events pose risks to aquaculture infrastructure and operations.	Carbon Sequestration: Integrating seaweed and other algae farming can act as carbon sinks, directly contributing to GHG reduction targets.
Sea-Level Rise: Coastal aquaculture operations face infrastructure risks, requiring adaptation in farm design and site selection.	Circular Economy Models: Utilizing aquaculture waste in Integrated Multi-Trophic Aquaculture (IMTA) systems can reduce environmental impacts and enhance resource efficiency.
Biodiversity Loss: Shifts in marine ecosystems due to warming waters can disrupt food chains critical for aquaculture species.	Green and Climate Finance: Access to funds through mechanisms like the Green Climate Fund or GCC's voluntary carbon markets can support sustainable aquaculture transitions.
Ocean Acidification: Higher levels of CO ₂ absorption by oceans degrade water quality and marine ecosystems, such as coral reefs that are essential for biodiversity and aquaculture.	Energy Transition: Incorporating renewable energy into aquaculture operations, such as solar-powered water pumps and aeration systems, aids in meeting NDCs and reducing operational emissions.
	Nature-Based Solutions: Mangrove and seagrass restoration alongside aquaculture development provides coastal protection, biodiversity enhancement, and carbon storage.

Climate Adaptation

The DPSIR framework helps identify vulnerabilities and adaptive measures in aquaculture practices:

1. **Adaptation to Temperature Changes:** Using thermotolerant species and adjusting farming schedules.
2. **Resilient Infrastructure:** Designing systems to withstand extreme weather events and sea-level rise.
3. **Ecosystem Restoration:** Integrating habitat restoration with aquaculture development to enhance coastal resilience.

Climate Mitigation

Aquaculture can contribute to climate mitigation through:

1. **Carbon Sequestration:** Seaweed farming acts as a natural carbon sink.
2. **Reducing Land Use Pressure:** Offshore aquaculture reduces reliance on terrestrial ecosystems.
3. **Energy Efficiency:** Transitioning to solar and wind energy for aquaculture operations.

Climate adaptation measures include utilizing thermotolerant species, designing resilient infrastructure to endure extreme weather and rising sea levels, and integrating habitat restoration with aquaculture to bolster coastal resilience. On the mitigation front, scaling seaweed farming for carbon sequestration, shifting towards offshore aquaculture to minimize land use pressure, and transitioning to renewable energy sources in aquaculture operations are key strategies.

Way Forward

Given the capabilities of King Abdullah University of Science and Technology (KAUST) and its existing collaborations with governmental and private sectors both locally and internationally, the following steps are recommended:

Leverage KAUST's Research Expertise

1. **Advanced Monitoring Systems:** Utilize KAUST's research in remote sensing, marine biology, and big data analytics to monitor and assess aquaculture impacts in real-time.
2. **Innovative Technologies:** Develop cutting-edge technologies, such as bioengineering for climate-resilient species and renewable energy integration into aquaculture systems.
3. **Ecosystem Restoration Projects:** Partner with KAUST's marine science experts to implement habitat restoration programs, such as coral and seagrass rehabilitation.

Strengthen Public-Private Partnerships

1. **Collaborative Projects:** Facilitate joint ventures between KAUST, Enerqa, and private sector players like Beacon to pilot sustainable aquaculture models.
2. **Policy Advocacy:** Leverage Enerqa's expertise to advocate for science-backed policies that promote sustainable aquaculture and align with ESG standards.
3. **Capacity Building Programs:** Develop training and education initiatives for stakeholders, emphasizing sustainability and innovation in aquaculture practices.

Support Climate Change Mitigation and Adaptation

1. **Carbon Footprint Assessment:** Collaborate with Enerqa to audit / conduct life cycle assessments (LCAs) for aquaculture systems, ensuring alignment with international carbon accounting standards (e.g., ISO 14040).
2. **Climate Finance Access:** Leverage KAUST's and Enerqa's expertise to secure climate finance through mechanisms such as the Green Climate Fund and voluntary carbon markets.
3. **Green Technology Deployment:** Utilize renewable energy technologies in aquaculture operations to reduce greenhouse gas emissions.

Drive Innovation Through International Collaboration

1. **Knowledge Sharing:** Partner with global institutions to exchange best practices and deploy scalable solutions in aquaculture and climate adaptation.
2. **Investment Attraction:** Showcase projects jointly developed by KAUST, Beacon, and Enerqa to attract funding for climate-smart aquaculture.
3. **Research on Nature-Based Solutions:** Explore natural approaches like mangrove and seagrass integration with aquaculture systems, leveraging Beacon's regional expertise.

Promote ESG and Sustainability Standards

1. **ESG Integration:** Introduce comprehensive ESG frameworks for aquaculture, incorporating metrics for environmental impact, social inclusion, and governance.
2. **MRV System Implementation:** Partner with Enerqa to establish robust measurement, reporting, and verification systems for aquaculture's climate impacts, aligning with UNFCCC requirements.
3. **Capacity Building:** Equip local stakeholders with tools and knowledge to implement sustainable aquaculture practices in alignment with international standards.

Given KAUST's capabilities and collaborations, several strategic steps are recommended. Leveraging research expertise involves advancing monitoring systems, developing bioengineered species, and restoring marine ecosystems. Strengthening partnerships between KAUST, Enerqa, and private entities can pilot sustainable aquaculture models, while accessing climate finance through global mechanisms and carbon markets can fund these initiatives. Promoting ESG integration requires introducing robust frameworks and measurement, reporting, and verification (MRV) systems to meet sustainability targets. Driving international collaboration enables the exchange of best practices and scaling of successful solutions globally.

References

1. Food and Agriculture Organization of the United Nations (FAO). (2020). "Sustainable Aquaculture Development." Retrieved from www.fao.org.
2. Saudi Vision 2030. (2023). "National Transformation Program Goals for Aquaculture." Retrieved from www.vision2030.gov.sa.
3. Duarte, C. M., et al. (2021). "The Role of Marine Aquaculture in Climate Mitigation." *Nature Communications*, 12(1), 1-10.
4. Red Sea Development Company. (2022). "Marine Biodiversity Conservation and Sustainable Practices in the Red Sea." Retrieved from www.theredsea.sa.
5. IPCC. (2019). "Special Report on the Ocean and Cryosphere in a Changing Climate." Retrieved from www.ipcc.ch.

The Imperative for ESG Readiness Tools: Unlocking Benefits for Companies, Institutions, and Society

By enerQA's development team



1. Introduction

Environmental, Social, and Governance (ESG) considerations have evolved from being a corporate trend to becoming essential criteria for sustainable and ethical growth. Across the globe, governments, companies, and institutions are increasingly aligning their operations with ESG principles to ensure compliance with global sustainability targets and the United Nations' Sustainable Development Goals (SDGs).

In 2022, over 90% of the world's largest companies published ESG reports (KPMG Survey of Sustainability Reporting 2022), yet less than 50% of SMEs globally had a clear ESG strategy. This gap underscores the need for readiness tools tailored to diverse organizational scales. On the other hand, ESG non-compliance increasingly translates to financial risks. The Global Sustainable Investment Alliance (GSIA) reported that \$35.3 trillion, or about one-third of global assets under management, were in ESG-integrated investments by 2020. Companies failing to meet ESG criteria risk exclusion from these funds.

ESG readiness tools have emerged as critical instruments in this transition, enabling entities to systematically assess, report, and improve their ESG performance. In the Middle East and North Africa (MENA) region, these tools are proving to be not only transformative for businesses but also pivotal in addressing national priorities related to sustainability, governance, and societal well-being.

2. Do we need to report our ESG actions?

ESG reporting has become an indispensable tool for companies and institutions, enabling them to measure and communicate their performance in environmental, social, and governance domains. This transparency fosters trust among stakeholders, including investors, customers, and regulators, while aligning businesses with societal values. As global challenges like climate change and inequality intensify, ESG reporting helps organizations demonstrate their commitment to sustainability and ethical governance, fulfilling growing demands for accountability and responsibility. One of the primary benefits of ESG reporting is its ability to attract investment. Sustainable funds and investors prioritize ESG-compliant businesses, with global assets under ESG management exceeding \$41 trillion in 2022. Additionally, it aids in risk management by identifying vulnerabilities in areas like climate impact, regulatory compliance, and social equity. By addressing these risks, companies not only safeguard their operations but also enhance their resilience and adaptability in a dynamic market landscape.

Beyond financial and operational advantages, ESG reporting plays a pivotal role in aligning organizations with global and national sustainability goals. It supports contributions to frameworks such as the United Nations' Sustainable Development Goals (SDGs) and national climate strategies, enabling measurable progress toward clean energy, responsible production, and social equity. Furthermore, reporting drives innovation by identifying inefficiencies, encouraging resource optimization, and fostering the adoption of sustainable practices and technologies.

In a rapidly evolving regulatory environment, ESG reporting ensures compliance with local and international standards, mitigating legal and reputational risks. It also enhances long-term value creation by integrating sustainability into corporate strategies, benefiting not only businesses but also communities and ecosystems. As more companies embrace ESG reporting, it will serve as a cornerstone for global progress, ensuring a future that balances economic growth with environmental and social stewardship.

3. Why ESG Readiness Tools Are Essential

ESG readiness tools serve as structured frameworks that allow companies and institutions to evaluate their ESG performance and readiness systematically. These tools typically encompass:

1. **Assessment Criteria:** Tools include detailed indicators across environmental, social, and governance dimensions, enabling organizations to benchmark their practices against global standards.
2. **Guidance and Strategy:** They provide actionable insights for improvement, helping entities align operations with sustainability and governance expectations.
3. **Transparency and Accountability:** By documenting practices, they foster trust among stakeholders, including investors, regulators, and communities.

4. The Role of ESG Readiness Tools in the MENA Region

The adoption of ESG frameworks in the MENA region has gained momentum in recent years. Countries such as Qatar, United Arab Emirates (UAE) and Saudi Arabia have introduced ambitious sustainability agendas aligned with their national visions. ESG reporting and readiness tools are playing an increasingly central role in translating these ambitions into actionable strategies for organizations.

Qatar, a major player in the global energy sector, has also embraced ESG principles. With its National Vision 2030 emphasizing sustainability, organizations in the country are increasingly turning to readiness tools to assess and improve their ESG standings. Companies like Qatar National Bank (QNB) have integrated ESG reporting into their operations, using structured tools to evaluate their carbon footprint and align their investments with green financing principles. These efforts not only contribute to Qatar's national climate commitments but also enhance the global appeal of its corporate sector.

In the UAE, the Securities and Commodities Authority (SCA) mandated ESG disclosure for all listed companies starting in 2022. This regulation has encouraged firms to utilize readiness tools to align their operations with the requirements. Companies like Majid Al Futtaim, a leading retail and hospitality group, have adopted ESG reporting mechanisms, helping them enhance energy efficiency across their properties and align with the UAE's Net Zero 2050 strategic

initiative. Similarly, in Saudi Arabia, the Vision 2030 framework emphasizes sustainability and good governance as cornerstones of the Kingdom's transformation. The Tadawul stock exchange introduced the ESG Disclosure Guidelines in 2021, compelling organizations to evaluate their environmental and social impact. The use of readiness tools in these contexts has enabled firms to identify gaps in compliance and develop targeted action plans.

4.1. Benefits for Organizations in the MENA Region

Organizations adopting ESG readiness tools in the MENA region are reaping significant benefits. For instance, compliance with ESG frameworks has proven to attract foreign investment. In 2021, the MENA region recorded a surge in sustainable financing, with green bonds and loans exceeding \$18 billion, according to the Climate Bonds Initiative. Firms with robust ESG reporting mechanisms were at the forefront of these opportunities, as international investors increasingly prioritize ESG-compliant markets.

4.2. Benefits for Nations and Society

In the MENA region, where economies are heavily reliant on hydrocarbons, these tools are instrumental in supporting the transition to greener industries. For example, the UAE's investments in renewable energy projects such as Masdar City demonstrate how ESG-aligned initiatives can catalyze economic growth while reducing dependency on fossil fuels. Similarly, Saudi Arabia's \$500 billion NEOM project integrates ESG principles to create a sustainable city powered entirely by renewable energy, showcasing how readiness tools can guide the implementation of visionary projects. The societal benefits of ESG adoption in the MENA region are equally profound. Improved corporate governance fosters trust in institutions, while environmental initiatives enhance public health by reducing pollution and promoting clean energy. Social criteria in ESG frameworks, such as gender diversity and labor rights, are driving positive change in workplaces, contributing to more equitable and inclusive societies. For instance, companies like Emaar Properties in the UAE have committed to enhancing gender diversity within their leadership teams, a commitment driven by ESG evaluations.

4.3. The Future of ESG in the MENA Region

The widespread adoption of ESG tools and reporting in the MENA region is likely to accelerate progress toward achieving national and global sustainability goals. As more companies and institutions embrace these frameworks, the region can position itself as a leader in sustainable development. Predictive models suggest that by 2030, nations that adopt ESG principles extensively could see a 10-15% increase in foreign direct investment (FDI) as global markets increasingly prioritize ethical and sustainable practices. This trend is particularly relevant for the MENA region, where diversification away from oil-based economies is a pressing priority.

In addition, the adoption of ESG readiness tools can pave the way for regional cooperation on sustainability issues. Shared frameworks and standards will enable countries to collaborate on cross-border challenges such as water scarcity and climate resilience. For example, the GCC countries could use ESG readiness as a platform to develop unified policies on renewable energy and carbon trading, creating a competitive and sustainable regional market.

5. EnerQA's ESG readiness tool

EnerQA has launched a free-to-download ESG Readiness Tool, designed on Microsoft Excel to provide an intuitive, user-friendly solution for assessing and improving ESG performance. The tool simplifies the evaluation of Environmental, Social, and Governance pillars through structured criteria and guiding questions, allowing users to input their practices, score their readiness, and receive tailored recommendations for improvement.

The tool's Excel format makes it accessible and customizable, enabling organizations of all sizes, especially SMEs, to adapt it to their specific needs without requiring additional software or expertise. By offering actionable insights based on scoring, it empowers users to address ESG gaps and align with sustainability and governance standards efficiently.

EnerQA's tool is an ideal resource for companies aiming to enhance transparency, meet stakeholder expectations, and align with global goals. Download it now and take the first step toward measurable ESG excellence.

ESG Readiness Checklist

Date: _____
Company: _____
Ref. No.: _____
Rep. Name: _____
Contract No.: _____

Pillar	Criteria	Questions to Consider	1	2	3	4	5
Environmental	Energy Management	Does the organization have an energy efficiency policy?					
	Renewable Energy Use	What percentage of energy comes from renewable sources?					
	Emissions	What are the total greenhouse gas emissions?					
	Waste	Is there a waste management strategy in place?					
	Water Usage	How is water usage monitored and reported?					
	Biodiversity Impact	Does the organization assess its impact on local biodiversity?					
Social	Sustainable Sourcing	Are materials sourced sustainably?					
	Regulatory Compliance	Is there compliance with local and international environmental laws?					
	Climate Risk Assessment	Has the organization conducted climate risk assessments?					
	Climate Mitigation and Adaptation	Has the firm applied any climate change mitigation or adaptation measures?					
	Labour Practices	Are labour practices in line with international labour standards?					
	Diversity and Inclusion	Does the organization have diversity and inclusion policies?					
Governance	Employee Engagement	How is employee engagement measured?					
	Community Engagement	What initiatives exist for community involvement?					
	Health and Safety	Are health and safety policies clearly defined and enforced?					
	Human Rights	Is there a commitment to uphold human rights in operations?					
	Training and Development	Are there programs for employee training and career development?					
	Board Diversity	Does the board reflect diversity in gender, race, and skills?					
Transparency and Reporting	Stakeholder Engagement	How are stakeholders engaged in decision-making processes?					
	Risk Management	Are there clear risk management processes in place?					
	Ethical Conduct	Is there a code of ethics that guides organizational behavior?					
	Compliance and Accountability	Are there mechanisms for ensuring compliance with laws and regulations?					
	Executive Compensation	Is executive compensation linked to sustainability performance?					
	Political Contributions	Does the organization disclose political contributions and lobbying activities?					

Score

1. No readiness. The organization has not yet begun addressing this ESG issue.
2. Low readiness. The organization has some awareness but has not implemented formal processes.
3. Moderate readiness. The organization has started addressing the issue, but there are gaps in consistency or depth.
4. High readiness. The organization has implemented most practices but could improve in certain areas.
5. Full readiness. The organization has fully integrated this issue into its strategy and practices, and can report comprehensively.

Initial Outcomes

Pillar	Score
Environment	35%
Social	60%
Governance	60%

Overall ESG Readiness Score 70 (%) 84%

Current Status Moderate Readiness

References:

1. **IBM Institute for Business Value.** (2021). Sustainability at a Turning Point. Available at: <https://www.ibm.com/thought-leadership/institute-business-value/report/sustainability>
2. **Bloomberg Intelligence.** (2022). ESG Assets May Hit \$41 Trillion by 2022 but Not Without Challenges. Available at: <https://www.bloomberq.com>
3. **United Nations Sustainable Development Goals.** (2015). Transforming our World: The 2030 Agenda for Sustainable Development. Available at: <https://sdgs.un.org/2030agenda>
4. **Harvard Business Review.** (2020). The Financial Benefits of ESG Integration. Available at: <https://hbr.org>
5. **Climate Bonds Initiative.** (2021). Green Bonds Market Report. Available at: <https://www.climatebonds.net>
6. **UAE Securities and Commodities Authority.** (2022). Mandatory ESG Disclosure Guidelines. Available at: <https://www.sca.gov.ae>
7. **Saudi Tadawul Group.** (2021). ESG Disclosure Guidelines. Available at: <https://www.saudiexchange.sa>
8. **United Nations Framework Convention on Climate Change (UNFCCC).** (2015). Paris Agreement. Available at: <https://unfccc.int/process-and-meetings/the-paris-agreement>
9. Climate Bonds Initiative, 2021 Green Bonds Market Report
10. UAE Securities and Commodities Authority, ESG Disclosure Mandates
11. Saudi Tadawul Group, ESG Disclosure Guidelines
12. Qatar National Vision 2030, Sustainability Goals
13. Harvard Business Review, The Financial Benefits of ESG Integration

Scope 4 Emissions: The Concept of Avoided Emissions

By: Yousif Ibrahim Hamza



In the global effort to combat climate change, the focus has traditionally been on reducing greenhouse gas (GHG) emissions from corporate operations and value chains.

The Greenhouse Gas Protocol, the most widely used framework for carbon accounting, categorizes emissions into three scopes:

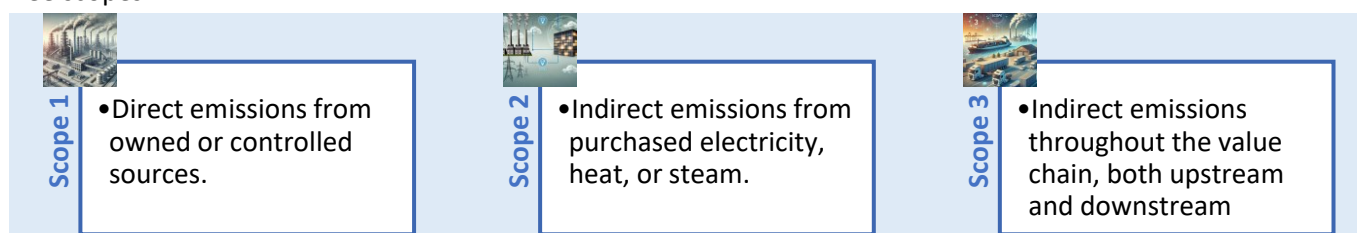


Figure 13: Categorizes Emissions According to the GHG Protocol

However, an additional concept often referred to as Scope 4 emissions has gained attention [10], [11]. While not officially recognized by the GHG Protocol, Scope 4 refers to avoided emissions, or emissions reductions that occur outside a company's direct operations or value chain as a result of the use of its products, services, or initiatives [12].

Scope 1

Direct Emissions



These are direct emissions from sources owned or controlled by the organization. They come from activities like burning fuel on-site or using company-owned vehicles. For example, emissions from a company's natural gas boiler or a delivery truck running on diesel fall under Scope 1.

Scope 3

Indirect Emissions Across the Value Chain



These are indirect emissions from the organization's value chain, both upstream (from suppliers) and downstream (from customers). They include emissions from activities like raw material production, product use, and disposal. For example, emissions from a supplier's manufacturing process or the fuel burned by customers driving a company's sold vehicles are part of Scope 3.

Scope 2

Indirect Emissions from Energy



These are indirect emissions from the generation of energy the organization purchases and uses. Although the emissions happen at the energy producer's site (e.g., power plant), they are attributed to the organization because it consumes the energy. For example, an office using electricity from a coal-fired power plant contributes to Scope 2 emissions.

Scope 4

Avoided Emissions



These refer to emissions avoided through the use of a product outside the organization's value chain. For instance, while an electric vehicle (EV) generates emissions during its production and through electricity consumption from the grid, it also prevents emissions that would have occurred if a traditional internal combustion engine vehicle were used instead.

Figure 14: Overview of Emission Categories: Scopes 1, 2, 3 and Scope 4, (Prepared by Yousif Mohammed)

What Is Scope 4 Emissions?

Avoided emissions occur when an innovation or intervention reduces GHG emissions compared to a baseline or business-as-usual scenario. Unlike Scopes 1, 2, and 3, which quantify existing emissions, Scope 4 highlights the positive impact of alternative solutions [11], [13]. Key examples include:

- Renewable Energy: Solar or wind energy displacing fossil fuel-based power generation.
- Low-Carbon Products: Electric vehicles replacing internal combustion engines, or energy-efficient appliances reducing electricity consumption.
- Circular Economy Initiatives: Recycling materials to avoid the emissions of producing virgin resources.
- Behavioral Programs: Promoting telecommuting or carpooling to reduce transportation emissions.

Why Do Scope 4 Emissions Matter?

Scope 4 emissions demonstrate how companies can extend their influence beyond their operational boundaries, contributing to global decarbonization efforts. The potential benefits include: [14]

1. Showcasing Climate Leadership: Highlighting how your organization's solutions address climate change at scale.
2. Enhancing Market Competitiveness: Differentiating your products and services as sustainable, appealing to environmentally conscious customers.
3. Strengthening Stakeholder Engagement: Attracting investors and partners who prioritize sustainability in their decision-making.
4. Fostering Innovation: Inspiring the development of climate-smart technologies and services.



Figure 15: Benefits of Addressing Scope 4 Emissions, (Created by Yousif Mohammed)

Challenges in Scope 4 Implementation

While the concept of Scope 4 offers significant potential, its adoption comes with challenges [10]:

1. **Standardization Gaps:** The absence of globally accepted frameworks can lead to inconsistent interpretations.
2. **Risk of Double Counting:** Overlapping claims of avoided emissions by multiple entities in the value chain may compromise reporting accuracy. [11]
3. **Complex Verification Processes:** Calculating avoided emissions requires robust data and meticulous life-cycle analysis.

Driving Change Through Scope 4

Although not formally part of the GHG Protocol, Scope 4 emissions provide a powerful narrative of how companies can act as catalysts for global emissions reductions. By embracing this concept, organizations can lead the charge in the transition to a low-carbon economy.

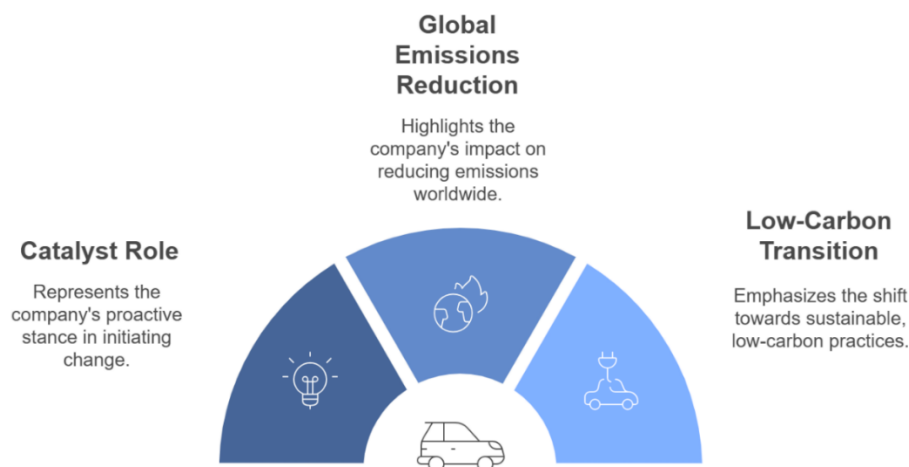


Figure 16: Driving Change Through Scope 4, (Prepared by Yousif Mohammed)

References

- [1] "Scoping Out: Tracking Nature Across The Supply Chain," CDP, 2023.
- [2] E. Thomson, "Droughts are creating new supply chain problems. This is what you need to know," World Economic Forum, 25 October 2023. [Online]. Available: <https://www.weforum.org/agenda/2023/10/drought-trade-rivers-supply-chain/>. [Accessed 25 September 2024].
- [3] Google, "Google," [Online]. Available: (DeepMind AI reduces energy used for cooling Google data centers by 40%).
- [4] Tesla, "Tesla," [Online]. Available: Best Tesla Powerwall in 2024: Here.
- [5] BCG, [Online]. Available: <https://www.bcg.com/publications/2023/how-ai-can-speedup-climate-action?utm>.

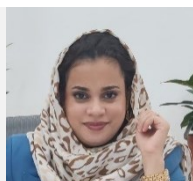
- [6] IBM, [Online]. Available: <https://newsroom.ibm.com/2023-03-14-IBM-launches-RFP-to-help-accelerate-global-water-management-solutions-for-vulnerable-populations?utm>.
- [7] Seoul. [Online]. Available: <https://english.seoul.go.kr/seoul-to-increase-recycling-rate-to-79-by-2026/?utm>.
- [8] Microsoft. [Online]. Available: <https://news.microsoft.com/apac/2021/10/11/microsoft-ai-for-earth-named-as-one-of-100-outstanding-biodiversity-positive-practices-and-actions-around-the-world/#:~:text=Launched%20in%202017%2C%20Microsoft%E2%80%99s%20%E2%80%9CAI%20for%20Earth%E2%80%9D%20is>.
- [9] A. M. Research. [Online]. Available: Artificial Intelligence (AI) Market Size, Share, Trends – 2033).
- [10] I. o. S. Studies, 14 October 2024. [Online]. Available: <https://instituteofsustainabilitystudies.com/insights/lexicon/what-are-scope-4-emissions/>.
- [11] W. B. C. f. S. D. (WBCSD), *Guidance on Avoided Emissions: Helping business drive innovations and scale solutions toward Net Zero*, Geneva, Amsterdam, New Delhi, London, New York, Singapore: World Business Council for Sustainable Development (WBCSD), 2023.
- [12] W. R. INSTITUTE, "WORLD RESOURCED INSTITUTE," 05 November 2013. [Online]. Available: <https://www.wri.org/insights/do-we-need-standard-calculate-avoided-emissions>.
- [13] M. D. G. G. C. L. M. A. Lauren Shwisberg, "How to Build Clean Energy Portfolios A Practical Guide to Next-Generation Procurement Practices," Rocky Mountain Institute, Basalt, CO, 81621 USA, 2020.
- [14] P. A. [Online]. Available: <https://plana.earth/glossary/scope-4-emissions>.
- [15] P. R. Attack, "Panama Canal Authority blames queues on drought and draft cut," *Ship-Technology.com*, 11 August 2023. [Online]. Available: <https://www.ship-technology.com/news/panama-canal-authority-blames-queues-drought-draft-cut/>. [Accessed 25 September 2024].
- [16] . J. Verschuur, E. E. Koks and J. W. Hall, "Systemic risks from climate-related disruptions at ports," *Nature Climate Change*, vol. 13, no. 8, p. 804–806, 2023.
- [17] L. Woetzel, D. Pinner, . H. Samandari, H. Engel, M. Krishnan, C. Kampel and J. Graabak, "Could climate become the weak link in your supply chain?," McKinsey Global Institute, 6 August 2020. [Online]. Available: <https://www.mckinsey.com/capabilities/sustainability/our-insights/could-climate-become-the-weak-link-in-your-supply-chain>. [Accessed 25 September 2024].
- [18] "Climate-Related Supply Chain Risk: Are You Prepared? - FTI Strategic Communications," FTI Consulting, 15 December 2022. [Online]. Available: <https://fticomunications.com/climate-related-supply-chain-risk-are-you-prepared/>. [Accessed 25 September 2024].
- [19] G. G. P. (. Protocol). [Online]. Available: <https://ghgprotocol.org/standards-guidance>.

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Dr. Quosay is an Assistant Professor of Petroleum Engineering, and an environmental and sustainability expert. He has extensive experience in greenhouse gas emissions reporting, environmental impact assessments, and sustainability initiatives. Dr. Quosay has collaborated with leading international organizations like the UNEP, UNDP, UNIDO, UNFCCC, and IPCC. He is skilled in preparing comprehensive greenhouse gas inventories, climate change mitigation strategies, and environmental assessments, aligning them with global standards. He has also number of publications in oil and gas, energy and transport sectors.



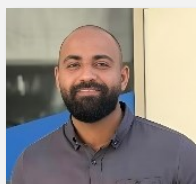
Islam Awad, PhD

A registered Technical Expert Reviewer in the UNFCCC Roster of Experts, and a certified climate and renewable energy finance expert. Dr. Islam brings specialized expertise in renewable energy finance, sustainable business models, and mainstreaming climate finance into national policies. She has contributed to key national-level initiatives, including the Capacity Building Initiative for Transparency (CBIT), and has collaborated with governments and international organizations to deliver impactful, results-driven projects.



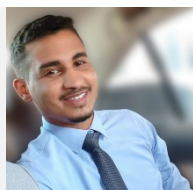
Yousif Ibrahim Hamza

Eng. Yousif Mohammed is an accomplished Environmental and Sustainability Specialist with over 10 years of experience in the oil and gas, cement, and mining industries. He has expertise in Environmental Impact Assessments, greenhouse gas management, and ESG reporting, with a strong focus on advancing sustainability in industrial operations. Currently spearheading initiatives at Enerqa Consultancy, Yousif is dedicated to driving environmental stewardship and fostering sustainable development in Qatar and beyond.



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Yasen Hatim is a skilled process engineer and environmental consultant specializing in energy management, renewable energy systems, GHG emission calculations, and sustainable industrial practices. A certified ISO 50001 Lead Energy Auditor, he has led projects in environmental assessments, feasibility studies, and energy optimization. Yasen is committed to sustainability, creative problem-solving, and advancing solutions that balance operational efficiency with environmental responsibility.



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Amr Nasreldin is a registered Technical Expert Reviewer in the UNFCCC Roster of Experts and a Certified Energy Manager (CEM) from the Association of Energy Engineers. His expertise encompasses technical analysis, climate change mitigation strategies, and policy development and assessment within the energy sector.



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A chemical engineer actively working to advance the renewable energy sector and reduce our reliance on fossil fuels. By educating the next generation, she aims to inspire a new wave of environmental stewards who will drive positive change for a sustainable future.

**Abdelaziz Ahmed**

Abdelaziz is a skilled data analyst with a strong background in engineering and sustainability. With a B.Eng. in Electrical and Electronics Engineering and a Data Analytics Professional Certificate, he has contributed to impactful sustainability projects with organizations like the UNDP and AFDP. He is passionate about leveraging data analytics to drive business growth and contribute to a sustainable future.

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Dr. Muzamil Hassan is an Instructor in Chemical Engineering at the University of Doha for Science and Technology. His research focuses on heterogeneous catalysis and sustainable energy conversion. He earned his Ph.D. from the University of Yamanashi in 2018, where his research on catalytic partial oxidation of methane under supercritical water conditions was funded by the Japanese government.

**Giovanni Fabbio**

Giovanni Fabbio is a seasoned financial professional with extensive experience in navigating complex financial landscapes across the Middle East, Europe, and Asia. He specializes in M&A, turnaround strategies, and financial restructuring. With a focus on sustainable growth and profitability, Giovanni has successfully led teams in various sectors, including manufacturing, recycling, IT, and construction. He is adept at identifying and addressing inefficiencies, optimizing operations, and driving financial performance.

**Mohamed
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Mohamed is a certified IBM data analyst with a strong background in engineering and business. He has experience in the energy and climate change sector, leveraging his skills in data analysis, wrangling, and modeling using tools like Excel, SQL, and Python. He is also proficient in data visualization and storytelling using Tableau and Power BI. His engineering background, particularly in oil and gas, coupled with his business acumen, makes him well-equipped to analyze complex data and provide valuable insights.

